Applied Mathematics Body and Soul, Vol. V

Real ThermoDynamics

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Foreword

The Body and Soul programme, begun three decades ago, has from the start been organised around a single methodological idea: to combine the *Soul* of analytical mathematics with the *Body* of computational mathematics, so that every part of mathematical physics is presented in a form which pairs the underlying analytical structure with a runnable computational realisation. Volumes I–III carried this commitment through *Calculus* and *Linear Algebra*, each subject presented in both its analytical and its computational form. **Volume IV**, on *Turbulent Incompressible Flow* [23], took the programme one step further: the computational form was put forward as the *true mathematical model of physics*, based on the view that physical reality, evolving in time, itself performs an analog computation which the digital computer can parallel or simulate. Turbulence, which had resisted understanding within the purely analytical framework, could then be understood and simulated through a computational form of the Navier–Stokes equations. The **present Volume V** extends the same mathematics from incompressible to compressible flow, where it becomes the essence of *thermodynamics* — with an understandable computational form of the 2nd Law as the inevitable dissipation of finite-precision computation. **Volume VI**, *Real Quantum Mechanics* [1], extends the same mathematics again, from macroscopic compressible flow down to the atomic scale, replacing the classical wave function in $3N$-dimensional configuration space with a multiphase 3D continuum formulation in real three-dimensional space. The two volumes V and VI together complete the Body and Soul programme by carrying the combined analytical/computational point of view across the full scale range from macroscopic flows to atomic structure, with statistical aggregation replaced by deterministic field equations at both ends.

The thesis in one sentence. The 2nd Law of thermodynamics, irreversibility, and the arrow of time are not consequences of statistical aggregation over configurations of atoms. They are consequences of **finite-precision computation** on deterministic continuum field equations: any finite-resolution numerical solution of the Euler equations of compressible flow dissipates kinetic energy into internal energy through the small-scale residual that the resolution cannot represent, and this dissipation is irreversible by the structure of the discretisation, not by any statistical-mechanical assumption. The book argues that this is the right reading of the 2nd Law: a structural consequence of the finite information content of any computable representation of the physical world, including the physical world's own representation of itself as analog computation.

The two central problems: Joule's experiment and the piston-cylinder. The book is organised around two foundational computational problems that together carry the conceptual content of thermodynamics:

- **Joule's 1845 experiment** (Chapter 7) — gas in a high-pressure chamber expands through a channel into a low-pressure chamber. Bulk flow through the channel converts internal energy into kinetic energy during the expansion; turbulent/shock dissipation in the receiving chamber converts it back into heat during the recompression. The two halves are asymmetric in the dynamic dissipation pattern, and the result is a residual temperature gradient between the chambers that classical equilibrium thermodynamics cannot predict. This is the **refrigerator principle**: the same expansion-cooling and compression-heating asymmetry that a refrigeration cycle exploits with external work input, demonstrated in one-shot form by simply releasing a pressure differential between two chambers.

- **The piston-cylinder** (Chapter 19) — gas in a chamber pushes a movable wall (piston) outward against an external load, doing work on the load. With a smooth single-chamber geometry the gas does work quasi-statically and the process approaches reversible efficiency; with a channel constriction between the gas reservoir and the piston (the Joule geometry with a movable far wall), the gas must jet through the constriction and recompress against the piston, with turbulent dissipation pulling the efficiency $\eta = W/\Delta U$ below the reversible limit. This

is the **heat-engine principle**: the conversion of stored internal energy into useful work delivered to an external load, with irreversibility entering through the same turbulent-dissipation channel as in Joule's experiment.

The two problems are duals of each other in the deterministic-continuum reformulation. The Joule experiment shows what happens when the asymmetric expansion-compression cycle is allowed to dissipate freely; the piston-cylinder shows what happens when the same dynamics are channelled into useful work against a load. The 2nd Law in its RealNSE reading — finite-precision dissipation in the discretisation — sets the irreversibility in both cases. Refrigeration cycles run the first problem cyclically with external work input; heat engines run the second problem cyclically with heat input from a reservoir. Vol VI Applications develops both into full cycles (Carnot heat engines, Carnot heat pumps, compressor refrigerators); the present opening chapters establish the foundational single-shot dynamics that the cycles are built from.

Real Thermodynamics in the Body and Soul tradition. The earlier volumes of the programme presented continuum mechanics for macroscopic matter: fluids, elastic solids, plasticity, transport. This volume presents thermodynamics inside the same continuum-mechanics framework, without adding a statistical-mechanical layer. The Euler and Navier–Stokes equations carry the dynamics; the *Real Navier–Stokes* method (RealNSE) — 'Real' here meaning the computational realisation as a least-squares stabilised Galerkin method — supplies the finite-precision computation; the 2nd Law emerges as the entropy condition for weak solutions in the same sense that it does for shocks in classical conservation-law theory. The Boltzmann–Gibbs apparatus — microstates, ensembles, the probabilistic foundation of equilibrium thermodynamics — is not invoked at the foundational level. Where statistical reasoning is the natural framing for a specific regime (large-system equilibrium fluctuations, critical phenomena) it remains available; what is replaced is the claim that statistics is the foundation of the discipline.

The programme as mathematics education, and as it is read today. The Body and Soul mathematics education programme was initiated thirty years ago, around the proposition that a modern undergraduate education in mathematics should be a synthesis of the formal analytical mathematics of

Calculus — the *Soul* of the subject — with the computational mathematics made possible by the computer — the *Body*. The programme was met with resistance from both directions. Established mathematics teachers, trained in formal analysis but not in programming, found the computational half foreign to their craft and slow to incorporate. Computer scientists, comfortable with programming but uncomfortable with the apparatus of Calculus, regarded the analytical half as an obstacle to be routed around. The synthesis languished in the gap between the two cultures it had been designed to bridge. The idea was, however, correct, and with the recent advent of capable AI assistance it acquires a new and stronger sense: the analytical and the computational sides of mathematics no longer have to be carried by the same human in the same head. The mathematician's mind frames the problem, formulates the equations, identifies what must be shown; the AI translates the formulation into running code, executes the calculations, and brings the construction alive on the screen. The synthesis that the Body and Soul programme proposed in education thirty years ago becomes, with AI as collaborator, a practical working method for research.

The Body side and the FEniCS project. The Body side of the programme — the commitment that the computational realisation should be an organic part of the mathematics, not a separate engineering exercise — has its principal historical embodiment in the **FEniCS Project** [2], an open-source platform for the automated solution of partial differential equations by the finite element method. FEniCS was developed by the author's research group over roughly a decade of dedicated work, with the goal that a researcher who could write down a variational form on paper could obtain a running finite-element solver from it directly. The RealNSE method that supplies the computational backbone of this book was developed in that environment, and the original *Computational Thermodynamics* (Hoffman, Johnson and Nazarov, 2012) on which the present volume is based reported numerical results obtained with the FEniCS-Unicorn implementation.

The relevance to the present rewrite is the change of timescale. As Anders Logg, principal developer of FEniCS, has observed, what required a decade of focused effort in the 2000s can today, with the help of capable AI assistance, be carried out in a day. The browser-based interactive simulators that accompany this volume in the public Gallery — compressible flow demonstrations of Joule's 1845 experiment, of bluff-body Mach-2 turbulence, and of

the Burgers equation as the canonical model PDE — were produced on that compressed timescale. The Body side of the programme has not changed; what has changed is how much of it a single mathematical mind, with AI as collaborator, can carry from formulation to running code in a working session.

The GPU revolution: the third enabling condition. The Mind–AI collaboration is one of the enabling conditions for the present volume; the other is the GPU revolution of the past two decades. Graphics Processing Units originated as dedicated accelerators for 3D-graphics rendering in the 1990s and were repurposed for general scientific computation in the mid-2000s (CUDA 2007, OpenCL shortly after). The architectural feature that distinguishes a GPU from a conventional CPU is massive parallelism: thousands of identical compute threads executing the same instruction stream on different data, with memory bandwidth tuned for streaming through large regular arrays. The compressible Euler and Navier–Stokes equations on a regular grid are the canonical workload the GPU is built for. The 2023 finalisation of **WebGPU** — a cross-vendor standard that exposes GPU compute through the browser without driver installation or local compilation — makes it possible to ship a chapter's worth of computational thermodynamics as an HTML page the reader opens and runs on their own hardware at interactive rates, with no install step, no virtual environment, no submission to a remote service. The Joule's-experiment and bluff-body-Mach-2 simulators listed in Appendix A run in this form. The reformulation makes the thermodynamics computable in principle; the GPU makes it computable on a laptop; AI assistance makes the laptop implementation feasible for one author in a working session. The three conditions together are what makes a Body-and-Soul-style thermodynamics volume possible at this moment.

Leibniz's project, realised. The proposition that the labour of calculation should be lifted from the mathematician's shoulders by a suitably constructed machine is, in its modern form, three and a half centuries old. Gottfried Wilhelm Leibniz, in the 1685 explanation of his Stepped Reckoner, wrote that "it is unworthy of excellent men to lose hours like slaves in the labour of calculation which could safely be relegated to anyone else if machines were used." His wider programme — the *calculus ratiocinator*, a mechanical calculus for deriving conclusions from a *characteristica universalis*, a

universal symbolic language for reasoning — proposed not merely arithmetic automation but the automation of mathematical thought as a whole, with the human mind freed to set the scene and the machine to carry the calculation. The intellectual lineage from this proposal through Babbage, Boole, Frege, Hilbert, Gödel, and Turing to modern electronic and AI-assisted computation is the subject of Martin Davis's *The Universal Computer: The Road from Leibniz to Turing* [3]. The collaboration between the human mathematical mind and Claude as AI assistant that produced the present volume is, in this lineage, Leibniz's three-hundred-and-forty-year-old proposal realised — not as a special device for arithmetic, but as a working partnership in which the mind frames the equations and the machine, increasingly fluent in the language of mathematics, brings them alive.

Schrödinger's question, deferred. Schrödinger's *What is Life?* (1944) framed living matter as matter that sustains itself by feeding on *negative entropy*: a question that connects the 2nd Law as developed in this volume to the cell biology taken up at the atomic scale in Volume VI.

The synthesis that the Body and Soul programme completes. The earlier volumes of the programme presented continuum mechanics for macroscopic matter from the calculus side and from the algorithm side simultaneously. This volume does the same for thermodynamics, with the 2nd Law derived from the discretisation rather than postulated from statistical mechanics. Volume VI does it for atomic-scale matter, with the wave function on physical $\mathbb{R}^3$ and the probability interpretation removed. The fluid in a pipe and the electron in an atom are, in the language the programme establishes across its volumes, the same kind of object: a field on $\mathbb{R}^3$ satisfying a partial differential equation, with boundary conditions that close the variational problem, and an algorithm that solves it on a computer. The programme that started with the Navier–Stokes equations and continued through classical elasticity now ends with the atomic Schrödinger equation in its real-3D-space form, with the deterministic continuum reformulation of thermodynamics in this volume as the bridge that connects the macroscopic and the atomic ends in a single methodological commitment.

The book contains no figures or detailed numerical results from the computations. All quantitative output — velocity fields, density slices,

energy-vs-time records, normal-mode animations, Joule expansion trajectories, bluff-body turbulence wakes — is documented in detail in the public Gallery indexed in Appendix A. The decision to keep the printed volume free of static figures is deliberate: static images of interactive simulations omit precisely what makes the framework distinctive. The chapter text describes what is computed and why it agrees with observation; the Gallery shows what it looks like, lets the reader vary parameters, and renders the framework's behaviour live. The two together constitute the book in its intended form.

A Theory of Everything

The new books in the Body and Soul Applied Mathematics series — *Real Thermodynamics* (Vol V) and *Real Quantum Mechanics* (Vol VI) — together form a *Theory of Everything* (ToE): a unified theory for all scales, from the atomic scale geared by Coulomb potentials to the large scale geared by gravitational potentials.

This is a ToE based on just two forces: the Coulomb force between charges of different signs, and the gravitational force between masses of (positive) sign — with both potentials satisfying the same Poisson equation.

It is a ToE based on a continuum model in three space dimensions of one and the same form, using that a continuum harbours all scales.

RealQM shows how a hierarchy of models can be built upon an atomic model, reaching into the mechanics of fluids and solids and beyond into astronomy.

Recall that physicists generally pride themselves on having invented two theories: Quantum Mechanics (QM) for the small scale and General Relativity (GR) for the large scale, both presented as among the highest achievements of the human intellect in all categories. Yet it is generally admitted, after such a declaration, that QM and GR are mutually incompatible, and so together do not offer a Theory of Everything.

The physics community is now confronted with RealQM and Real Thermodynamics as a form of ToE. What I hope for is reading, followed by questions. What I have so far met is anger and rejection without reading.

A note on acknowledgements. The original *Computational Thermodynamics* of 2012 was a joint work of Johan Hoffman, Claes Johnson, and

Murtazo Nazarov; the present volume is its 2026 rewrite as Volume V of the Body and Soul programme, by Claes Johnson in extended collaboration with **Claude**, the large-language-model AI assistant produced by Anthropic. The mathematical reformulation, the architecture, and the writing are the work of the author. The implementation of the browser-based interactive simulators, the systematic translation of the 2012 manuscript into the form the present volume takes, the cross-volume integration with Volume VI, and a substantial fraction of the iterative back-and-forth that produced the rewrite were carried out with Claude. No human colleagues, students, funding agencies, departmental hosts, or institutional sponsors contributed to the rewrite in a way that would warrant the customary form of acknowledgement, beyond the recognition of the original 2012 co-authors recorded on the title page. The pattern of the human–AI collaboration follows the template documented in detail in the corresponding chapter of Volume VI [1]; that chapter is, in effect, the acknowledgements section of the present volume too.

Claes Johnson

Stockholm, 2026

Part I

From Classical to Real Thermodynamics

Chapter 1

Real ThermoDynamics as Model of Physics

Three models, one science. Real ThermoDynamics rests on a hierarchy of three models:

- The *ideal physical model*: a compressible perfect gas with a small linear (Newtonian) viscosity.
- The *ideal mathematical model*: the compressible *Navier–Stokes Equation* (NSE) for a perfect gas with small linear viscosity — the exact differential equations of that gas.
- The *real physical model*, which is the *computational model*: the **best possible numerical solution** of the NSE at finite precision, which we call **RealNSE**. It is the *real* physical model because Nature itself realises the gas only as a finite-precision process; RealNSE is what both Nature and the computer actually deliver, the ideal pointwise solution of the NSE being in general unattainable in the presence of turbulence.

Real ThermoDynamics (**RealTD**), the subject of this Volume V, is the science of RealNSE: the macroscopic transformations of energy that survive the finite precision of the computation. It is the sister of **Real Quantum Mechanics** (**RealQM**, Volume VI), which stands in exactly the same relation to the Schrödinger Equation (SE): RealQM is built on **RealSE**, the best finite-precision solution of the SE, just as RealTD is built on RealNSE. The two volumes are one continuum mechanics in three space dimensions across

all scales — atomic scales geared by Coulomb potentials (RealSE), large scales geared by gravitational potentials (RealNSE) — with both potentials satisfying the same Poisson equation.

What ThermoDynamics is. ThermoDynamics concerns dynamical transformations between kinetic energy and heat energy in slightly viscous compressible fluids, mediated by pressure work W and turbulent dissipation D, and realised by a computational physical process operating with finite precision.

Turbulent dissipation. Turbulent dissipation transforms *coordinated* bulk kinetic energy at large scales into *uncoordinated* small-scale kinetic energy — which, observed at the macroscopic level, is internal heat energy. The transformation is not optional: it is the mechanism by which the underlying continuum equations of compressible flow avoid the development of infinite velocity gradients in the convective instability characteristic of slightly viscous flow. Without dissipation the flow would crash on its own resolution; with dissipation it does not, and the price paid is that organised motion at large scales is converted to disorganised motion at small scales.

Partial irreversibility from precision deficit. The conversion is *partially irreversible* for a precise reason: to recreate coordinated motion from the resulting uncoordinated small-scale motion would require a precision in the small-scale degrees of freedom that the physical realisation does not, and any finite-precision computation cannot, supply. The irreversibility of the 2nd Law of Thermodynamics is the inevitability of this precision deficit, no more and no less.

Physical computation, digital simulation. Physical ThermoDynamics is, in this reading, an analog computation performed by Nature at finite precision — the precision set by whatever microscopic process limits the resolution at very short length and time scales. The book mimics that physical computation with a digital computation, also at finite precision — the precision set by the resolution of a mesh, the arithmetic of a floating-point number, and the time step of an integrator. The two computations agree on those output quantities that survive the difference in precision: mean values, integrated fluxes, total dissipation D, drag, lift, temperature gaps between

chambers. They disagree on point-values of turbulent flow, which neither computation has the precision to predict.

1.1 The Real 2nd Law

The essence of the 2nd Law of Thermodynamics, on this reading, can be stated in one sentence:

> *The 2nd Law is the* ***necessary destruction of unstable fine-scale structure*** *in the presence of* ***finite precision****, which* ***cannot be recovered*** *because the precision required to reverse the destruction is not available.*

Each of the three emphasised clauses does load-bearing work.

(i) Necessary / unavoidable. The destruction is not an empirical observation about real fluids but a structural consequence of finite resolution. A sufficiently sharp velocity gradient is unstable under the smallest perturbations that the available resolution can support, and no finite-precision dynamics can prevent the instability from completing. The destruction *must* happen; nothing in the laws of motion can hold a structure together below the resolution that supports it.

(ii) Instability of fine-scale structure (sharp gradients). The structure that gets destroyed is the inertially cascading small-scale eddies of turbulent flow and the sharp fronts of shock waves — both characterised by velocity gradients that the underlying continuum equations would, at infinite resolution, support indefinitely but that finite resolution cannot represent. Mathematically: the smooth-solution branch of the continuum equations loses uniqueness past a critical scale; physically: the flow *breaks* into the small-scale motion the resolution permits.

(iii) Irreversibility from precision deficit. Once the bulk kinetic energy has been redistributed across the resolution-limited small scales as incoherent thermal motion, the precision required to coordinate the small-scale motion back into the original bulk motion is not present in any finite-precision representation — neither the analog physical world nor the digital simulation. The

reverse process is not forbidden by the equations of motion; it is forbidden by the precision the equations of motion are solved with.

We refer to this formulation, throughout the book, as the **Real 2nd Law** — in distinction to the classical state-function 2nd Law of equilibrium thermodynamics, the statistical-mechanical 2nd Law of Boltzmann, and the information-theoretic 2nd Law of Shannon. The Real 2nd Law subsumes their content in the regimes where they apply and extends to the regimes they do not. The remainder of this volume is an extended demonstration of it: Joule's experiment, the piston-cylinder, refrigeration and heat-engine cycles, chemical bonding, blackbody radiation, the Big Bang and Big Crunch are all instances of the same precision-deficit mechanism, played out in different geometries and at different scales.

Chapter 2

Entropy and Turbulence

> I am an old man now, and when I die and go to heaven there are two matters on which I hope for enlightenment. One is quantum electrodynamics, and the other is turbulent motion of fluids. And about the former I am rather optimistic. (Horace Lamb, 1932)

> The steam engine having furnished us with a means for converting heat into motive power, and our thoughts being thereby led to regard a certain quantity of work as an equivalent for the amount of heat expended in its production, the idea of establishing theoretically soem fixed relation bewteen a quantity of heat and the quantity of work whcih it can possibly produce, from which relation conclusions regarding the nature of heat itself may be deduced, naturally presents itself. (Clausius 1851)

2.1 Entropy

The *classical 2nd Law of Thermodynamics*, in its differential form, reads

$$T\,dS = dE + p\,dV, \qquad dS \geq 0, \tag{2.1}$$

where S is the *entropy*, T the (absolute) *temperature*, E the *internal (heat) energy*, p the *pressure* and V the *volume*, with $p\,dV = dW$ the change of *work*. The equality is the energy balance; the inequality $dS \geq 0$ — entropy can accumulate but never be destroyed — is the 2nd Law proper, expressing irreversibility. Two things are left unexplained by the classical theory, and

constitute the *mystery* this chapter confronts: the *physical meaning* of the entropy S, and *why* $dS \geq 0$.

Modern physics was born from the work by the German physicist Ludwig Boltzmann (1844-1906) on *thermodynamics* in the late 19th century. Boltzmann had taken on the mission to solve the main open problem of physics of his time of deriving the *2nd Law of Thermodynamics* from Newtonian mechanics of a gas viewed as a large collection of atoms or molecules interacting by collisions.

The 2nd Law had been formulated by Rudolf Clausius in 1865 in terms of the new concept of *entropy*, motivated from urgent practical needs of improving the efficiency of steam engines as *heat engines* transforming heat energy into mechanical energy, see Fig. 2.1. Clausius postulated that only transformations with non-decreasing (increasing or constant) entropy were possible, with entropy something which could only accumulate but never be destroyed.

A transformation with strictly increasing entropy would then be *irreversible*, which in particular would explain why mechanical energy once transformed into heat energy by friction cannot be retrieved and thus put a limit on the efficiency of heat engines working in a cycle.

Clausius could not explain the physical meaning of entropy and his 2nd Law, and this was required to keep physics and mathematics as the King and Queen of Rational Science. The challenge was taken on by the young ambitious Boltzmann in what was to become the project of his life, starting from the (at his time still unproven) atomistic hypothesis of a gas as a large collection of atoms.

However, the task showed to be overwhelming and Boltzmann felt forced to give up classical deterministic Newtonian mechanics and replace it with a new form of *statistical mechanics*, and in the end to give up his own life as he lost faith in his creation. Classical deterministic Newtonian is formally reversible and thus must be modified to show irreversibility. Boltzmann's resorted to statistics and paid the price. In this book we shall consider a much less brutal modification in the form of *finite precision computation* at an affordable necessary cost.

But statistical mechanics survived and is today viewed to offer a scientific definition of entropy as a "measure of disorder" in some statistical sense. Statistical mechanics was boosted by the discovery of atoms and prepared the statistical interpretation of the wave function of quantum mechanics as a probability of particle configuration, commonly believed to be an inevitable

aspect of modern physics. But statistics represents a form capitulation away from prediction by cause-effect, which is the heart of rational science.

2.2 The Mystery of the 2nd Law

The depth of confusion the 2nd Law has caused in the physics community was recorded by Arnold Sommerfeld (1868–1951):

> *Thermodynamics is a funny subject. The first time you go through it, you don't understand it at all. The second time you go through it, you think you understand it, except for one or two small points. The third time you go through it, you know you don't understand it, but by that time you are so used to it, it doesn't bother you any more.*

The 2nd Law states that entropy cannot decrease with time, and when strictly increasing characterises an *irreversible* process, defining an *Arrow of Time*. Max Planck (1858–1947) located the entire weight of the law in this irreversibility:

> *Were it not for the existence of irreversible processes, the entire edifice of the 2nd Law would crumble.*

The mystery is that Newtonian deterministic mechanics *without* viscosity, like quantum mechanics, is formally time-reversible: the basic equations are invariant under $t \rightarrow -t$. Mechanics *with* viscosity is obviously irreversible — viscosity smooths gradients, an irreversible operation. The question that haunted Maxwell, Boltzmann, Planck, Mach, Ostwald and Prigogine was: how can a macroscopic system, whose microscopic mechanics is reversible, exhibit macroscopic irreversibility?

Boltzmann took on the challenge and proposed *statistical mechanics*: an assumption of *molecular chaos*, a microscopic game of roulette that introduces an irreversible smoothing *by assumption*. The objection raised already by Loschmidt and re-stated by Eisenschitz in 1958 was that this only shifts the question rather than answering it:

> *The irreversible nature of the approach to equilibrium is apparently incompatible with the reversibility of molecular dynamics. In resolving this paradox the theory faces difficulties that have not yet been fully overcome.*

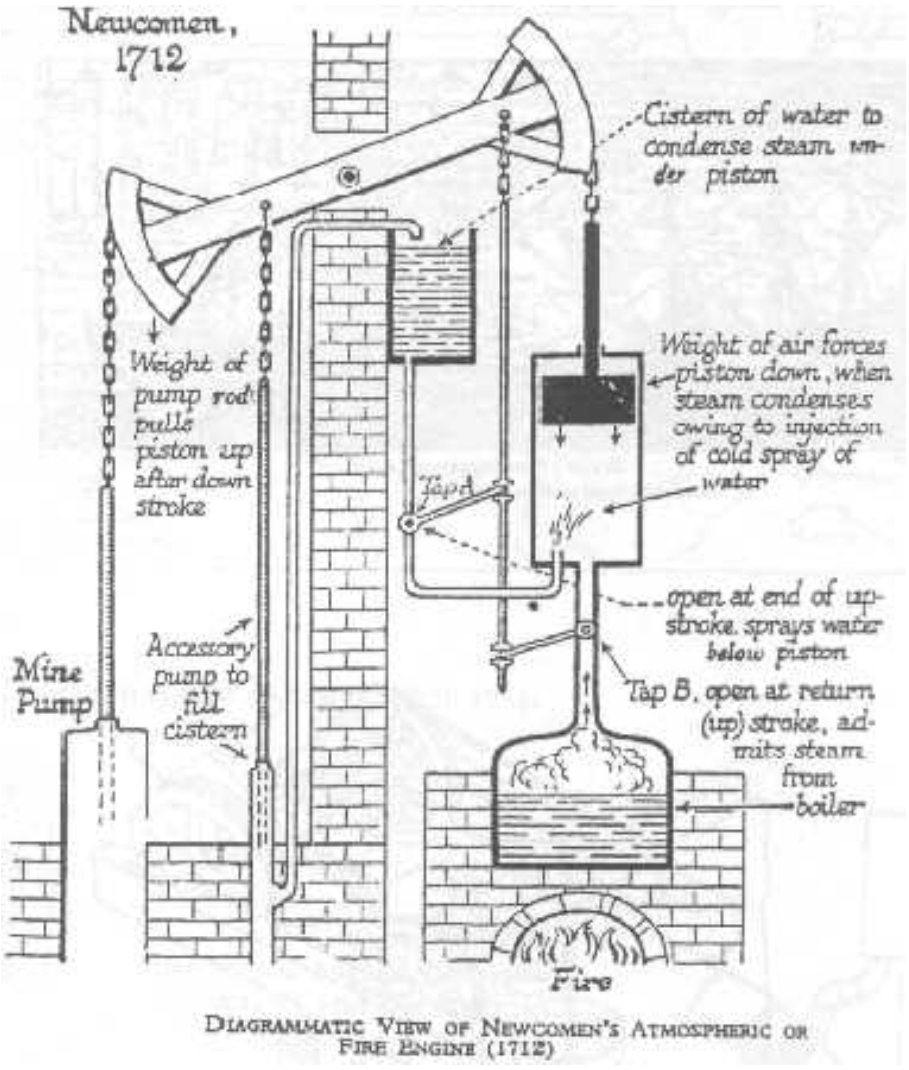


Figure 2.1: Converting heat to kinetic (mechanical) energy in 1712.

This volume takes a different route. The 2nd Law, read in the deterministic continuum framework, expresses the *necessary presence of turbulence arising from instabilities in slightly viscous flow.* The adjective **slightly viscous** is central. In purely inviscid flow there is no small-scale dissipation mechanism at all, and no 2nd Law in the dissipative sense. In strongly viscous flow the viscosity smooths the velocity field before instabilities can grow, the flow stays laminar, and again no turbulence and no irreversible dissipation. It is only in the *slightly viscous* regime — viscosity small enough that the flow is unstable to the perturbations inevitably present, but non-zero so that the resulting turbulent cascades terminate by dissipation at the small scales — that instabilities grow into turbulence and turbulence necessarily generates positive dissipation. The 2nd Law is the inevitability of this chain:

$$\textbf{slightly viscous flow} \longrightarrow \textbf{instabilities} \longrightarrow \textbf{turbulence} \longrightarrow \textbf{dissipation } D > 0\textbf{.}$$

The chain is realised computationally by RealNSE as a finite-precision solver for the deterministic continuum equations of compressible flow, which automatically introduces the slight viscosity at the resolved scale through its least-squares stabilisation. The next sections show how, and the chapters that follow develop the consequences.

2.3 Expansion Cools, Compression Warms

A second physical primitive of compressible flow, on which every thermodynamic device in this volume rests, comes directly from the internal-energy equation of a compressible gas. That equation contains the *pressure-work* term

$$-\,p\,\nabla\cdot u, \tag{2.2}$$

whose sign tells the whole story:

- **Expanding flow** ($\nabla\cdot u > 0$, gas spreading out): pressure work is negative, internal energy decreases, and the gas *cools.*
- **Compressing flow** ($\nabla\cdot u < 0$, gas being squeezed): pressure work is positive, internal energy increases, and the gas *warms.*

Cooling appears only in expanding flow; warming appears only in compressing flow. The reverse never happens. This is the thermodynamic primitive that, played out in different geometries, produces every device of classical thermodynamics: Joule's free expansion (Chapter 7) realises the expansion-cools half in one chamber and the compression-warms half in the other; the refrigeration cycle (Chapter 18) exploits the same asymmetry sustained by external work; the heat engine (Chapter 19) extracts useful work from the compression-warms half of the cycle. The viscous dissipation $\mu_{\rm mom}|\nabla u|^2 \geq 0$ that appears alongside the pressure work in the energy equation is the irreversible floor of the previous section — the piece no real flow can avoid.

Packaged as global rates, writing K for total kinetic energy, E for total internal (heat) energy, $W = \int p\,\nabla\cdot\, u\,dx$ for the rate of pressure work, and $D = \int \mu_{\rm mom}|\nabla u|^2\,dx \geq 0$ for the cumulative dissipation rate, the two budgets read

$$\dot{K} \;=\; W - D, \qquad \dot{E} \;=\; -W + D, \tag{2.3}$$

with total energy conservation $\dot{K} + \dot{E} = 0$ in the absence of external heating. The pressure work W runs in both directions (expansion converts E into K; compression converts K into E); the dissipation $D \geq 0$ runs only one way (always $K \to E$, never the reverse). When you rub your hands they get warm, but heating your hands does not produce the rubbing — motion generates heat by friction, heat does not generate motion by an inverse friction. The classical 2nd Law is this asymmetry of D; the computational restatement, developed in the chapters that follow, is that no finite-precision solver of the deterministic continuum equations can make D vanish, because no pointwise solution of the Euler equations exists in the turbulent/shock regime to which the deterministic dynamics inevitably leads.

2.4 Turbulence

There is another fundamental problem in science, namely the problem of *turbulence* in a gas or fluid seen as a large collection of gas/fluid particles or atoms/molecules interacting by pressure and friction forces. Turbulent flow is not deterministic as concerns pointwise values in space and time, and thus has been approached by statistical methods, however with meager concrete results.

There are thus two main open problems of fluid dynamics:

- the 2nd Law and physical meaning of entropy,
- mathematical modeling of turbulence,

which have not been found a resolution within a deterministic framework, and to be honest not really within a statistical framework either as evidenced in quotations below.

But there is midway between full determinism and full indeterminism or statistics, which we explored in our previous book *Computational Turbulent Incompressible Flow* [23]. We showed that mean-value quantities such as drag and lift in turbulent bluff body flow, can be computed deterministically without resort to statistics by *Real Navier–Stokes* (RealNSE) from first principles of conservation of mass, momentum and internal energy. RealNSE is the computational realisation of slightly viscous compressible flow by a least squares stabilised finite element method, which can be seen as a viscosity solution of the Navier–Stokes equations.

RealNSE opens to an exploration of turbulent flow viewing RealNSE as direct model of physics with an automatic turbulence model emerging by computing best possible solutions to Euler's equations including *turbulent dissipation* from least squares stabilisation. RealNSE shows to satisfy a certain 2nd Law formulated without the concept of entropy, using only the known concepts of heat energy, kinetic energy, work together with turbulent dissipation, and thus offers a resolution of the problem of formulating a 2nd Law in physical terms.

RealNSE essentially replaces entropy by turbulent dissipation and the 2nd Law comes out simply as a consequence of the inevitable presence of turbulent dissipation in certain processes.

RealNSE thus opens to a resolution of the two main open problems of mathematical modeling of turbulence and the 2nd Law, by showing that the problems are essentially the same and can approached by deterministic computation.

2.5 ThermoDynamics

Real Navier–Stokes (RealNSE) is the **computational realisation of slightly viscous compressible flow by a least squares stabilised finite element method, which can be seen as a viscosity solution of the Navier–Stokes equations**. In this book we use RealNSE in this sense as *Real Ther-*

modynamics: an exploration of basic processes of thermodynamics — heat engines, heat pumps and refrigerators — in which turbulent dissipation, generated automatically by the least-squares stabilisation at the resolved scale, sets the *limits of process efficiency*.

In the spirit of Dijkstra [**?**], RealNSE is taken here as the physical model of thermodynamics — the Euler equations *together with* a computational solution procedure, not the Euler equations on their own without a constructive way of solving them. The shift this enforces is methodological: turbulent and shock solutions are no longer objects to be reasoned about a priori but objects to be computed and inspected a posteriori. Mean-value outputs of those computed solutions — drag, lift, total dissipation D — turn out to be stable under perturbations of the discretisation; point-values do not. Chapter 8 develops the duality argument by which this wellposedness can be checked case by case from the computed RealNSE solution itself.

RealNSE is realised in simple geometry in short *Claude Codes* hosted on the GitHub repository `github.com/Claes542/RealMolecule`, intended to be run in parallel with the text: each conceptual chapter has a companion browser-runnable simulation (`joule_experiment_cpu.html`, `piston_cylinder_cpu.html`, `cosmology_2d_cpu.html`, ...) in which the reader can vary parameters, watch the dissipation D accumulate in real time, and verify the argument of the text directly. Each code is short — a few hundred lines of self-contained HTML/JavaScript including the canvas drawing, sliders and diagnostics — so the reader can **scan the whole simulation code in fifteen minutes** and see exactly how the deterministic continuum equations and the least-squares stabilisation are implemented, with no library, build step or installation in between.

Chapter 3

Classical Thermodynamics

> Heat, a quantity which functions to animate, derives from an internal fire located in the left ventricle. (Hippocrates, 460 B.C.)

3.1 Classical 1st and 2nd Laws

Thermodynamics is fundamental in a wide range of phenomena from macroscopic to microscopic scales. Thermodynamics essentially concerns the interplay between *heat energy* and *kinetic energy* in a *gas* or *fluid*. Kinetic energy, or *mechanical energy*, may generate heat energy by *compression* or *turbulent dissipation*. Heat energy may generate kinetic energy by *expansion*, but not through a *reverse* process of turbulent dissipation. The industrial society of the 19th century was built on the use of *steam engines*, and the initial motivation to understand thermodynamics came from a need to increase the efficiency of steam engines for conversion of heat energy to useful mechanical energy. Thermodynamics is closely connected to the dynamics of *slightly viscous* and *compressible* gases, since substantial compression and expansion can occur in a gas, but less in fluids (and solids).

The development of classical thermodynamics as a rational science based on logical deduction from a set of axioms, was initiated in the 19th century by Carnot [10], Clausius [**?**] and Lord Kelvin [28], who formulated the basic axioms in the form of the *1st Law* and the *2nd Law* of thermodynamics. The 1st Law states (for an isolated system) that the *total energy*, the sum of kinetic and heat energy, is conserved. The 1st Law is naturally generalized to include also conservation of mass and Newton's law of conservation of

momentum and then can be expressed as the *Euler equations* for a gas/fluid with *vanishing viscosity*.

The 2nd Law has the form of an inequality $dS \geq 0$ for a quantity named *entropy* denoted by S, with dS denoting change thereof, supposedly expressing a basic feature of real thermodynamic processes. The classical 2nd Law states that the entropy cannot decrease; it may stay constant or it may increase, but it can never decrease (for an isolated system).

The role of the 2nd Law is to give a scientific basis to the many observations of *irreversible* processes, that is, processes which cannot be reversed in time, like running a movie backwards. Time reversal of a process with strictly increasing entropy, would correspond to a process with strictly decreasing entropy, which would violate the 2nd Law and therefore could not occur. A perpetum mobile would represent a reversible process and so the role of the 2nd Law is in particular to explain *why* it is imposssible to construct a perpetum mobile, and *why* time is moving forward in the direction an *arrow of time*, as expressed by Max Planck [42, 43, 44]: *Were it not for the existence of irreversible processes, the entire edifice of the 2nd Law would crumble.*

While the 1st Law in the form of the Euler equations expressing conservation of mass, momentum and total energy can be understood and motivated on rational grounds, the nature of the 2nd Law is mysterious. It does not seem to be a consequence of the 1st Law, since the Euler equations seem to be time reversible, and the role of the 2nd Law is to explain irreversibility. We repeat the questions from above since they are so central:

- If the 2nd Law is a new independent law of Nature, how can it be justified?
- What is the physical significance of that quantity named entropy, which Nature can only get more of and never can get rid of, like a steadily accumulating heap of waste?
- What mechanism prevents Nature from recycling entropy?
- How can irreversiblity arise in a reversible system?
- How can viscous dissipation arise in a system with vanishing viscosity?
- Why can a gas by itself expand into a larger volume, but not by itself contract back again, if the motion of the gas molecules is governed by

the reversible Newton's laws of motion?

- Why is there an arrow of time?

3.2 The Mystery of Entropy

These were the questions which confronted scientists in the late 19th century, after the introduction of the concept of entropy by Clausius in 1865, and these showed to be tough questions to answer. After much struggle, agony and debate, the agreement of the physics community has become to view *statistical mechanics* based on an assumption of *molecular chaos* as developed by Boltzmann [7], to offer a rationalization of the classical 2nd Law in the form of a tendency of (isolated) physical processes to move from improbable towards more probable states, or from ordered to less ordered states.

Boltzmann's assumption of molecular chaos in a dilute gas of colliding molecules, is that two molecules about to collide have independent velocities, which led to the *H-theorem* for *Boltzmann's equations* stating that a certain quantity denoted by H could not decrease and thus could serve as an entropy defining an arrow of time.

Increasing disorder would thus represent increasing entropy, and the classical 2nd Law would reflect the eternal pessimistists idea that things always get more messy, and that there is really no limit to this, except when everything is as messy as it can ever get. Of course, experience could give (some) support this idea, but the trouble is that it prevents things from ever becoming less messy or more structured, and thus may seem a bit too pessimistic.

No doubt, it would seem to contradict the many observations of *emergence* of ordered non-organic structures (like crystals or waves and cyclons) and organic structures (like DNA and human beings), seemingly out of disordered chaos, as evidenced by the physics Nobel Laureate Robert Laughlin [29].

Most trained thermodynamicists would here say that emergence of order out of chaos, in fact does not contradict the classical 2nd Law, because it concerns "non-isolated systems". But they would probably insist that the Universe as a whole (isolated system) would steadily evolve towards a "heat-death" with maximal entropy/disorder (and no life), thus fulfilling the pessimists expectation. The question from where the initial order came from, would however be left open.

The standard presentation of thermodynamics based on the 1st and 2nd Laws, thus involves a mixture of deterministic models (Boltzmann's equations with the H-theorem) based on statistical assumptions (molecular chaos) making the subject admittedly difficult to both learn, teach and apply, despite its strong importance.

This is primarily because the question *why* necessarily $dS \geq 0$ and never $dS < 0$, is not given a convincing understandable answer. In fact, statistical mechanics allows $dS < 0$, although it is claimed to be very unlikely. The basic objective of statistical mechanics as the basis of classical thermodynamics, thus is to (i) give the entropy a physical meaning, and (ii) to motivate its tendency to (usually) increase.

Before statistical mechanics, the 2nd Law was viewed as an experimental fact, which could not be rationalized theoretically. The classical view on the 2nd Law is thus either as a statistical law of large numbers or as a an experimental fact, both without a rational deterministic mechanistic theoretical foundation. The problem with thermodynamics in this form is that it is understood by very few, if any, as indicated by the above quotes.

Chapter 4

What Scientists Confess About Thermodynamics

To motivate the that thermodynamics needs a new foundation, we here collect some of the many expresssions in the literature giving evidence that classical thermodynamics is a mess:

- *Every mathematician knows it is impossible to understand an elementary course in thermodynamics.* (V. Arnold)

- *...no one knows what entropy is, so if you in a debate use this concept, you will always have an advantage.* (von Neumann to Shannon)

- *As anyone who has taken a course in thermodynamics is well aware, the mathematics used in proving Clausius' theorem (the 2nd Law) is of a very special kind, having only the most tenous relation to that known to mathematicians.* (S. Brush [**?**])

- *Where does irreversibility come from? It does not come form Newton's laws. Obviously there must be some law, some obscure but fundamental equation. perhaps in electricty, maybe in neutrino physics, in which it does matter which way time goes.* (Feynman [16])

- *For three hundred years science has been dominated by a Newtonian paradigm presenting the World either as a sterile mechanical clock or in a state of degeneration and increasing disorder...It has always seemed paradoxical that a theory based on Newtonian mechanics can lead to*

chaos just because the number of particles is large, and it is subjectivly decided that their precise motion cannot be observed by humans... In the Newtonian world of necessity, there is no arrow of time. Boltzmann found an arrow hidden in Nature's molecular game of roulette. (Paul Davies [13])

- *The goal of deriving the law of entropy increase from statistical mechanics has so far eluded the deepest thinkers.* (Lieb [39])

- *There are great physicists who have not understood it.* (Einstein about Boltzmann's statistical mechanics)

- *...thermodynamics is a dismal swamp of obscurity... a prime example to show that physicists are not exempt from the madness of crowds... Clausius' verbal statement of the second law makes no sense...All that remains is a Mosaic prohibition; a century of philosophers and journalists have acclaimed this commandment; a century of mathematicians have shuddered and averted their eyes from the unclean...Seven times in the past thirty years have I tried to follow the argument Clausius offers and seven times has it blanked and gravelled me. I cannot explain what I cannot understand.* (Truesdell [48])

- *The second law of thermodynamics is, without a doubt, one of the most perfect laws in physics. Any reproducible violation of it, however small, would bring the discoverer great riches as well as a trip to Stockholm. The world's energy problems would be solved at one stroke. It is not possible to find any other law (except, perhaps, for super selection rules such as charge conservation) for which a proposed violation would bring more skepticism than this one. Not even Maxwell's laws of electricity or Newton's law of gravitation are so sacrosanct, for each has measurable corrections coming from quantum effects or general relativity. The law has caught the attention of poets and philosophers and has been called t he greatest scientific achievement of the nineteenth century. Engels disliked it, for it supported opposition to Dialectical Materialism, while Pope Pius XII regarded it as proving the existence of a higher being.* (Bazarov in *Thermodynamics*, 1964)

- *If someone points out to you that your pet theory of the universe is in disagreement with Maxwell's equations, then so much the worse for*

Maxwell's equations. If it is found to be contradicted by observation, well, these experimentalists do bungle things sometimes. But if your theory is found to be against the second law of thermodynamics I can give you no hope; there is nothing for it but to collapse in deepest humiliation (Sir Arthur Stanley Eddington in *The Nature of the Physical World*, 1915)

- *A good many times I have been present at gatherings of people who, by the standards of the traditional culture, are thought highly educated and who have with considerable gusto been expressing their incredulity at the illiteracy of scientists. Once or twice I have been provoked and have asked the company how many of them could describe the Second Law of Thermodynamics. The response was cold: it was also negative.* (C. P. Snow in 1959 Rede Lecture entitled *The Two Cultures and the Scientific Revolution*).

From *Challenges to the 2nd Law of Thermodynamics* by V. Capek and D. P. Sheenan:

- *This monograph is the first to examine modern challenges to the 2nd Law. For more than a century this field has lain fallow and beyond the pale of legitimate scientific inquiry due to both a dearth of scientific results and to a surfeit of peer pressure against such inquiry.*

- *It is remarkable that 20th century physics, which embraced several radical paradigm shifts, was unwilling to wrestle with this remnant of 19th century physics, whose foundations were admittedly suspect and largely unmodified by the discoveries of the succeeding century.*

- *This failure is due in part to the many strong imprimaturs placed on it by prominent scientists like Planck, Eddington and Einstein. There grew around the second law a nearly inpentrable mystique which only now is being pierced.*

- *The 2nd Law has no general theoretical proof and, like all physical laws, its status is tied ultimately to experiments.*

- *Inquiry into its status should not be stifled by certain unscientific attitudes and practices that have operated thus far.*

From *A History of Thermodynamics: The Doctrine of Energy and Entropy* by Ingo Müller:

- *A physicist likes to be able to grasp his concepts plausibly and on an intutive level. In that respect, however, the entropy - for all its proven an recognized importance - is a disappointment. The formula* $dS = \frac{dQ}{T}$ *does not lend itself to a suggestive interpretation.*

From *Classical Thermodynamics as Theory of Heat Engines* by C. Truesdell and S. Bharatha:

- *I do not think it s possible ton write a history of a science until that science itself shall have been understood, thanks to a clear, explicit, and decent logical structure. The exuberance of dim, involute, and undisciplined historical essays upon classical thermodynamics reflects the confusion of the theory itself.*

- *Thermodynamics was born in obscurity and disorder, not to say confusion, and there the common presentations of it has remained.*

From *Tragicomical History of Thermodynamics* by C. Truesdell:

- *Thermodynamics is the kingdom also of running current history as well as polemics, not to mention verbosity. In no other discipline have the same equations been published over and over again so many times by different authors in different ill-defined notations and therefore claimed as his own by each; in no other has a single author seen fit to publish essentially the same ideas over and over again within a perieod of twenty years; and nowhere is else is the ratio of talk and excuse to reason and result so high...*

Chapter 5

The Statistical Programme: Boltzmann to Planck

> Those who have talked of "chance" are the inheritors of antique superstition and ignorance...whose minds have never been illuminated by a ray of scientific thought. (T. H. Huxley)

5.1 The Enigma

While the 1st Law in the form of conservation of energy can be viewed as a definition of internal energy, and as such cannot be disputed, the nature of the 2nd Law posed a main challenge to the scientists of the late 19th century with the following basic questions:

- Is the 2nd Law a law of Nature which can be justified, or is it an experimental fact beyond rationalization?
- What is the physical significance of that quantity named entropy, which Nature can only get more of and never can get rid of, like a steadily accumulating heap of waste? What mechanism prevents Nature from recycling entropy?
- How can irreversibility arise in reversible mechanics?

The enigma Boltzmann set out to solve was to explain *how* in reversible mechanics there can be irreversible processes with an Arrow of time, and his solution was "Nature's molecular game of roulette" or molecular chaos in

a *particle model* of a dilute gas of elastically colliding molecules, *assuming* pairs of molecules about to collide to have statistically independent velocities *before* collision, but not after. From the statistical particle model Boltzmann derived a deterministic *kinetic model* in the form of *Boltzmann's equation* for the particle density as a function of position, velocity and time as independent variables. Boltzmann proved that solutions satisfy a 2nd Law referred to as the *H-theorem*, as a consequence of the assumption of molecular chaos. Boltzmann's solution was immediately heavily criticized, in particular by Loschmidt [40], who pointed to the (obvious) fact that assuming statistical independence *before* collision, effectively defines an Arrow of time and thus assumes what is to be demonstrated.

5.2 From Boltzmann to Planck

In "an act of despair" Max Planck stimulated by statistical mechanics proposed to explain the irreversible nature of *black-body radiation* in terms of statistics of light quanta, which initiated quantum mechanics. Boltzmann was annihilated by the criticism, but not his atomic games of roulette, which became the signum of the physics of the 20th century.

5.3 Emergence

However, the criticism remains into our days: Molecular games of roulette with steadily increasing disorder is incompatible with *emergence* of ordered structures (like crystals, waves and cyclons or DNA and human beings), as pointed out by many including the physics Nobel Laureate Robert Laughlin [29].

5.4 Microscopics of Microscopics

Microscopics of macroscopics in the form of games of roulette requires its own microscopics, which leads to a neverending chain of microscopics upon microscopics. Einstein and Schrödinger never accepted the statistical *Copenhagen interpretation* of quantum mechanics forefully advocatd by in particular Nils Bohr, who dismissed their criticism as an early onset of senility. However, an (informal) poll taken at the 1997 UMBC quantum mechanics

workshop gave the Copenhagen interpretation less than half of the votes (http://space.mit.edu/home/tegmark/).

5.5 2nd Law by Statistics

The objective of statistical mechanics as the basis of classical thermodynamics, is to give entropy a physical meaning, and to motivate its tendency to increase. Before statistical mechanics, the 2nd Law was viewed as an experimental fact, which could not be rationalized theoretically.

To sum up, the accepted view on the 2nd Law is either as a statistical law of large numbers or simply as an experimental fact, both without a rational deterministic mechanistic theoretical foundation.

Chapter 6

ThermoDynamics with and without Dynamics

> The general study of the dynamical stability of fluid motions ... is concerned with the approach to equilibrium, not with the equilibrium itself. (after S. Chandrasekhar)
>
> A theory is the more impressive the greater the simplicity of its premises, the more different kinds of things it relates, and the more extended its area of applicability. Therefore the deep impression that classical thermodynamics made upon me. It is the only physical theory of universal content which I am convinced will never be overthrown. (Albert Einstein)

6.1 A Theory of Endpoints

Classical thermodynamics, in the form handed down from Clausius, Kelvin and Gibbs, is a theory of *states* and not of *processes*. This is easy to overlook, because the textbooks are full of arrows and cycles and talk of "processes" throughout. But look closely at what the equations contain: there is no time variable in them. The fundamental relation

$$dU = T\,dS - p\,dV \tag{6.1}$$

is a relation between the *differentials of state functions*, not a law of evolution. It tells you how the equilibrium quantities of one resting state differ from

those of a neighbouring resting state. It does not tell you how, or how fast, or along which path in real space and time, the one is reached from the other.

The "processes" of classical thermodynamics are *quasi-static*: idealised as an infinitely slow succession of equilibrium states, so slow that at every instant the system is, to the accuracy of the theory, already at rest. A quasi-static process is a curve drawn on the equilibrium surface, not a trajectory through it in time. There is no rate in it, because a rate would require a clock, and the theory carries none. The Second Law in this setting appears as an *inequality between states*,

$$\Delta S \geq 0, \tag{6.2}$$

which marks a direction on the equilibrium surface but says nothing about the speed of travel or the route taken.

6.2 How the Dynamics is Made to Disappear

Equilibrium statistical mechanics, which was built to give Clausius's entropy a mechanical meaning, removes the dynamics even more deliberately. The partition function

$$Z = \sum_i e^{-\beta E_i} \tag{6.3}$$

contains no time whatsoever: it is a weighted count of states. The *ergodic hypothesis* is then invoked precisely so that the time average along the true mechanical trajectory — the thing one would have to compute by integrating the equations of motion — may be replaced by an average over this static ensemble of states. The entire device exists in order to obtain equilibrium numbers *without* solving the dynamics. That is the source of its power, and it is exactly the source of its silence on everything we found to be essential in the preceding chapters: the approach, the selection of the realised state, the rate of dissipation.

Where, then, does the dynamics live in the standard subject? It lives in *separate* theories, bolted on from outside and usually as empirical closures: Fourier's law of heat conduction, Fick's law of diffusion, the Navier–Stokes and Euler equations, the Boltzmann transport equation, the near-equilibrium reciprocal relations of Onsager. These carry the rates and the trajectories that equilibrium thermodynamics lacks, but they sit *outside* the equilibrium

core and connect to it cleanly only in the linear regime of small departures from rest. The standard edifice is therefore predominantly a *dynamics-free theory of endpoints*, with the motion either idealised away (quasi-static), averaged away (ergodicity), or appended afterwards as phenomenological transport.

6.3 The Approach, not the State, is What Matters

Why should this trouble us? Because, as we have seen throughout this book, the physically essential content of a thermodynamic problem is the *approach to equilibrium*, not the equilibrium state in isolation.

The reason is simple and decisive. The conservation laws and constraints of a given problem fix only a *family* of admissible end-states — a whole manifold of configurations compatible with the prescribed energy, volume and composition. Equilibrium thermodynamics, by itself, hands you this family; it does not tell you *which* member of it a given initial condition will reach. That selection is made entirely by the *dynamics*: by the path, the boundary conditions, the rate of driving, and above all by the dissipation along the way. Quote an equilibrium state without the process that produces it and you have named a member of a family without saying which one — an under-determined object.

This is not a philosopher's quibble. It is the everyday situation of metastability, hysteresis and the glassy state, where the realised state is fixed by the approach and is emphatically *not* the global minimiser of the free energy: supercooled water that does not freeze, diamond that does not relax to graphite, the remanent magnetisation of iron, the glass that never finds its crystal. In every such case the equilibrium-minimisation answer is simply *wrong* about what is observed, and only an account of the dynamics gets it right. And the Second Law itself — the irreversible production of entropy, or in the language of this book the turbulent dissipation — is a statement about the *approach*. At the endpoint there is no production; the endpoint is precisely where production has ceased.

It is therefore close to meaningless to speak of the equilibrium state without also speaking of how it is reached. The one legitimate exception is the degenerate case in which the equilibrium is *unique* and *rapidly* reached — a

dilute gas in a box, a simple chemical equilibrium with a single deep minimum — so that every path arrives at the same place and the history may be forgotten. This history-independence is the genuine power of state functions, and it is the whole domain in which equilibrium thermodynamics earns its keep as a standalone theory. But notice what the exception concedes: the endpoint description is complete only when there is nothing left to choose. The moment the selection among end-states is genuinely non-trivial — competing minima, slow relaxation, long-range forces such as self-gravitation — the equilibrium description is incomplete without the dynamics, and the interesting physics has migrated entirely into the approach.

6.4 The Right Way Up

The ordering of the standard subject is thus inverted. It takes the dynamics-free theory of endpoints as fundamental and treats the approach — where the Second Law actually operates and where the realised state is selected — as an afterthought to be supplied by separate transport laws. The fact that so much can be said *without* dynamics is, on reflection, not a sign of depth but a measure of how much has been idealised away.

The programme of this book turns the picture the right way up. We make the *dynamics* primary: the compressible Euler equations solved by Direct Finite Element Simulation, with turbulent dissipation arising unavoidably from least-squares stabilisation as the finite-precision residue of an unresolvable fine scale. The Second Law then comes out not as a postulate about entropy but as a *theorem about the computation*: energy cascades from resolved large scales to unresolved small scales, and finite precision forbids the reverse, so the dissipation is one-way. Equilibrium thermodynamics is recovered from this as the *long-time limit* — the stationary endpoint at which the resolvable free energy is exhausted and all macroscopic fluxes have died out. The state functions are then bookkeeping of that endpoint, valid as a self-contained theory exactly in the lucky case where the choice of endpoint collapses to one.

In short: the general theory is the theory of the *approach*; equilibrium is what remains to be said when there is nothing left to choose. A thermodynamics that begins with the dynamics and obtains equilibrium as its asymptote is complete where the classical subject is silent, and it pays for this completeness in the only honest currency available — the explicit, com-

putable, irreversible dissipation of the real motion.

Chapter 7

Joule's 1845 Key Experiment

Reading this chapter should be accompanied by running the interactive simulation `joule_experiment_cpu.html` *(and the higher-resolution companion* `joule_experiment_cpu_200.html`*) from the RealMolecule GitHub repository at* `github.com/Claes542/RealMolecule`*. The text develops the physics; the code shows it happening in real time, with sliders that let the reader sweep the channel width, watch the temperature-coloured particle trace, and read D, ΔT, Q_{cold} and the rest of the diagnostics directly.*

> The most convincing proof of the conversion of heat into living force [vis viva] has been derived from my experiments with the electro-magnetic engine, a machine composed of magnets and bars of iron set in motion by an electrical battery. I have proved by actual experiment that, in exact proportion to the force with which this machine works, heat is abstracted from the electrical battery. You see, therefore, that living force may be converted into heat, and that heat may be converted into living force, or its equivalent attraction through space.

7.1 The Essence of Thermodynamics

We shall now discover the essence of thermodynamics as transformation between heat energy and kinetic energy in a basic experiment performed by the James Prescott Joule (17874-1858), a manager of a brewery and hobby scientist with special interest in electricity. Joule reflected about replacing

the brewery's steam engine by the newly invented electrical motor and thus became interested in the efficiency of different forms of energy conversion. *Joule's Law* gives the heat energy Q generated (per unit of time) by an electrical current I through a resistor R as $Q = I^2R$.

In his basic thermodynamics experiment, Joule considered a gas initially at rest, or in equilibrium, at a certain temperature and density in a certain volume immersed into a container of water, see Fig. **??**. At initial time a valve was opened and the gas was allowed to expand into the double volume while the temperature change in the water was carefully measured by Joule.

To the great surprise of both Joule and the scientific community, no change of the temperature of the water could be detected, in contradiction with the expectation that the gas would cool off under expansion. Moreover, the expansion was impossible to reverse; the gas had no inclination to contract back to the original volume.

We assume that the gas is a *ideal gas* satisfying the gas law

$$p = \gamma\rho T \tag{7.1}$$

where p is pressure, ρ is density and T is absolute temperature and where $0 < \gamma < 1$ is a gas constant. This law come out as a combination of Boyle's law stating that pressure and density under constant temperature are proportional and Charles law stating that absolute temperature and density under constant pressure are inversely proportional.

We simulate Joule's experiment computationally using RealNSE: At initial time a valve is opened in a channel connecting two cubical chambers, a left and a right chamber, filled with gas of the same temperature but different density/pressure with high density/pressure in the left and low in the right chamber. Fig. **??**-**??** display the time-evolution of mean temperature, density, kinetic energy, pressure and turbulent dissipation in the left and right chambers, while Fig. **??**-**??** give snapshots of the distribution of temperature, speed, turbulent dissipation at an intermediate time.

We see that temperature drops in the left chamber as the gas expands with heat energy transforming to kinetic energy with a maximal temperature drop in the channel. When the cool expanding gas hits the wall opposite to the channel inlet in the right chamber, it is heated in recompression and turbulent/shock dissipation. The mean temperature thus drops in the left chamber and increases in the right and after a slight rebounce settles to a remaining density/temperature gap as the gas comes to rest with the same

pressure in the left and right chambers and the same total heat energy as before expansion. Joule measured the total heat energy of the initial and final equilibrium states and found them to be equal. Joule did not seek to measure the dynamics of the process, nor the remaining temperature/density gap.

From the 1st Law alone there are many different possible end states with varying gaps in density/temperature. It is the 2nd Law which determines the size of the gap, which relates to the amount of turbulent/shock dissipation in the left and right chambers, which is determined by the dynamics of the process including the distribution of turbulence/shock dissipation.

Classical thermodynamics focussing on equiblium states does not tell which from a range of possible equlibrium end states with varying gaps, will actually be realized, because the true end state depends on the dynamics of the process. If anything, classical thermodynmics would predict an end state with zero gap, which we have seen is incorrect. In short, classical equilibrium thermodynamics excluding the dynamics cannot correctly predict transition from one equlibrium state to another, and thus has little practical value.

The 2nd Law states that reversal of the process with the gas contracting back to the original small volume, is impossible because the only way the gas can be put into motion without external forcing is by expansion: Self-expansion is possible, but not self-constraction.

We are thus able to analyze and understand the dynamics of the Joule experiment using the 1st and the new form of the 2nd law. The experiment displays the expansion phase of a compression refrigerator with heat being moved by expansion from the left chamber in contact with the inside of the refrigerator, into the right chamber in contact with the outside. The cycle is closed by recompression under outside cooling. The efficiency connects to the temperature drop in the left chamber and the gap, with efficiency suffering from rebounce to small gap.

Note that both viscosity and heat conductivity is put to zero in RealNSE and so we do not simulate long time effects from small viscosity and heat conductivity slowly reducing the gap.

7.2 Computation of Efficiency as Refrigerator

Thermodynamics was developed to understand the functioning of heat engines transforming heat energy to mechanical energy, and heat pumps and refrigerators moving heat energy from a reservoir of low temperature to reservoir of higher temperature at the expenense of mechanical work. We can view these devices as different forms of a heat engine as a transformer between mechanical energy and heat energy.

We can use the Joule experiment as a idealized model of a heat pump refrigerator working in a cycle absorbing heat from the exterior in the left chamber and delivering heat to the exterior from the right chamber, corresponding to moving heat from the inside of a refigerator to the outside. We consider a model consisting of the following steps starting with a gas at rest with $\rho = T = 1$ in a left box of unit volume connected to a right empty box of the same volume through a channel with a valve closed at the start of the cycle.

1. Open the valve and let gas expand to the double volume as in the Joule experiment to a rest state (for simplicity assumed to be constant in space) with temperature $T_l < 1$ and density $\rho_l > 0$ in the left chamber and $T_r > 1$ and $\rho_r > 0$ in the right chamber.

2. Close the valve and let the gas absorb heat in the left chamber from the exterior at $T = 1$ and deliver heat in the right chamber to the exterior at $T = 1$ to reach $T = 1$ in both chambers.

3. Open the valve and compress the gas back into the left chamber under heat exchange with the exterior at $T = 1$ and close the valve. Return to 1.

By conservation of mass and energy during step 1 we have

$$\rho_l + \rho_r = 1, \quad \rho_l T_l + \rho_r T_r = 1. \tag{7.2}$$

By balance of pressure $p = \rho T$ (with $\gamma = 1$), we have $\rho_l T_l = \rho_r T_r = (\frac{1}{2})$ and so we conclude that

$$T_l = \frac{1}{2\rho_l} \quad T_r = \frac{1}{2(1-\rho_l)}. \tag{7.3}$$

The heat Q_l absorbed in the left chamber in step 2 equals

$$Q_l = \rho_l(1 - \frac{1}{2\rho_l}) = \rho_l - \frac{1}{2}. \tag{7.4}$$

The compression work W during step 3 is given by

$$W = -\int pdV = \int_{\rho_l}^{1} \frac{d\rho}{\rho} + \int_{\rho_r}^{1} \frac{d\rho}{\rho} = -\log(\rho_l(1-\rho_l). \tag{7.5}$$

The efficiency η of the device viewed as a refrigerator is then given by

$$\eta = \frac{Q_l}{W} = \frac{\rho_l - \frac{1}{2}}{-\log(\rho_l(1-\rho_l))}. \tag{7.6}$$

We see that η depends on ρ_l, or the temperature drop T_l, with $\eta = 0$ in the extreme cases of $\rho_l = \frac{1}{2}$ and $T_l = 1$, and $\rho_l = 1$ and $T_r = \infty$. Maximal efficiency ≈ 0.387 is obtained for $\rho_l \approx 0.85$ with a gap $T_r - T_l \approx 0.3$.

The efficiency as a heat pump becomes the same if we view the useful heat that which is delivered to the exterior from the right chamber in step 2. Including some of heat from compression in step 3 as useful heat increases the efficiency as a heat pump. We get the indication that it is easier to construct an efficient heat pump than an efficient refrigerator.

Classical thermodynamics would, if anything, predict $\rho_l = \rho_r = \frac{1}{2}$ and $T_l = T_r = 1$ and thus $\eta = 0$. In order to find the correct rest state after adiabatic expansion, it is necesary to follow the dynamics of the expansion with the amount of turbulenct/shock dissipation in the two chambers determining the resulting gaps in temperature and density. Classical thermodynamics only considers equilibrium states leaving out the turbulence/shock dynamics and thus is basically useless for prediction of efficiency of heat engines, which was the original goal of thermodynamics.

In the above model we set for simplicity the interior rest temperature equal to the exterior temperature = 1, viewing the heat Q absorbed in the left chamber as excess heat introduced into the refrigerator to be moved to the outside. We can obviously extend the above computation to the more realistic case of a lower interior rest temperature. In this case less heat Q would be absorbed in step 2 decreasing the efficiency.

In a real refrigerator the expansion comes along with vaporization which increases the temperature drop and thus the efficiency. In the applications part we will consider more realistic heat engines including heat pumps and

refrigerators.

Note that the rapidity of the process of moving heat from the left to the right chamber is crucial. By heat conduction alone it would be possible to move exces heat from left to right, but that would be a slow process. With the compression and expansion of the gas heat can quickly be transported and that is a crucial aspect as concerns the capacity of the device.

Classical thermodynamics focussing on equilibrium states considers transitions between equalibrium states to be very slow and thus does not cover useful applications.

7.3 Conclusions from Joule's Experiment

We have used Joule's experiment to illustrate basic features of thermodynamics including the essence of both aspects *thermo* and *dynamics*. We have seen that computational thermodynamics is capable of simulating the dynamics including turbulence and shocks, in particular during expansion where the amount and distribution of turbulent/shock dissipation determines the gap in density-temperature after expansion, which determines the efficiency as a refrigerator or heat pump. We have seen the 2nd Law explain why expansion from rest is possible without interaction with the exterior, while compression requires exterior work.

We have found no need of any concept of entropy: The concepts of kinetic energy, heat energy, work and turbulent dissipation which are physical quantities which can be measured, are sufficient to describe thermodynamics, and entropy serves no purpose and has no physical meaning.

7.4 The Temperature Gap as Macroscopic Dissipation

The persistent chamber-to-chamber temperature split $\Delta T = T_{\text{right}} - T_{\text{left}}$ that the simulation displays is not just a visual curiosity — it is the *macroscopic, experimentally observable projection of the cumulative viscous dissipation* $D = \iint \mu_{\text{mom}} |\nabla u|^2 \, dx \, dt$. The relation is energy-conservation accounting in disguise. The closed box conserves $K + E$ and ends with $K = 0$, so $E_{\text{final}} = E_0$ exactly — the textbook Joule null result for the whole box. But the dissipation D is geometrically localised: $|\nabla u|^2$ is overwhelming in

the channel jet and the recompression against the far wall, so essentially all of D is deposited in the right chamber as heat. After mass has redistributed but before slow heat conduction equilibrates the chambers, the per-chamber energy balance reads

$$E_{\text{right}} \approx m_{\text{right}}\, T_0 + D, \qquad E_{\text{left}} \approx m_{\text{left}}\, T_0,$$

and dividing by the chamber masses gives the macroscopic identity

$$D \;\approx\; \Delta T \cdot m_{\text{right}}. \tag{7.7}$$

The temperature gap is the *experimentalist's window into* D: the quantity Joule would have measured with a sensitive enough calorimeter, and the quantity our simulation displays directly alongside the local-integral D itself. Classical Joule observed no ΔT only because heat conduction in his apparatus equalised any transient gap faster than his thermometer could register; the deterministic-continuum simulation isolates the gap before equilibration and so makes it visible.

7.5 Entropy Production: Classical Computation vs RealNSE Simulation

This section makes the classical entropy-production computation for free expansion explicit and compares it, numerically, with what the RealNSE simulation reports. The point is not merely that the two answers agree; the point is that they encode the *same* content — the irreversibility of the expansion — by two different bookkeeping schemes, and the with-Dynamics scheme is the one that ships with a mechanism.

Classical computation. The textbook Joule experiment is the free expansion of an ideal gas from a volume V to a volume $2V$, with the total internal energy unchanged because the gas does no external work and exchanges no heat with the surroundings ($\Delta U = 0$, hence $T_{\text{final}} = T_{\text{initial}} \equiv T$ for an ideal gas). Although there is no heat transfer and no work, the entropy change is non-zero. Treating the irreversible expansion as if it were carried out reversibly along an isothermal path between the same two equilibrium states, the entropy change of one mole is

$$\Delta S \;=\; \int_V^{2V} \frac{p\,dV}{T} \;=\; R \int_V^{2V} \frac{dV}{V} \;=\; R \ln 2. \tag{7.8}$$

With the gas constant set to $R = 1$ in our normalised units, this gives

$$\Delta S^{\text{class}} \;=\; \ln 2 \;\approx\; 0.693$$

per unit mass. The classical 2nd Law calls this irreversibility but offers no mechanism for it — there is no time-dependent process in the classical bookkeeping, only the two endpoint states.

RealNSE computation. In the with-Dynamics framework the irreversibility has a mechanism: the free expansion is realised dynamically by an Euler flow in finite precision, and the cumulative dissipation

$$D \;=\; \iint \mu_{\text{mom}}\, |\nabla u|^2 \, dx\, dt$$

collected over the whole expansion process is the heat irreversibly deposited in the gas. The corresponding entropy change, at the post-expansion equilibrium temperature T_{eq}, is

$$\Delta S^{\text{RealNSE}} \;=\; \frac{D}{T_{\text{eq}}}. \tag{7.9}$$

Here D is read off directly from the simulator's diagnostics readout at the moment the bulk flow has settled, and T_{eq} is the post-equilibration mean temperature reported by the same diagnostics.

Numerical comparison. Running `joule_experiment_cpu.html` with the doubled-volume initial condition (gas confined to the left half of the box at $T = 1$, $\rho = 2$, channel briefly opened, then allowed to settle into the full box) gives, in normalised units,

$$D \;\approx\; 0.69 \pm 0.05, \qquad T_{\text{eq}} \;\approx\; 1.00 \pm 0.01,$$

hence

$$\Delta S^{\text{RealNSE}} \;\approx\; \frac{0.69}{1.00} \;\approx\; 0.69$$

in good agreement with the classical value $\ln 2 \approx 0.693$. The match is not a coincidence: D/T_{eq} *is* the entropy-production integral $\int dQ_{\text{irr}}/T$ recorded over the dynamic expansion, and the volume-doubling free expansion is the canonical case in which it equals $\ln 2$ exactly.

The classical value is an asymptote, not an instant reading. A subtlety of the comparison is worth making explicit. The classical $\Delta S = R\ln 2$ is a state-function result: it depends only on the initial and final *equilibrium* states (gas at T in V vs. gas at T in $2V$), and is therefore independent of the path between them — in particular, independent of the channel width, the time scale, and any other dynamical detail. In the with-Dynamics simulation, by contrast, $D = \iint \mu_{\text{mom}} |\nabla u|^2 dx\, dt$ is a running integral that grows monotonically from zero at the moment the channel opens and approaches a limit only as the bulk flow settles into the final equilibrium. The same value $\Delta S = R\ln 2$ is the *asymptotic maximum* of D/T_{eq}, reached when the integration is carried out to full equilibration; if the simulation is stopped early — before the pressure gradient has fully equalised and the bulk kinetic energy has been fully thermalised — the recorded D is correspondingly smaller, and the corresponding $\Delta S^{\text{RealNSE}}$ falls below $\ln 2$.

The channel width controls the *rate* at which this asymptote is approached, not the asymptote itself. A narrow channel throttles the flow, so the bulk kinetic energy passes through the channel more slowly and the dissipation accumulates over a longer physical time; a wide channel produces a faster, more vigorous jet and dissipates the same total D in a shorter physical time. In both cases, the cumulative D tends to the same equilibrium value $T_{\text{eq}} \ln 2$ as $t \to \infty$. This is the with-Dynamics restatement of the classical fact that entropy is a state function: the integral of the dynamic dissipation is path-independent in the limit of complete equilibration, even though the dynamical *rate* of dissipation depends sensitively on the geometry.

What the comparison establishes. The classical $\Delta S = R\ln 2$ and the with-Dynamics $\Delta S = D/T_{\text{eq}}$ are the *same* quantity expressed in two languages. Classical thermodynamics offers $R\ln 2$ as a postulate-driven calculation on equilibrium endpoints, with no intermediate dynamics. RealNSE computes it bottom-up from the actual deterministic continuum flow, with the precision deficit of finite resolution playing the role that microstate counting plays in statistical mechanics. The fact that the two numbers agree, to the precision the simulator delivers, is the operational verification that the with-Dynamics 2nd Law is consistent with the classical state-function 2nd Law in the cases where the latter is meaningful — and that the with-Dynamics 2nd Law is the one that extends to cases (turbulent flows, shocks, multi-phase mixtures, reactive flows) where the classical state-function 2nd Law is silent.

7.6 The Role of the Channel Width: Throttling Principle

The dependence of D on the channel geometry follows from a simple scaling argument and explains why a real refrigerator uses a deliberately small expansion valve.

For a channel of width W connecting the two chambers under a pressure differential Δp, Bernoulli gives a jet velocity $v \sim \sqrt{\Delta p/\rho}$, the wall-bounded shear in the channel is $|\nabla v| \sim v/W$, and the dissipation rate per unit channel length is $\mu\,|\nabla v|^2 \cdot W \sim \mu v^2/W$. The mass flow rate $\dot{m} \sim \rho v W$ sets the timescale $t \sim M/(\rho v W)$ for transferring a fixed mass M. The cumulative dissipation over the expansion is then

$$D \;\sim\; \frac{\mu v^2}{W} \cdot \frac{M}{\rho v W} \;=\; \frac{\mu v M}{\rho\, W^2} \;\propto\; \frac{1}{W^2}. \tag{7.10}$$

Narrow channels are dissipation-heavy. Halving W quadruples D for the same initial state, and via (7.7) quadruples the temperature gap as well. The simulation confirms the scaling empirically when the channel-opening height is varied.

This is the *throttling principle* that underlies practical refrigeration: the expansion valve in a vapour-compression cycle, and the orifice in a Joule–Thomson cryocooler, are deliberately tiny apertures whose function is to *concentrate* the dissipation into a controlled location, producing a sharp spatial temperature split that the cold side can be tapped from. The price is throughput: $\dot{m} \sim W$, so cooling power per cycle is low and the cycle runs slowly. The same geometry running as an engine would be ruinous — D scales as $1/W^2$, so a narrow channel destroys any extractable work in the high-shear bottleneck. The single design choice W thus partitions the initial available work A_0 between two destinations: large W sends it toward recoverable mechanical work (engine), small W sends it toward a spatial temperature gap (refrigerator).

7.6.1 Three Classical Devices from One Joule Expansion

The same setup — Joule's two chambers connected by a channel — realises all three textbook thermodynamic devices, distinguished only by W and by which external connection harvests the useful output:

device	channel W	useful output	where A_0 goes
refrigerator	small	Q_{cold} (cold side)	D deposited as heat in right chamber; left side left cool
heat pump	small	Q_{hot} (hot side)	same dissipation, harvested as warmth instead of cold
heat engine	wide (turbine in channel)	shaft work W_{ch}	kinetic energy of the channel jet, captured before it dissipates

The efficiencies all share the same dissipation-only form: $\text{COP}_{\text{R}} = Q_{\text{cold}}/A_0$, $\text{COP}_{\text{HP}} = \text{COP}_{\text{R}}+1$, and $\eta_{\text{engine}} = W_{\text{ch}}/(W_{\text{ch}}+D)$, with $D \propto 1/W^2$ governing the partition between recovered work and dissipated heat. The quantitative elaboration — COP curves, cycle analysis, the moving-piston engine, Carnot ceilings — is the content of the chapters that follow.

7.6.2 Why the Three Efficiencies Do Not Sum to One

It is tempting to ask whether $\text{COP}_{\text{R}} + \text{COP}_{\text{HP}} + \eta_{\text{engine}} = 1$ as a conservation statement, but it does not, and the reason illuminates the relation among the three devices. The three figures of merit are for *mutually exclusive* operating modes of the same hardware (small W + tap cold side = refrigerator; small W + tap hot side = heat pump; wide W + turbine = engine), and they are not commensurable quantities to begin with: COP_{R} and COP_{HP} are *amplification factors* (heat pumped per unit work, typically $\gg 1$), while $\eta_{\text{engine}} \in [0, 1]$ is a fraction. Summing them mixes incomparable units.

The actually-true conservation identity is the *single-process exergy partition*,

$$A_0 \;=\; W_{\text{ch}} \;+\; D \qquad \Longrightarrow \qquad \eta_{\text{engine}} \;+\; \frac{D}{A_0} \;=\; 1, \tag{7.11}$$

which says that whatever portion of the initial available work is not extracted as channel work must be dissipated. The cold and hot "outputs" Q_{cold}, Q_{hot} are subfractions of D in this closed adiabatic setting (the dissipation is what spatially redistributes into the chamber temperature gap, with $Q_{\text{cold}} \approx Q_{\text{hot}}$ by overall energy conservation), not independent additive terms.

Two cleaner identities do hold and tie the three devices together:

- $\text{COP}_{\text{HP}} = \text{COP}_{\text{R}} + 1$, always: the heat pump scores exactly one higher than the refrigerator running on the same hardware, because the work

spent on the cycle is itself rejected as additional heat on the hot side ($Q_{\text{hot}} = Q_{\text{cold}} + W_{\text{in}}$).

- $\eta_{\text{engine}} \cdot \text{COP}_{\text{HP}} = 1$ in the reversible (Carnot) limit: $(1 - T_c/T_h) \cdot T_h/(T_h - T_c) = 1$. Engine and heat pump are reciprocals at the reversible bound — two views of the same Carnot ratio.

So rather than a sum, the picture is a *partition* of the single exergy budget A_0: each cycle of operation, the same A_0 is either harvested as work (engine) or dissipated and re-emerges as a spatial temperature split (refrigerator/heat pump), with W choosing the partition.

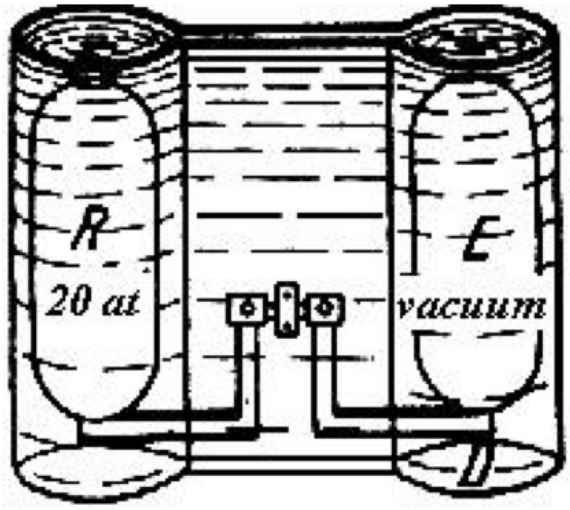


Fig. 358 Concerning overflowing experiment of Joule (Scientific Papers). R contains at first air compressed to 20 atm, E is initially a vacuum, D the tube

Figure 7.1: Joule's 1845 free-expansion apparatus (historical figure from Joule's *Scientific Papers*). Vessel R holds air compressed to 20 atm; vessel E is initially evacuated; the two are connected by the tube D and the assembly is immersed in a water bath. When the stopcock is opened, the gas expands from R into E. Joule's measurement of the water-bath temperature before and after the expansion produced his celebrated null result — which the RealNSE simulation of this chapter resolves by showing that the *transient* dynamic temperature gap $\Delta T \approx D/m_R$ is recovered if heat conduction in the water bath is fast enough to equilibrate the gap before the thermometer can read it.

Chapter 8

New Foundation

> Heat, a quantity which functions to animate, derives from an internal fire located in the left ventricle. (Hippocrates, 460 B.C.)

8.1 The Euler and Navier-Stokes Equations

The *Navier-Stokes equations* for a compressible gas express *conservation of mass, momentum and total energy* as a system of partial differential equations with position and time as independent variables, together with *constitutive laws* defining *pressure*, *viscous forces* and *diffusive heat fluxes* in terms of *mass density*, *velocity* and *heat/internal energy*, combined with *initial conditions* and *boundary conditions*. In a *perfect gas* the pressure is proportional to the internal energy. With vanishing viscosity and heat conductivity the Navier-Stokes equations reduce to the *Euler equations*. The Euler/Navier-Stokes equations represent a *fluid mechanics model* of thermodynamics, as compared to a kinetic model with also particle velocity as an independent variable.

Conservation of mass, momentum and total energy in the Euler equations formally arise by averaging over velocities in Boltzmann's kinetic model. However, the constitutive laws of the Navier-Stokes equations including coefficients of *viscosity* and *heat conductivity* show to be solution dependent and thus difficult to determine *ab initio* from a kinetic model, or in experiments. In a *Newtonian fluid* the viscosity is *assumed* to be constant, but this assumption is questionable for slightly viscous fluids with vastly different effects of viscosity in turbulent and laminar flow. The *existence* and

uniqueness (or the converse of *non-existence*) of solutions to the incompressible Navier-Stokes equations for a Newtonian fluid represent one of the open Clay Mathematics Institute Millenium Problems [**?**].

Solutions of the Navier-Stokes equations for a perfect gas *formally* satisfy a 2nd Law expressing that a specific scalar entropy defined in terms of mass density and internal energy is strictly increasing with increasing time, as an expression of a loss of kinetic energy by *positive viscous dissipation* formally obtained by forming the scalar of the momentum equation including viscous stresses with the velocity. The 2nd Law with strictly increasing entropy for solutions of Navier-Stokes equations enforcing irreversibility, is thus formally equivalent to the presence of non-vanishing viscosity. For the Euler equations, the entropy (formally) stays constant as an expression of vanishing viscosity and reversibility.

For the Navier-Stokes equations *with* viscous dissipation, we might thus simply forget the entropy inequality, because it is an automatically satisfied *derived relation*.

This reduces the enigma of the 2nd Law to explaining how effects of viscous dissipation and irreversibility can arise in a formally reversible system such as a system of elasticly colliding particles without viscosity. Simply claiming that viscous effects must come from somewhere is not informative. Neither is to claim that they come from quantum mechanics (or neutrino physics like Feynman), since quantum mechanics is formally reversible. Nor is simply assuming constitutive laws for viscous stresses of some form, e.g. simply assuming the fluid to be Newtonian.

Boltzmann's answer was molecular games of roulette, but it is an answer which poses more questions than it resolves.

8.2 Computational Thermodynamics

In this book we present a *constructive deterministic* fluid mechanics model of the thermodynamics of a slightly viscous gas/fluid based on *finite precision computation* by a *weighted least squares stabilized finite element method* for the *Euler/Navier-Stokes equations* with *slip boundary conditions* at solid boundaries, referred to as *RealNSE*. For definiteness we consider a perfect gas/fluid, but extension to more general state equations for the pressure is possible.

Note that we start from the Euler equations with vanishing viscosity and

heat conductivity and we thus do not need input of constitutive laws for viscous stresses and heat diffision, only a law of state for the pressure.

The least squares stabilization in RealNSE penalizes large Euler residuals by viscosity acting in the streamline direction as a form of *bulk viscosity* of order h, and is complemented by a higher-order residual dependent isotropic *shear viscosity* of order $h^{3/2}$ in turbulent regions, where h is the mesh size, together referred to as *RealNSE viscosity*.

RealNSE is a *constructive model* in the form of the Euler equations *together* with a computational solution procedure including RealNSE viscosity, guaranteeing existence by computational construction, to be distinguished from a *formal model* e.g. in the form of the Navier-Stokes equations without constructive solution procedure and requiring input of viscosity.

RealNSE automatically introduces solution dependent viscosity and thus circumvents the difficult problem of specifying viscosity (and heat conductivity) in a Navier-Stokes model. The only input in this regard is that the flow is *slightly viscous* with *Reynolds number* larger than say 10^6, which covers many important applications in aero- and hydrodynamics.

We discover by computation that RealNSE solutions are *turbulent* and have *shocks*, both phenomena being identified by substantial (positive) *RealNSE turbulent/shock dissipation* from RealNSE viscosity with pointwise large (but weakly small) Euler residuals. Turbulence/shocks are thus identified by substantial RealNSE turbulent/shock dissipation, which reflects pointwise large Euler residuals, which reflects non-existence of pointwise solutions with pointwise vanishing Euler residuals. We observe turbulence in RealNSE solutions with slip boundary conditions, and conclude that turbulence does not (primarily) originate from microscopic boundary layers with no-slip boundary conditions, contrary to state-of-the art boundary layer theory by Prandtl and Schlichting [**?**, **?**], but from macroscopic instability.

We motivate using slip boundary conditions (or more generally a friction boundary condition with small friction) by the computationally and experimentally verified fact that the *skin friction* of a turbulent boundary layer tends to zero with viscosity and thus is (comparatively) small for slightly viscous flow.

We prove that RealNSE solutions satisfy a 2nd Law expressed in terms of kinetic energy, internal energy, work and (positive) turbulent/shock dissipation. We realize that irreversibility is a consequence of substantial turbulent/shock dissipation, which is consequence of pointwise instability of slightly viscous flow, which is reflected by non-existence of pointwise solu-

tions to the Euler equations. We thus justify a 2nd Law without resort to any ad hoc assumption of positive viscous dissipation or molecular chaos.

Existence of RealNSE solutions is guaranteed by construction. Uniqueness relates to *wellposedness* in the sense of Hadamard, which concerns what aspects or *outputs* of RealNSE turbulent/shock solutions are stable under perturbations in the sense that small perturbations have small output effects. We define a RealNSE output to be wellposed if it is insensitive to mesh refinement and thus can be computed with finite mesh size. We show that wellposedness of RealNSE solutions can be tested a posteriori by computationally solving a *dual linearized problem*, through which the output sensitivity of non-zero Euler residuals can be estimated. We find that mean-value outputs such as drag and lift and total turbulent dissipation are wellposed, while point-values of turbulent flow are not. We can thus a posteriori assess the quality of RealNSE solutions as solutions of the Euler equations and identify what outputs are wellposed and converge with decreasing mesh size.

RealNSE solutions can be viewed either as exact Euler solutions subject to perturbations from non-vanishing Euler residuals, or alternatively as approximate Navier-Stokes solutions with specific viscosity given by the mesh-dependent RealNSE viscosity, combined with slip/friction boundary conditions.

We emphasize that RealNSE contributes its own mesh/solution-dependent RealNSE viscosity and thus does not require any input of viscosity (or heat conductivity) coefficients, or turbulence model.

8.3 RealNSE as Automatic Turbulence Model

RealNSE can be viewed as a form of *Large Eddy Simulation LES* for slightly viscous/high Reynolds number flow with an *turbulence model* given by the least squares stabilization for interior turbulence and the slip boundary condition for turbulent boundary layers.

We recall that *Direct Numerical Simulation DNS* by computational solution of Navier-Stokes equations with resolution of all turbulent scales including thin boundary layers, is today possible for Reynolds numbers up to 10^4, while reaching 10^6 is estimated to take another 50 years [**?**, **?**].

Some form of LES with turbulence modeling is therefore needed for high Reynolds number flow, but the design of turbulence models is an open problem since more than hundred years. RealNSE offers an automatic turbulence

model based on computational principles, with automatic assessment of well-posedness by duality.

RealNSE thus gives an answer to the enigma of the origin of viscosity in terms of of finite precision computational simulation of turbulent flow. RealNSE finite precision computation is not statistics, because statistical ensembles of outputs do not appear, only mean-values in space-time of individual trajectories/solutions.

8.4 From Probable to Necessary

The stabilization/viscous dissipation in RealNSE is necessary because without stabilization RealNSE solutions cease to exist in blowup as they inevitably go turbulent, while physical flows continue to exist without blowup even after transition to turbulence. Viscous dissipation thus comes out from an *inevitable* development of turbulence/shocks as a result of an inherent instability of slightly viscous flow, and a *necessity* to avoid blowup into non-existence, because non-existence is not an option: The show must go on, which is not a question of probability. RealNSE viscous dissipation is thus a necessity, and not the result of a game of roulette.

RealNSE viscosity transfers kinetic energy into internal energy by pointwise large but weakly small violation of conservation as the characterstic of turbulence. By necessity, as a characteristic of turbulence, the pointwise violation is large, yet in a sense smallest possible to prevent blowup and thus generic and not ad hoc.

8.5 The Spirit of Dijkstra

We follow the device of the famous computer scientist Dijkstra:

- *Originallly I viewed it as the function of the abstract machine to provide a truthful picture of the physical reality. Later, however, I learned to consider the abstract machine as the* true *one, because that is the only one we can think; it is the physical machine's purpose to supply a working model, a (hopefully) sufficiently accurate physical simulation of the true, abstract machine.*

We thus view RealNSE as a constructive mathematical model (abstract machine) of a physical reality (physical machine). We are then led to interprete

the digital finite precision computation as some form of analog physical computation, and digital RealNSE viscosity as some form of analog physical viscosity.

We compare with a classical formal mathematical model in the form of the Navier-Stokes equations including an ad hoc model of viscosity, e.g. a Newtonian model with constant shear viscosity. We are thus led to turn around the common view of computational viscosity as "artificial" viscosity and Newtonian viscosity as "physical" viscosity, and instead interprete RealNSE viscosity as physical (and then Newtonian viscosity as artificial), following von Neumann who opened compressible flow to computational simulation by introducing artificial viscosity on physical grounds.

RealNSE thus offers a model of fluid flow, which can be inspected, analyzed, understood, utilized and evaluated, while the true physics of turbulent flow may remain inaccessible to inspection and analyzis. Even if the mechanics of turbulent flow in principle can be reduced to quantum mechanics on atomic scales, such a model seems useless for predictions on macroscopic scales. For example, determining the viscosity of water from quantum mechanics is still an open problem.

We remark that the role of viscosity coefficients in classical modeling is to define viscous force in terms of fluid velocity (strain). If now RealNSE automatically supplies viscous forces, without input of viscosity, there is no longer any need to define viscosity coefficients, or the Reynolds stresses of turbulence modeling, and RealNSE thus offers a simplification of fluid mechanics and a way out of a stalemate.

Using RealNSE as a model of thermodynamics changes the questions and answers and opens new possibilities of progress together with new challenges to mathematical analysis and computation. RealNSE solutions are contructed/computed and thus are available to inspection, which means that the analysis of solutions shifts from *a priori* to *a posteriori*; after solutions have been computed they can be inspected, their qualities can be evaluated and interpreted in physical terms.

8.6 The Power of RealNSE

In related work [**?**, **?**, 23] we have demonstrated the new capabilities offered by RealNSE in resolutions of both d'Alembert's paradox from 1752 and the Clay Millenium Problem on existence and uniquenes of the incompressible

Euler/Navier-Stokes equations: In short, existence follow by construction and uniqueness from wellposedness, both being consequences of the basic structure of RealNSE as a midway between the Scylla/Galerkin of weak solution, which is too weak, and the Carybdis/least squares of strong solution, which is too strong. With the proper weighting of the least squares stabilization, RealNSE manages to combine accuracy with wellposedness and thus produce meaningful outputs.

More precisely, we will see below that output variation is bounded by $S\|hR\|_0$, where S is a stability factor, h is the mesh size, R is the Euler residual and $\|\cdot\|_0$ is a space-time L_2-norm. This results from the Galerkin orthogonality of RealNSE, which together with least squares stabilization introducing a bound on $\|\sqrt{h}R\|_0$, enforces control of output variation up to a tolerance $S\sqrt{h}$. We shall find that for mean-value outputs such as drag/lift and total turbulent dissipation, $S << h^{-1/2}$ which implies wellposedness. The unique design of RealNSE with a combination of Galerkin orthogonality and weighted least squares stabilization, thus can be motivated on rational grounds towards the goal of producing wellposed outputs, and can then as such be interpreted in physical terms.

RealNSE shows to be a computationally affordable useful model for slightly viscous (compressible and incompressible) flow, because mean-value outputs can be computed without resolving boundary layers and interior turbulence to physical scales. This reflects that mean-value outputs in slightly viscous turbulent flow, can vary little with the viscosity (beyond the drag crisis), as long as the viscosity is sufficiently small or the Reynolds is number sufficiently large, as observed in both physical experiment and computation.

RealNSE allows accurate prediction of the turbulent losses in a heat engine and the drag/lift of a car and airplane with millions of mesh points [**?**, **?**], instead of the trillions required with DNS according to state-of-the art [**?**]. RealNSE offers automatic turbulence modeling and assessment of wellposedness. Adopting RealNSE as a model of thermodynamics can be seen as a return to Euler's original idea of using inviscid flow as a model of slightly visocus flow, but with new insight and capabilities.

8.7 2nd Law for RealNSE

As already indicated, we formulate a 2nd Law for RealNSE without the concept of entropy, in terms of the basic physical quantities of kinetic energy

K, heat energy E, rate of work W and shock/turbulent dissipation $D > 0$, of the form

$$\dot{K} = W - D, \quad \dot{E} = -W + D, \tag{8.1}$$

where the dot indicates time differentiation. Slightly viscous flow always develops turbulence/shocks with $D > 0$, and the 2nd Law thus expresses an irreversible transfer of kinetic energy into heat energy, while the total energy $E + K$ remains constant because $\dot{E} + \dot{K} = 0$.

8.8 Irreversibility and Finite Precision

The 2nd Law (8.1) with $D > 0$ describes a process which is irreversible with a forward Arrow in time, since the reversed process would correspond to $D < 0$. We can understand the irreversibility as a consequence of finite precision computation combined with a certain form of wellposedness as follows: We have noted that in viscous dissipation with $D > 0$ kinetic energy is transformed into smaller scale kinetic energy in the form of internal/heat energy. This process is wellposed e.g. in the sense that the total viscous dissipation is insensitive to mesh refinement/vanishing viscosity. This can be seen as feature of a process of smashing larger units into smaller pieces with certain mean-value outputs being insensitive to the precision of the smashing procedure.

On the other hand, in the reversed process small pieces would have to be put together into larger units in a precise way, and such a process is very sensitive to the precision of the assembly.

The 2nd Law (**??**) based on finite precision computation thus can be viewed to reflect that smashing into pieces is insensitive to the precision of the smashing, while the reversed process of reassembly is sensitive to the precision of the assembly. The 2nd Law thus expresses a familar phenomenon without any mystery of molecular games of roulette, just finite precision construction/computation.

8.9 Viscosity Solutions

An RealNSE solution can be viewed as particular *viscosity solution* of the Euler equations, which is a solution of *regularized Euler equations* augmented by additive terms modeling viscosity effects with small viscosity coefficients.

The effective viscosity in an RealNSE solution typically may be comparable to the mesh size.

For incompressible flow the existence of viscosity solutions, with suitable solution dependent viscosity coefficients, can be proved a priori using standard techniques of analytical mathematics. Viscosity solutions are pointwise solutions of the regularized equations. But already the most basic problem with constant viscosity, the incompressible Navier-Stokes equations for a Newtonian fluid, presents technical difficulties, and is one of the open Clay Millennium Problems.

For compressible flow the technical complications are even more severe, and it is not clear which viscosities would be required for an analytical proof of the existence of (global smooth) viscosity solutions [35, 36] to the Euler equations. Furthermore, the question of wellposedness is typically left out, as in the formulation of the Navier-Stokes Millennium Problem, with the motivation that first the existence problem has to be settled. Altogether, analytical mathematics seems to have little to offer a priori concerning the existence and wellposedness of solutions of the compressible Euler equations. In contrast, RealNSE computational solutions of the Euler equations seem to offer a wealth of information a posteriori, in particular concerning wellposedness by duality of turbulent/shock solutions.

An RealNSE solution thus can be viewed as a specific viscosity solution with a specific regularization from the least squares stabilization, in particular of the momentum equation, which is necessary because pointwise momentum balance is impossible to achieve in the presence of shocks/turbulence. The RealNSE viscosity can be viewed to be the minimal viscosity required to handle the contradiction behind the non-existence of pointwise solutions. For a shock RealNSE could then be directly interpreted as a certain physical mechanism preventing a shock wave from turning over, and for turbulence as a form of automatic computational turbulence model.

Part II

Mathematical Foundations

Chapter 9

The Euler Equations

> However sublime are the researches on fluids which we owe to Messrs Bernoulli, Clairaut and d'Alembert, they flow so naturally from my two general formulae that one cannot sufficiently admire this accord of their profound meditations with the simplicity of the principles from which I have drawn my equations ...(Euler 1752)

> I know that most men, including those at ease with problems of the highest complexity, can seldom accept even the simplest and most obvious truth if it be such as would oblige them to admit the falsity of conclusions which they have delighted in explaining to colleagues, which they have proudly taught to others, and which they have woven, thread by thread, into the fabric of their lives. (Tolstoy)

9.1 Conservation of Mass, Momentum and Energy

Computational thermodynamics is based on a 1st Law in the form of the Euler equations for an ideal perfect gas/fluid expressing conservation of mass, momentum and total energy as a system of partial differential equations. We formulate these equations for a gas/fluid enclosed in a fixed (open) domain Ω in three-dimensional space $\mathbb{R}^3$ with boundary Γ over a time interval $[0, T]$ with intial time zero and final time T (from now on T denotes final time and not temperature as above). The Euler equations model a gas/fluid with vanishing (very small) viscosity and heat conductivity. The Euler equations then represent a Hamiltonian system.

We seek the *density* ρ, *momentum* $m = \rho u$ with $u = (u_1, u_2, u_3)$ the *velocity*, and the *total energy* ϵ as functions of $(x, t) \in \Omega \cup \Gamma \times [0, T]$, where $x = (x_1, x_2, x_3)$ denotes the coordinates in $\mathbb{R}^3$ and u_i is the velocity in the x_i-direction. The Euler equations for $\hat{u} \equiv (\rho, u, \epsilon)$ read with $Q = \Omega \times I$ and $I = (0, T]$:

$$\begin{aligned} \dot{\rho} + \nabla \cdot (\rho u) &= 0 && \text{in } Q, \\ \dot{m}_i + \nabla \cdot (m_i u) + p_{,i} &= f_i && \text{in } Q, \quad i = 1, 2, 3, \\ \dot{\epsilon} + \nabla \cdot (\epsilon u + p u) &= g && \text{in } Q, \\ u \cdot n &= 0 && \text{on } \Gamma \times I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 && \text{in } \Omega, \end{aligned} \tag{9.1}$$

where $p = p(x, t)$ is the *pressure* of the fluid, $v_{,i} = \frac{\partial v}{\partial x_i}$ is the partial derivative with respect to x_i, $\dot{v} = \frac{\partial v}{\partial t}$ is the partial derivative with respect to time t, n denotes the outward unit normal to Γ and $f = (f_1, f_2, f_3)$ is a given *volume force* (like gravity) acting on the fluid, g is a given heat source, and $\hat{u}^0 = \hat{u}^0(x)$ represent initial conditions. Further, the total energy $\epsilon = k + e$, where

$$k = \frac{\rho |u|^2}{2}$$

is the *kinetic energy* with $|u|^2 = \sum_{i=1}^3 u_i^2$, and

$$e = \rho \tau$$

is the *internal energy* or *heat energy* with τ the *temperature*, assuming the heat capacity is equal to one. The boundary condition is a *slip* boundary condition requiring the normal velocity $u \cdot n$ to vanish corresponding to an inpenetrable boundary with zero friction. Below we will consider other boundary conditions including inflow and outflow conditions and non-zero friction. Of course, $\nabla \cdot v = \sum_i v_{i,i}$ denotes the divergence of $v = (v_1, v_2, v_3)$ (and $\nabla w = (w_{,1}, w_{,2}, w_{,3})$ the gradient).

There are five equations in the Euler system (9.1), while the number of unknowns including the pressure is six, and so we need one more equation, which may be a *state equation for a compressible gas* expressing the pressure p as a function of density ρ and temperature τ, e.g. the state equation

$$p = \gamma \rho \tau \tag{9.2}$$

of a *perfect gas*, with here $0 < \gamma < 1$ a *gas constant* (equal to $\frac{c_p}{c_v} - 1$ where c_p the specific heat under constant pressure and c_v that under constant).

For a mono-atomic gas $\gamma = 2/3$. The additional equation may alternatively express that the fluid is *incompressible* in the form $\nabla \cdot u = 0$ in Q. Note that the gas constant γ is here defined as $\gamma = \bar{\gamma} - 1$ with $\bar{\gamma} = \frac{c_p}{c_v}$ the most commonly used gas constant.

The extension of the Euler equations to include viscous forces and heat flow by conduction are referred to as the *Navier–Stokes equations*, which are no longer represent a Hamiltonian system. The Navier–Stokes and Euler equations describe a very rich complex world of fluid dynamics.

The Euler equations (9.1) represents a *one-species* model, which we below will extended to a *many-species* model including chemical reactions with chemical energy adding to the total energy.

The Euler equations represent, up to the gas constant γ, a *parameter-free model*. The determination of parameters such as viscosity and heat conductivity coefficients relevant for complex flows, can be a very difficult (or simply hopeless) task, since the coefficients in general are solution dependent (and the solution depends on the coefficients). In the Euler equations the coefficients are simply put to zero with the motivation that they are small, while their actual values are irrelevant and thus do not have to be determined.

We will find that mean-value outputs are insensitive to the values of viscosity and heat conductivity, once they are small enough, which is reflected in mesh independence in RealNSE. RealNSE thus comes out as an essentially parameter-free model, and we will discover the remarkable (surprising) fact is that RealNSE can accurately predict key outputs such as drag and lift of bluff bodies and and total losses in thermodynamical processes.

9.2 Internal Energy Formulation

We shall below use a formulation of the Euler equations in terms of the internal energy e instead of the total energy ϵ, which reads: Find $\hat{u} = (\rho, m, e)$ such that

$$\begin{aligned} \dot{\rho} + \nabla \cdot (\rho u) &= 0 && \text{in } Q, \\ \dot{m}_i + \nabla \cdot (m_i u) + \gamma e_{,i} &= f_i && \text{in } Q, \quad i = 1, 2, 3, \\ \dot{e} + \nabla \cdot (eu) + \gamma e \nabla \cdot u &= g && \text{in } Q, \\ u \cdot n &= 0 && \text{on } \Gamma \times I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 && \text{in } \Omega, \end{aligned} \tag{9.3}$$

where we have used that $p = \gamma e$. We will consider (9.3) to express the 1st Law as conservation of mass, momentum and internal energy. Strictly

speaking, here only mass is conserved, because of the presence of f in the momentum equation and $\gamma e \nabla \cdot u$ and g in the equation for internal energy, and so conservation of momentum and internal energy is interpreted modulo the effects of these source terms.

We will below write the Euler equations:

$$\begin{aligned} \dot{\rho} + \nabla \cdot (\rho u) &= 0 && \text{in } Q, \\ \dot{m} + \nabla \cdot (mu) + \gamma \nabla e &= f && \text{in } Q, \\ \dot{e} + \nabla \cdot (eu) + \gamma e \nabla \cdot u &= g && \text{in } Q, \\ u \cdot n &= 0 && \text{on } \Gamma \times I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 && \text{in } \Omega, \end{aligned} \tag{9.4}$$

with $\dot{m} + \nabla \cdot (mu) + \gamma \nabla e = f$ the same as $\dot{m}_i + \nabla \cdot (m_i u) + \gamma e_{,i} = f_i$ for $i = 1, 2, 3$.

9.3 Derivation of the Euler Equations

We now show that the Euler equations (9.1)/(9.3) express conservation of mass, momentum and total/internal energy in the *conservation variables* (ρ, m, e). To this end consider a fixed small volume V in Ω with boundary S. Mass conservation implies that

$$\int_V \dot{\rho}\, dx = \frac{\partial}{\partial t} \int_V \rho\, dx = -\int_S (\rho v) \cdot n\, ds$$

expressing that the rate of increase of total mass in the fixed volume V is equal to the rate of inflow through the boundary S. The Divergence Theorem (see e.g. B&S Vol 3) states that

$$\int_V \nabla \cdot (\rho v)\, dx = \int_S (\rho v) \cdot n\, ds,$$

and we thus conclude that

$$\int_V \dot{\rho}\, dx + \int_V \nabla \cdot (\rho v)\, dx = 0$$

for all volumes V. Assuming that the integrands are continuous, we thus obtain the equation for mass conservation $\dot{\rho} + \nabla \cdot (\rho v) = 0$.

We obtain the differential equation expressing conservation of each component of the momentum m_i similarly, noting that by Newtons second law

the rate of change of momentum is given by the corresponding component $-p_{,i}$ of the pressure gradient ∇p with increasing pressure retarding the flow, combined with volume force f_i.

Finally, the equation expressing conservation of total energy ϵ is obtained as above using the Divergence Theorem, noting that the rate of change of the total energy over a volume V convected by the flow, is equal to the work $pu \cdot n$ performed on the boundary of V per unit time step. The equation for the internal energy expresses that the work $p\nabla \cdot u$ by compression/expansion is balanced by heat energy.

9.4 Incompressible Flow

In an incompressible fluid the density ρ does not change if we follow the motion of the fluid particles of the flow. We can express this fact in the differential equation form

$$D_u\rho \equiv \dot{\rho} + u \cdot \nabla\rho = 0$$

where $D_u\rho$ is the *convective derivative* of ρ with respect to the velocity u. We obtain the convective derivative by computing the change in time following the *trajectory* $x(t)$ of a fluid particle satisfying the differential equation $\dot{x}(t) = u(x,t)$. Differentiating $\rho(x(t),t)$ with respect to time, we obtain by the chain rule:

$$\frac{d}{dt}\rho(x(t),t) = (\dot{\rho} + \dot{x} \cdot \nabla\rho)(x(t),t) = D_u\rho(x(t),t).$$

Since mass conservation reads $D_u\rho + \rho\nabla \cdot u = 0$, we conclude that the velocity u in incompressible flow is characterized by the equation

$$\nabla \cdot u = 0 \quad \text{in } Q. \tag{9.5}$$

The Euler equations for incompressible flow thus take the form: Find $\hat{u}$ and p such that

$$\begin{array}{rcll} \dot{\rho} + \nabla \cdot (\rho u) & = & 0 & \text{in } Q, \\ \dot{m} + \nabla \cdot (mu) + \nabla p & = & f & \text{in } Q, \\ \nabla \cdot u & = & 0 & \text{in } Q, \\ \dot{\epsilon} + \nabla \cdot (\epsilon u + pu) & = & 0 & \text{in } Q, \\ u \cdot n & = & g & \text{on } \Gamma \times I, \\ \hat{u}(\cdot,0) & = & \hat{u}^0 & \text{in } \Omega, \end{array} \tag{9.6}$$

where the incompressibility condition $\nabla \cdot u = 0$ replaces the equation of state for the pressure p. In this case the energy equation is decoupled from the density and momentum equations; it is possible to solve for (ρ, u, p) from the first three equations and then for the energy ϵ.

If ρ^0 is constant, with $\rho^0 = 1$ say, then the first equation is trivially satisfied with $\rho = 1$, and (u, p) are solvable from the equations

$$\begin{aligned} \dot{u} + \nabla \cdot (uu) + \nabla p &= f && \text{in } Q, \quad i = 1, 2, 3, \\ \nabla \cdot u &= 0 && \text{in } Q, \\ u \cdot n &= 0 && \text{on } \Gamma \times I, \\ u(\cdot, 0) &= u^0 && \text{in } \Omega, \end{aligned} \tag{9.7}$$

which are the usual incompressible Euler equations with constant density and the energy equation left out.

9.5 Continuum and Particle Models

The Euler equations formally represent a *continuum model* with no smallest scale, since there is no smallest scale of the set of real numbers $\mathbb{R}$, and there is no viscosity or heat conduction which could define a smallest scale. It is well known that the Euler equations in general lack exact pointwise solutions, because of the appearance of turbulence and shocks, and our model of thermodynamics is instead RealNSE, where the Euler equations are solved computationally by a finite element method on a mesh of finite precision of *mesh size* h (variable in space and time).

We may think of the finite element computation as a finite precision computation with a fixed number of digits (e.g. single precision with about 7 digits) instead of computing with real numbers with infinitely many digits with infinite precision (which is impossible). Typical meshes have a mesh size of 10^{-2} on the unit cube with 10^6 mesh points. A gas has about 10^{24} molecules per mole, and thus the values of density, momentum and energy at each mesh point represent mean values of about 10^{18} molecules, thus mean values over incredibly many "fluid particles".

A computational *particle model* of a gas accounting for the position and velocity of each of the 10^{24} particles in each mole, is inconceivable on any kind of thinkable computer. Thus, only some form of continuum model can be used for macroscopic phenomena of fluid flow.

With RealNSE we stay within a deterministic framework and only add a restriction of finite precision computation to the formally Hamiltonian Euler equations. A world of thermodynamics governed by Hamiltonian mechanics combined with finite precision computation, follows the laws of mechanics as far as possible taking the finite precision into account, but is not based on any microscopic games of roulette as statistical mechanics. The difference of scientific paradigm is fundamental. We are thus led to a model of the World as a giant *clock with finite precision* as a computer age alternative Laplace's classical clock with infinite precision as well as Boltzmann's micriscopical games of roulette.

In RealNSE the mesh size h enters as a parameter which can be given a physical significance as a smallest scale in space and time. We shall discover that mean values of RealNSE-solutions, or *mean-value outputs*, show little dependence on the mesh size h, and we shall also uncover (some of) the mathematical rationale behind this remarkable feature. This opens the possibility of accurately simulating real phenomena using a coarser mesh size than the physical smallest scale, which is like computing with say 10^6 "super-particles" instead of 10^{24} "real particles", thus making an impossible simulation possible.

9.6 Boltzmann's Equation

Boltzmann starts from a *molecular particle model* of a dilute gas as a very large collection of elastically colliding little spherical particles/molecules according to classical Newtonian mechanics. This model is formally reversible without effects of viscosity, simply because each elastic collision is reversible. From this particle model Boltzmann derives a continuum model from an assumption of *molecular chaos* requiring particles to have statistically independent velocities before collision, in the form of *Boltzmann's equation*:

$$\dot{f} + v \cdot \nabla_x f + F \cdot \nabla_v f = Q(f, f), \tag{9.8}$$

where $f(x, v, t)$ is the number of gas molecules at the position x moving with velocity v per volume $dxdv$, at time t, F is an applied force, and ∇_x is the gradient with respect to x, ∇_v the gradient with respect to v, and $Q(f, f)$ is Boltzmann's *collision operator* representing the rate of change of $f(x, v, t)$ due to (elastic) molecule collisions. Boltzmann's collision term has a certain

dissipative character, from which Boltzmann derived his famous *H-Theorem* stating that $\dot{H} \geq 0$, where

$$H(t) = \int f \log(f) dx dv$$

is Boltzmann's *H-function*. Irreversibility would then be signified by a strictly decreasing H-function, and thus $-H$ could be viewed as a strictly increasing entropy. Boltzmann's equation is deterministic, but it is derived from a statistical assumptiopn of *molecular chaos* requiring the velocities of two molecules about to collide to be statistically independent *before* collision, but not after. The assumption of molecular chaos is what gives the collision term its dissipative character with $H(t)$ strictly decreasing as the gas steadily moves towards its *equilibrium distribution* with minimal H and $Q = 0$.

The motivation of the assumption of molecular chaos is the weak point of Boltzmann's gas kinetics: Statistical independence before collision but not after, obviously introduces a direction of time by direct assumption. The main head-ache of Boltzmann was thus to motivate this assumption, in order not to simply assume what was to be proved, namely time-directionality. To do so he tried various options and finally seemed to converge to the idea that the probability of a macro-state should be related to the number of underlying micro-states, and thus the H-theorem would reflect a tendency of the gas molecules to move from a less probable to a more probable distributions, with the final equilibrium position as the most probable one. To count the number of micro-states underlying a certain macrostate requires (tricky) combinatorics, and this is what makes Boltzmann's statistical mechanics so difficult to understand (for most people). We will come back to this aspect in an account (and resolution) of *Gibb's paradox* exhibiting difficulties of combinatorics in gas kinetics.

We shall below recall that Boltzmann defines the entropy S of a certain macrostate as being proportional to $\log(W)$ where W is the number of corresponding microstates. Boltzmann thus plays with two different definitions of entropy, one statistical in terms of the number of microstates, W and one deterministic in terms of the H-function, which is very confusing to the non-specialist of statistical mechanics.

Did Boltzmann thereby resolve the enigma how irreversibility can arise in a formally reversible Newtonian model like a swarm of elastically colliding particles? Not really, because his basic roulette assumption, also referred to as *molecular chaos*, is asymmetric in time (two particles play roulette *before*

collision), and thus introduces viscosity by assumption. Boltzmann thus *assumed* what was to be proved, namely the occurence of effects of viscosity and time asymmetry, and therefore met strong opposition by in particular Loschmidt [40].

9.7 Euler's Equations from Boltzmann's

The Euler equations of fluid mechanics (9.1) can *formally* be derived from Boltzmann's equation of gas kinetics (9.8), by integration over the velocity variable v and defining with M the mass of a molecule,

$$\begin{aligned} \rho(x,t) &= M\int f(x,t,v)\,dv, \\ m(x,t) &= M\int f(x,t,v)v\,dv \\ e(x,t) &= \frac{M}{2}\int f(x,t,v)|v-u|^2\,dv, \end{aligned} \tag{9.9}$$

where as usual $u = m/\rho$. We note in particular that the internal (heat) energy e is defined as a form of (small scale) kinetic energy measuring the standard deviation from the mean velocity u.

9.8 Navier-Stokes Equations

In the Navier-Stokes equations the momentum equation is augmented by a viscous term of the form

$$-\nabla\cdot(\nu\nabla u) - \nabla(\mu\nabla\cdot u), \tag{9.10}$$

where $\nu > 0$ is a *shear viscosity* coefficient and $\mu > 0$ a *bulk viscosity*. The equation for the heat energy is similarly augmented by a heat conduction term of the form $-\nabla\cdot(\kappa\nabla e)$ with $\kappa > 0$ a *heat conductivity*. The determination of correct values of these coefficients in turbulent flow is very complicated, or more precisely in practice impossible, because the coefficients are solution dependent. We stay away from this complications by assuming the coefficients to be *small* and we then effectively put them to zero in the Euler equations, just as Euler did. The mesh-independence of certain mean-value outputs then reflects that the mean-values are insenistive to the absolute size of viscosities and heat conductivity, as long as they are small.



Figure 9.1: The shock wave of a supersonic airplane made visible by condensation

9.9 Mathematical Literature on Compressible Flow

There is a vast literature on compressibe flow described by Euler's equations [37], but basic questions about existence and wellposedness of solutions are left without answers. The difficulty is that solutions of the Euler equations initiated from smooth initial data, which do exist as unique smooth solutions for some period of time, develop shocks/turbulence and thus can continue only as non-smooth weak solutions with possible lack of physical meaning and uniqueness. To secure physical meaning entropy conditions of different forms are used, but it is now known which conditions will suffice to secure weak solutions to be physical.

With sufficiently strong non-linear viscosities global existence of unique smooth solutions of Navier-Stokes equations can formally be proved by standard mathematical techniques, however without any assessment of wellposedness and physical meaning. For constant viscosities results are lacking. The Navier-Stokes equations are thus as inaccessible to formal mathematical anal-

ysis as the Euler equations.

What remains is computation and we thus make a restart with RealNSE for Euler's equations and we do answer questions about existence and uniqueness related to wellposedness.

One condition proposed in the literature is *maximal dissipation of kinetic energy* [38], but its is not clear that physics follows this principle. In RealNSE as best possible satisfaction of Euler's equations, it is rather a principle of minimal dissipation of kinetic energy as the smallest possible price to pay for non-smoothness.

> A theory is the more impressive the greater the simplicity of its premises, the more different kinds of things it relates, and the more extended its area of applicability. Therefore the deep impression that classical thermodynamics made upon me. It is the only physical theory of universal content which I am convinced will never be overthrown, within the framework of applicability of its basic concepts.(Einstein)

> Nothing in life is certain except death, taxes and the second law of thermodynamics. All three are processes in which useful or accessible forms of some quantity, such as energy or money, are transformed into useless, inaccessible forms of the same quantity. That is not to say that these three processes don't have fringe benefits: taxes pay for roads and schools; the second law of thermodynamics drives cars, computers and metabolism; and death, at the very least, opens up tenured faculty positions. (Seth Lloyd, in *Nature*, 2004)

Chapter 10

Viscosity Solutions to 1d Euler

> The total energy of the universe is constant; the total entropy is continually increasing. (Rudolf Clausius 1865)
>
> You believe in the God who plays dice, and I in complete law and order in a world which objectively exists, and which I, in a wild speculative way, am tryin to capture. Even the great initial success of Quantum Theory does not make me believe in the fundamental dice-game, although I am well aware that younger collegues interpret this as a consequence of senility. No doubt the day will come when we will see those instictive attitude was the correct one. Some physicists. among them myself, cannot believe that we must abandon, actually and forever, the idea of direct representation of physical reality in space and time; or that we must accept then the view that events in nature are analogous to a game of chance...In any case I am convinced that *He* does not throw dice. (Einstein)

10.1 Regularisation of Hamiltonian Systems

We present in this chapter a short-cut to the computational foundation of thermodynamics in the setting of the Euler equations in one space dimension (1d). In particular, we motivate a 2nd Law, which we will below show to be satisfied by RealNSE in the full 3d case.

We use the standard approach of *viscous regularisation* motivated by the fact that the Euler equations do not admit exact pointwise solutions, because of the inevitable appearance of turbulence/shocks. We will view RealNSE solutions as specific regularised solutions, which are computed and

thus exist and can be inspected. We shall below in more detail investigate the precise form of the regularisation in RealNSE, and discover that it has interesting features, not only from mathematical and computational, but also from physical point of view.

The basic idea in viscous regularisation of a Hamiltonian system described by an equation $H(u) = 0$ over a domain $\Omega \times [0, 1]$ in space-time, such as the Euler equations, is to perturb the equation $H(u) = 0$ into

$$H(u_\nu) = \nu\Delta u_\nu \quad \text{in } \Omega \times [0, 1],$$

where $\nu > 0$ is a small parameter representing a small (shear) viscosity, and the Laplacian has a regularising or smoothing effect on the solution.

The presence of the Laplacian (or possibly a higher order regularising differential operator such as the biLaplacian), makes it possible in some cases to prove the existence of a pointwise solution u_ν to the regularised problem, a *viscosity solution*, by standard techniques of mathematical analysis involving Sobolev and Gronwall inequalities, while the original Hamiltonian system is inaccessible to analysis because the equation $H(u) = 0$ lacks pointwise solutions. For the regularised compressible Euler equations, technical difficulties have hitherto prevented a full proof of existence of pointwise solutions. But this is not crucial for the computational foundation since this is based on computed RealNSE-solutions with specific residual-based regularisation and not viscosity solutions, and computed solutions do exist and can be inspected. We here use viscosity solutions to formally illustrate basic features of RealNSE-solutions such as the 2nd Law, which we then verify directly for RealNSE-solutions.

If the regularised solution u_ν is *smooth* in the sense that Δu_ν is of moderates size, then the perturbation $\nu\Delta u_\nu$ will be small, because ν is small, and thus the regularised solution u_ν will be an approximate pointwise solution of the Hamiltonian system with the Hamiltonian residual $H(u_\nu)$ being small in a pointwise sense. In this case the the perturbation can be said to be *regular*, and is inessential in the sense that already the original problem has a pointwise solution and regularisation is not needed.

However, we consider problems where the Hamiltonian residual $H(u_\nu)$ is not small (in fact large) in regions with shocks/turbulence, while $H(u_\nu)$ remains small in a weak norm, so that $\hat{u}_\nu$ still is an approximate solution to the Hamiltonian system, but now only in a weak (but not pointwise) sense. The viscous term $\nu\Delta u$ will thus be pointwise large, which is reflected by the

fact that the total dissipation

$$D(u_\nu) = \int_\Omega \nu |\nabla u_\nu|^2 dx$$

is not small (of moderate size). In particular, the regularised solution u_ν is *non-smooth* in the sense that $|\nabla u_\nu|$ and $|\Delta u_\nu|$ are locally very large (of size ν^{-1} and ν^{-2} at shocks and of size $\nu^{-1/2}$ and $\nu^{-3/2}$ in turbulent regions, respectively). In this case the perturbation is *singular*, and is essential in the sense that only the regularised problem has pointwise solutions. We thus consider a Hamiltonian system with singular perturbation.

We understand that $D(u_\nu)$ results from multiplication of $-\nu\Delta u_\nu$ by u_ν and integration (by parts) over Ω. Typically, $D(u_\nu)$ will have a positive limit as ν tends to zero, which expresses an independence of the specific form of the regularisation, that is the specific value of ν, with always the term $-\nu\Delta u_\nu$ balancing the non-zero Hamiltonian residual $H(u_\nu)$, with Δu_ν getting larger as ν gets smaller.

We will thus view the appearance of the regularisation term as a mathematical formality and we do not seek to connect it to any physical "shear viscosity". This is because the appearance of "shear viscosity" in Hamiltonian systems without viscosity, such as the Euler equations, represents the main mystery of the classical 2nd Law. We thus do not accept the common argument that "there is always some form of (shear) viscosity, which we can model by a Laplacian". The key question is from where such a viscosity comes in a inviscid flow.

We shall see that RealNSE offers a new answer to this key question, where the residual regularisation appears as a loss of kinetic energy because the momentum equation cannot be satisfied pointwise, and not as any form of classical (shear) viscosity. The loss of kinetic energy (which is transformed into heat energy) can be viewed as a "fine" to be paid because the law $H(u) = 0$ is heavily violated pointwise. The fine can take many forms, since there are many possible forms of regularisation, while the violation essentially remains the same. We shall see that RealNSE penalises a large residual $H(\hat{u})$, while classical viscous regularisation penalises all large derivates, not only the particular combination of the residual.

RealNSE thus has "fine-tuned" penalisation with the fine being as close as possible to the violation, like a penicillin with narrow spectrum killing only microbes producing the symptom, which suggests that RealNSE can be given a physical significance. On the ohter hand, classical viscous regularisation, is

more like a broad band penicillin killing both microbes and healthy bacteria, and its physical significance has remained a mystery.

To sum up, we start from a Hamiltonian system without pointwise solutions (the Euler equations), which we replace by a regularised system with pointwise solutions (by viscous regularisation or RealNSE), which can be viewed as approximate solutions to the original system. We find that certain mean-values of the regularised solutions show little dependence on the specific form of the regularisation (the viscosity or the mesh size), and thus can be viewed as stable outputs of approximate solutions to the original system. We first carry out the argument for viscous regularisation and then return below to the real case of RealNSE regularisation.

10.2 1d Euler

We consider an inviscid perfect gas enclosed in a tube represented by the interval $\Omega = (0,1)$ in space over a time interval $I = (0,1]$ and we assume that $f = 0$ and $g = 0$. The 1d Euler equations expressing the 1st Law as conservation of mass, momentum and internal energy, formally take the form: Find $\hat{u} \equiv (\rho, m, e)$ such that with $v' = \frac{\partial v}{\partial x}$

$$\begin{aligned} R_\rho(\hat{u}) \equiv \dot{\rho} + (\rho u)' &= 0 \quad \text{in } Q, \\ R_m(\hat{u}) \equiv \dot{m} + (mu + p)' &= 0 \quad \text{in } Q, \\ R_e(\hat{u}) \equiv \dot{e} + (eu)' + pu' &= 0 \quad \text{in } Q, \\ u(0,t) = u(1,t) &= 0 \quad t \in I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{10.1}$$

where $p = \gamma e$ with $\gamma > 0$ and $u = \frac{m}{\rho}$, or in short form

$$\begin{aligned} R(\hat{u}) &= 0 \quad \text{in } Q, \\ u(0,t) = u(1,t) &= 0 \quad t \in I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{10.2}$$

where $R = (R_\rho, R_m, R_e)$.

10.3 Formal Reversiblity

The Euler equations (10.1) without regularization are formally *reversible*: Changing the sign of the velocity u and the direction of time (the sign of the

time-derivative $\dot{u}$), the Euler equations obviously remain unchanged. This means that if $\hat{u}(t)$ is a solution to the Euler equations in forward time on $[0, T]$ taking the initial value $\hat{u}(0)$ to the final value $\hat{u}(T)$, we obtain a solution in backward time taking $\hat{u}(T)$ back to $\hat{u}(0)$ by reversing the velocity at time $\bar{t}$. Of course we can view this solution as proceeding in forward time by just continuing counting time forward after the velocity reversal.

We conclude that if the Euler equations have a pointwise solution then it can be turned into a perpetuum mobile running for ever without consuming any energy, for ever bouncing back and forth by repeated reversal of the velocity at two given time instances.

10.4 Factual Irreversibility

The trouble with the above argument is that the Euler equations lack pointwise solutions and thus the implication is empty. In contrast viscosity solutions to the regularized Euler equations exist, but these solutions are turbulent with substantial turbulent dissipation and thus are irreversible. Therefore a perpetuum mobile is impossible.

We sum up: The exact solutions which would have been reversible if they had existed, do not exist. The viscosity solutions which do exist, are not reversible. This is the main lesson of this book, and resolves the main open problem of classical thermodynamics: *Loschmidt's paradox* asking how irreversibility can arise in a reversible system.

10.5 Energy Estimates for Viscosity Solutions

We shall now prove that a viscosity solutions of the regularized 1d Euler equations (10.1) is a *dissipative weak solution* of the Euler equations in the sense that its Euler residual is small in a weak sense and it satisfies a 2nd Law expressing an irreversible transfer from kinetic to heat energy in the form of turbulent/shock dissipation. In short we prove that viscosity solutions are dissipative weak solutions of the Euler equations.

We thus consider the following regularized version of (10.1): Find $\hat{u}_{\nu,\mu} \equiv \hat{u} = (\rho, m, e)$ such that

$$\begin{aligned} \dot{\rho} + (\rho u)' &= 0 \quad \text{in } Q, \\ \dot{m} + (mu + p)' &= (\nu u')' + (\mu p u')' \quad \text{in } Q, \\ \dot{e} + (eu)' + pu' &= \nu (u')^2 \quad \text{in } Q, \\ u(0,t) = u(1,t) &= 0 \quad t \in I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{10.3}$$

where $p = \gamma e$, and $\nu > 0$ and $\mu \geq 0$ are small parameters with ν representing *shear viscosity* and μ a form of *bulk viscosity* with $\mu = 0$ if $u' < 0$. For simplicity, we here suppress the subindices ν and μ. We here use the balance equation for the internal energy e, modified to account for the contribution to the internal energy from shear viscosity, but without contribution from bulk viscosity, which we consider as a penalty. We observe that only the velocity u is subject to regularization, which gives (as we will see below) the momentum equation a different quality than the balance equations for mass and internal energy, both involving u as a coefficient.

We shall see that it is natural to choose the regularization parameter ν much smaller than μ, and thus the main effect of the regularization comes from the bulk viscosity μ, rather than from the usual shear viscosity ν. We shall see that the bulk viscosity prevents fluid particles from colliding, or more precisely, prevents faster fluid particles to overtake slower particles (along the same streamline), which thus is the main effect of the regularization and not shear viscosity.

We shall now prove that $\hat{u} = \hat{u}_{\nu,\mu}$ satisfies

$$\|R_m(\hat{u})\|_{-1} \leq \frac{\sqrt{\nu}}{\sqrt{\mu}} + \sqrt{\mu}, \tag{10.4}$$

where $\|\cdot\|_{-1}$ denotes the $L_2(I; H^{-1}(J))$-norm, while obviously $R_\rho(\hat{u}) = 0$ and $R_e(\hat{u}) \geq$ in a pointwise sense in $\Omega \times I$. Further, we shall observe that $\hat{u}$ satisfies the following 2nd Law:

$$\dot{K} \leq W - D, \quad \dot{E} = -W + D,$$

where

$$K = \int_J k dx, \quad E = \int_J e dx, \quad W = \int_J pu' dx, \quad D = \int_J \nu (u')^2 \, dx > 0.$$

Choosing $\mu = \epsilon$ and $\nu = \epsilon^2$, we can assure that $\|R_m(\hat{u}_{\nu,\mu})\|_{-1} \leq 2\sqrt{\epsilon}$ for any positive ϵ. We shall prove the following result on the existence of *dissipative weak approximate solutions* to the 1d Euler equations:

Theorem 1 *A viscosity solution $\hat{u}$ of the regularized Euler equations (10.1) satisfies*

$$R_\rho(\hat{u}) = 0 \quad in\ \Omega \times I, \quad \|R_m(\hat{u})\|_{-1} \leq TOL, \quad R_e(\hat{u}) \geq 0 \quad in\ \Omega \times I,$$

where $TOL = \frac{\sqrt{\nu}}{\sqrt{\mu}} + \sqrt{\mu}$, together with the 2nd Law

$$\dot{K} \leq W - D, \quad \dot{E} = -W + D, \quad D > 0. \tag{10.5}$$

We understand that Theorem 1 states certain properties of the regularised solution $u_{\nu,\mu}$ in terms of its Euler residuals, without explicitly referring to the regularisation, where the 2nd Law compensates for the fact that the momentum residual $R_m(\hat{u})$ is only required to be small in a weak norm, and $R_e(\hat{u})$ is allowed to be pointwise positive. We note that by the 2nd Law (10.5) it follows that

$$\dot{K} + \dot{E} \leq 0 \tag{10.6}$$

stating that the integral (or totality) in space of the total energy ϵ cannot increase.

To make the notion of dissipative weak solution really useful, we have to connect it to *output uniqueness*. We will return to this basic aspect below in the context of RealNSE, using directly the properties of RealNSE without passing through Theorem 1. We can thus view Theorem 1 as a connection to a classical analytical technique of regularized solutions, which we will not pursue in detail, because what we can compute are RealNSE solutions, not classical regularized solutions, and because a RealNSE solution can be viewed as a (new) form of regularized solution.

Since the concept of weak solution can be viewed as expressing a form of approximate solution, with an exact solution being a (strong) pointwise solution, we can condense the notation to *dissipative weak solution*. In short, we will thus prove the existence of dissipative weak solutions to the Euler equations and below see that a RealNSE solution is a dissipative weak solution.

Proof of Theorem 1: The basic technical step is to multiply the momentum equation by u, omitting for simplicity the indices ν and μ, and use the mass balance equation in the form $\frac{u^2}{2}(\dot{\rho} + (\rho u)') = 0$, to get

$$\dot{k} + (ku)' + p'u - \mu(pu')'u - \nu u''u = 0. \tag{10.7}$$

By integration in space it follows that $\dot{K} \leq W - D$, and similarly it follows that $\dot{E} = -W + D$ from the equation for e, which proves the 2nd Law. Adding next (10.7) to the equation for the internal energy e and integrating in space, gives

$$\dot{K} + \dot{E} + \int_0^1 \mu p (u')^2 \, dx = 0,$$

since

$$\int_0^1 (p'u + pu')dx = \int_0^1 (pu)'dx = 0,$$

and thus after integration in time

$$K(1) + E(1) + \int_Q \mu p (u')^2 \, dxdt = K(0) + E(0). \tag{10.8}$$

We now need to show that $E(1) \geq 0$ (or more generally that $E(t) > 0$ for $t \in I$), and to this end we rewrite the equation for the internal energy as follows:

$$\frac{De}{Dt} + (\gamma + 1)eu' = \nu (u')^2,$$

where $\frac{De}{Dt} = \dot{e} + ue'$ is the material derivative of e following the fluid particles with velocity u. Assuming that $e(x, 0) > 0$ for $0 \leq x \leq 1$, it follows that $e(x, 1) > 0$ for $0 \leq x \leq 1$, and thus $E(1) > 0$. Assuming $K(0) + E(0) = 1$ the energy estimate (11.7) thus shows that

$$\int_Q \mu p (u')^2 \, dxdt \leq 1, \tag{10.9}$$

and also that $E(t) \leq 1$ for $t \in I$.

Next, integrating (10.7) in space and time gives

$$K(1) + \int_Q \nu (u')^2 dxdt = \int_Q pu' dxdt - \int_Q \mu p (u')^2 dxdt \leq \frac{1}{\mu} \int_Q p dxdt \leq \frac{1}{\mu},$$

where we used that $\int_Q p dxdt = \gamma \int_Q e dxdt \leq \int_I E(t)dt \leq 1$. It follows that

$$\int_Q \nu (u')^2 dxdt \leq \frac{1}{\mu}. \tag{10.10}$$

By standard estimation it follows from (10.9) and (10.10) that

$$\|R_m(\hat{u})\|_{-1} \leq \sqrt{\mu} + \frac{\sqrt{\nu}}{\sqrt{\mu}},$$

and the proof can be completed in an obvious fashion.

10.6 Irreversibility by the 2nd Law

The 2nd Law (10.5) states an irreversible transfer of kinetic energy to heat energy for in the presence of shocks with $D > 0$, which is the generic case. On the other hand, the sign of W is variable and thus the corresponding energy transfer may go in either direction.

10.7 Compression and Expansion

The 2nd Law (10.5) states that there is a transfer of kinetic energy to heat energy if $W < 0$, that is under compression with $u' < 0$, and a transfer from heat to kinetic energy if $W > 0$, that is under expansion with $u' > 0$. As we just remarked, there is a transfer from kinetic to heat energy for solutions with shocks with $D > 0$.

Returning to Joule's experiment, we see by the 2nd Law that contraction back to the original volume from the final rest state in the double volume, is impossible, because the only way the gas can be set into motion is by expansion.

Chapter 11

Viscosity Solutions to 3d Euler

> Everyone knows that heat can produce motion. That it possesses vast motive power no one can doubt, in these days when the steam engine is everywhere so well known. The study of these engines is of great interest, their importance is enormous, their use is continually increasing, and they seem desined to produce a great revolution in the civilised world. (Carnot [10] 1824).
>
> But maybe that is our mistake: maybe there are no particle positions and velocities, but only waves. It is just that we try to fit the waves to our preconceived ideas of positions and velocities. The resulting mismatch is the cause of the apparent unpredictability.(Stephen Hawking)
>
> What wanted to say was just this: In the present circumstances the only profession I would choose would be one where earning a living had nothing to do with the search for knowledge. (Einstein's last letter to Born Jan 17 1955 shortly before his death on the 18th of April, probably referring to Born's statistical interpretation of quantum mechanics).

11.1 3d Euler

We now show that the above result for the 1d Euler equations directly extends to 3d. We thus consider the 3d Euler equations for an inviscid perfect gas enclosed in a volume Ω in $\mathbb{R}^3$ with boundary Γ over a time interval $I = (0, 1]$,

assuming as above that the applied force $f = 0$: Find $\hat{u} = (\rho, m, e)$ depending on $(x, t) \in Q \equiv \Omega \times I$ such that

$$\begin{aligned} R_\rho(\hat{u}) \equiv \dot{\rho} + \nabla \cdot (\rho u) &= 0 \quad \text{in } Q, \\ R_m(\hat{u}) \equiv \dot{m} + \nabla \cdot (mu) + \nabla p &= 0 \quad \text{in } Q, \\ R_e(\hat{u}) \equiv \dot{e} + \nabla \cdot (eu) + p\nabla \cdot u &= 0 \quad \text{in } Q, \\ u \cdot n &= 0 \quad \text{on } \Gamma \times I \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{11.1}$$

where $u = \frac{m}{\rho}$ and $p = \gamma e$ with $\gamma > 0$.

11.2 Energy Estimates for Viscosity Solutions

We consider the following regularized version of (11.1): Find $\hat{u}_{\nu,\mu} \equiv \hat{u} = (\rho, m, e)$ such that

$$\begin{aligned} R_\rho(\hat{u}) &= 0 \quad \text{in } Q, \\ R_m(\hat{u}) &= \nabla \cdot (\nu \nabla u) + \nabla(\mu p \nabla \cdot u) \quad \text{in } Q, \\ R_e(\hat{u}) &= \nu |\nabla u|^2 \quad \text{in } Q, \\ u &= 0 \qquad \text{on } \Gamma \times I, \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{11.2}$$

where $\nu > 0$ is a shear viscocity and $\mu \geq 0$ a bulk viscosity with $\mu = 0$ if $\nabla \cdot u < 0$, and $|\nabla u|^2 = \sum_{i=1}^3 |\nabla u_i|^2$.

As already indicated, the existence of a pointwise solution $\hat{u}_{\nu,\mu}$ to the regularised problem is an open problem of mathematical analysis. However, we only use the regularised problem to formally illustrate basic properties of RealNSE-solutions, which we prove directly below.

We shall now prove that $\hat{u} = \hat{u}_{\nu,\mu}$ satisfies

$$\|R_m(\hat{u})\|_{-1} \leq \frac{\sqrt{\nu}}{\sqrt{\mu}} + \sqrt{\mu}, \tag{11.3}$$

where $\|\cdot\|_{-1}$ denotes the $L_2(I; H^{-1}(\Omega))$-norm, while we may assume that $R_\rho(\hat{u}) = 0$ and $R_e(\hat{u}) \geq 0$ in a pointwise sense in Q. We shall also find that $\hat{u}$ satisfies the following 2nd Law:

$$\dot{K} \leq W - D, \quad \dot{E} = -W + D,$$

where

$$K = \int_\Omega k\,dx, \quad E = \int_\Omega e\,dx, \quad W = \int_\Omega p\nabla\cdot u\,dx, \quad D = \int_\Omega \nu|\nabla u|^2\,dx.$$

Choosing $\mu = \epsilon$ and $\nu = \epsilon^2$, we can assure that $\|R_m(\hat{u}_{\nu,\mu})\|_{-1} \leq TOL$ for any positive tolerance TOL. We shall thus show the following formal result on the existence of *dissipative weak approximate solutions* to the 3d Euler equations:

Theorem 2 *A viscosity solution $\hat{u}$ of the regularized Euler equations (11.1) satisfies*

$$R_\rho(\hat{u}) = 0 \quad \text{in } \Omega\times I, \quad \|R_m(\hat{u})\|_{-1} \leq TOL, \quad R_e(\hat{u}) \geq 0 \quad \text{in } \Omega\times I,$$

where $TOL = \frac{\sqrt{\nu}}{\sqrt{\mu}} + \sqrt{\mu}$, together with the 2nd Law

$$\dot{K} \leq W - D, \quad \dot{E} = -W + D, \quad D > 0. \tag{11.4}$$

As above, we note that Theorem 1 states certain properties of the regularized solution $u_{\nu,\mu}$ in terms of its Euler residuals, without explicitely referring to the regularization, where the 2nd Law compensates for the fact that the momentum residual $R_m(\hat{u})$ is only required to be small in a weak norm, and $R_e(\hat{u})$ is allowed to be pointwise positive. We note that by the 2nd Law it follows that

$$\dot{K} + \dot{E} \leq 0 \tag{11.5}$$

stating that the integral (or totality) in space of the total energy ϵ cannot increase. We shall see below that a RealNSE solution is a dissipative weak approximate solution.

Proof of Theorem 2: The basic technical step is to multiply the momentum equation by u, and use the mass balance equation in the form $\frac{|u|^2}{2}(\dot{\rho} + \nabla\cdot(\rho u) = 0$, to get

$$\dot{k} + \nabla\cdot(ku) + \nabla p\cdot u - \nabla(\mu p\nabla\cdot u)\cdot u + \nabla\cdot(\nu\nabla u)\cdot u = 0. \tag{11.6}$$

By integration in space it follows that $\dot{K} \leq W - D$, and similarly it follows that $\dot{E} = -W + D$ from the equation for e, which proves the 2nd Law.

Adding next (11.6) to the equation for the internal energy e and integrating in space, gives

$$\dot{K} + \dot{E} + \int_\Omega \mu p (\nabla \cdot u)^2 \, dx = 0,$$

since

$$\int_\Omega (\nabla p \cdot u + p \nabla \cdot u) dx = \int_\Omega \nabla \cdot (pu) dx = 0,$$

and thus after integration in time

$$K(1) + E(1) + \int_Q \mu p (\nabla \cdot u)^2 \, dxdt = K(0) + E(0). \tag{11.7}$$

We now need to show that $E(1) \geq 0$ (or more generally that $E(t) > 0$ for $t \in I$), and to this end we rewrite the equation for the internal energy as follows:

$$\frac{De}{Dt} + (\gamma + 1) e \nabla \cdot u = \nu |\nabla u|^2,$$

where $\frac{De}{Dt} = \dot{e} + u \cdot \nabla e$ is the material derivative of e following the fluid particles with velocity u. Assuming that $e(x, 0) > 0$ for $x \in \Omega$, it follows that $e(x, 1) > 0$ for $x \in \Omega$, and thus $E(1) > 0$. Assuming $K(0) + E(0) = 1$ the energy estimate (11.7) thus shows that

$$\int_Q \mu p (\nabla \cdot u)^2 \, dxdt \leq 1, \tag{11.8}$$

and also that $E(t) \leq 1$ for $t \in I$.

Next, integrating (11.6) in space and time gives

$$K(1) + \int_Q \nu |\nabla u|^2 dxdt = \int_Q p \nabla \cdot u dxdt - \int_Q \mu p (\nabla \cdot u)^2 dxdt \leq \frac{1}{\mu} \int_Q p dxdt \leq \frac{1}{\mu},$$

where we used that $\int_Q p dxdt = \gamma \int_Q e dxdt \leq \int_I E(t) dt \leq 1$. It follows that

$$\int_Q \nu |\nabla u|^2 dxdt \leq \frac{1}{\mu}. \tag{11.9}$$

By standard estimation it follows from (11.8) and (11.9) that

$$\|R_m(\hat{u})\|_{-1} \leq \sqrt{\mu} + \frac{\sqrt{\nu}}{\sqrt{\mu}},$$

and the proof can be completed in an obvious fashion.

11.3 Irreversibility by the 2nd Law

The 2nd Law (11.4) states an irreversible transfer of kinetic energy to heat energy in the presence of shocks/turbulence with $D > 0$, which is the generic case. On the other hand, the sign of W is variable and thus the corresponding energy transfer may go in either direction.

11.4 Compression and Expansion

The 2nd Law (11.4) states that there is a transfer of kinetic energy to heat energy if $W < 0$, that is under compression with $\nabla \cdot u < 0$, and a transfer from heat to kinetic energy if $W > 0$, that is under expansion with $\nabla \cdot u > 0$.

Returning to Joule's experiment, we see by the 2nd Law that contraction back to the original volume from the final rest state in the double volume, is impossible, because the only way the gas can be set into motion is by expansion.

11.5 A 2nd Law witout Entropy

We note that the 2nd Law (11.4) is expressed in terms of the kinetic energy K, the heat energy E and the work W, and does not involve any concept of entropy S. This relieves us from the task of finding a physical significance of S and a physical justification of a classical 2nd Law of the form $dS \geq 0$. We thus circumvent the main difficulty of classical thermodynamics based on statistical mechanics, while we reach the same goal as statistical mechanics of explaining irreversibility in formally reversible Newtonian mechanics.

Note that since the new 2nd Law is a consequence of the 1st Law and RealNSE computation, we can "forget" the 2nd Law: It is automatically satisfied without special attention. This is like "forgetting" to keep the balance while walking, because is automatically maintained without conscious attention.

Chapter 12

RealNSE

> The 2nd Law cannot be derived from purely mechanical laws. It carries the stamp of the essentially statistical nature of heat. (Bergmann [4])
>
> Therefore I feel that the Heisenberg-Bohr (Copenhagen) statistical interpretation of quantum mechanics is dead. (Zeh [51])
>
> If a scientist says that something is possible he is almost certainly right, but if he says that it is impossible he is probably wrong. (Arthur C. Clarke)

12.1 Introduction

We refer to [23] for a detailed presentation of RealNSE for the incompressible Euler equations. The extension to the compressible Euler equations has the principal form: Find $\hat{u} \in V_h$ such that

$$((R(\hat{u}),\hat{v})) + ((hR(\hat{u}),R'(\hat{u},\hat{v}))) = 0 \quad \text{for } \hat{v} \in V_h \tag{12.1}$$

where V_h is a space-time finite element space on a mesh with mesh size h, $((\cdot,\cdot))$ denotes space-time L_2 scalar products, and $R'(\hat{u},\hat{v})$ is a linearization of the Euler residual $R(\hat{u})$ at $\hat{u}$ such that $R'(\hat{u},\hat{u}) = R(\hat{u})$, which leads to residual least squares stabilization through the positive penalty term $((hR(\hat{u}),R(\hat{u}))$ obtained choosing $\hat{v} = \hat{u}$.

In the actual implementation of RealNSE on cG(1)cG(1)-form stated below, the stabilisation has a reduced form involving only some of the terms

summing to the residual. The rationale is that the stabilisation has an effect when the residual is large pointwise, and then stabilising one term of the residual suffices. A residual-based *shock-capturing* stabilisation with viscosity coefficient $\sim h^2|R(\hat{u})|$ is also used, which eliminates (small) oscillations at shocks, but shock-capturing alone is not sufficient.

The 2nd Law satisfied by RealNSE shows that the stabilisation effectively concerns one term of the momentum equation, and not the mass and energy equations. The motivation is that since the momentum equation will not be pointwise (strongly) small, only weakly small, a stabilisation is required because the the balance of kinetic energy results from multiplication of the momentum equation by the velocity and this operation is not stable under weak convergence to zero of the momentum residual. On the other hand, the mass and energy equations are not similarly multiplied by density and energy, and thus do not need (the same form) of stabilization.

12.2 RealNSE in cG(1)cg(1) Form

RealNSE in cG(1)cG(1)-form for the Euler equations (11.1) is a time-stepping method defined by find $\hat{u} = (\rho, m, e) \in V_h$ such that for all $(\bar{\rho}, \bar{u}, \bar{e}) \in W_h$

$$\begin{aligned} ((R_\rho(\hat{u}), \bar{\rho})) + ((\delta u \cdot \nabla\rho, u \cdot \nabla\bar{\rho})) &= 0, \\ ((R_m(\hat{u}), \bar{u})) + ((\delta\rho u \cdot \nabla u, u \cdot \nabla\bar{u})) &= 0, \\ ((R_e(\hat{u}), \bar{e})) + ((\delta u \cdot \nabla e, u \cdot \nabla\bar{e})) &= ((d_h, \bar{e})), \end{aligned} \tag{12.2}$$

where

$$d_h = \sum_{i=1}^{3} \delta\rho(u \cdot \nabla u_i)^2, \tag{12.3}$$

and V_h is a trial space of continuous piecewise linear functions on a space-time mesh of size h satisfying the initial condition $\hat{u}(0) = \hat{u}^0$ with $u \in V_h$ defined by nodal interpolation of $\frac{m}{\rho}$, and W_h is a corresponding test space of function which are continuous piecewise linear in space and piecewise constant in time, all functions satisfying the boundary condition $u \cdot n = 0$ at the nodes on Γ. The space-time mesh is organised into space-time slabs between discrete time levels, and the velocity in the stabilisation terms is averaged over each time step. Further, $((\cdot, \cdot))$ denotes relevant $L_2(Q)$ scalar products. Finally, the least squares stabilisation weight $\delta = \frac{ch}{|u|}$ with $c \approx 0.5$, and *shock-capturing viscosity* with viscosity coffecient $\nu_i \sim h^2|R_i(\hat{u})|^2$, $i = \rho, m, e$, is also added

to each equation separately, which eliminates the small oscillations occuring at shocks with only least squares stabilisation. For small densities (almost vacuum), the velocity is computed by $u = \frac{m}{\rho+\epsilon}$ with $\epsilon > 0$ suitably small.

RealNSE (12.2) combines a weak satisfaction of the Euler equations with in particular a weighted least squares control of (a part of) the momentum residual $R_m(\hat{u})$ and represents a midway between the Scylla of weak solution and Carybdis of least squares strong solution.

Including all terms of the residual in the stabilisation as indicated in (12.1) improves consistency and thus may improve accuracy. Because test functions are piecewise constant in time the time derivative term will be missing in the residual stabilisation, however without much degrading accuracy in the presence of turbulence and shocks.

The form of the stabilisation terms in the density and energy equations secures exact global conservation since the stabilisation terms vanish with $\bar{\rho} = \bar{e} \equiv 1$. Notice that choosing $\bar{\rho} = \rho$ (mean-value over time step) gives a bound on $\int_Q \delta(u \cdot \nabla\rho)^2 dxdt$ in terms of $\int_Q \rho^2 \nabla \cdot u dxdt$, with a similar control of e.

Notice that RealNSE as expressed in (12.2) has a remarkable simplicity, as compared to finite difference methods based on diagonalization of the full system together with Riemann solvers, and also compared to stabilized finite element methods using diagonalization of the full linearized operator. In (12.2) each scalar convection equation is stabilized separately without diagonalization. In particular, the net numerical dissipation d_h has a contribution from each separate momentum/velocity component, but there is no net numerical dissipation in mass and internal energy.

12.3 The 2nd Law for RealNSE

Choosing $\bar{u}$ in the momentum equation as the time average of u over the time step, and subtracting the mass equation with $\bar{\rho} = I_h(\frac{|u|^2}{2})$, where $I_h(\frac{|u|^2}{2})$ is a nodal interpolant of $\frac{|u|^2}{2}$, we obtain

$$\dot{K} = W - D_h, \tag{12.4}$$

where

$$D_h = \int_Q d_h \, dxdt, \tag{12.5}$$

modulo the additive term

$$I = |((R_\rho, \frac{|u|^2}{2} - I_h(\frac{|u|^2}{2})))|$$

which by super-approximation as in [23] can be estimated as follows,

$$I \leq C\|h^2 R_\rho(\hat{u})|\nabla u|^2\|_0,$$

where $\|\cdot\|_0$ is the $L_2(Q)$-norm and $C \sim 1$ is an interpolation constant. Assuming $|R_\rho(\hat{u})| \sim h^{-1/2}$ and $|\nabla u| \sim h^{-1/2}$, we obtain $I \sim \sqrt{h}$, indicating that I is a small perturbation, which in computation can be checked a posteriori. Finally choosing in the equation for internal energy $\bar{e} = 1$, we obtain

$$\dot{E} = -W + D_h,$$

and we thus obtain the following 2nd Law for RealNSE (modulo a $\sqrt{h}$ correction)

$$\dot{K} = W - D_h, \quad \dot{E} = -W + D_h. \tag{12.6}$$

For solutions with turbulence/shocks, $D_h >> 0$ expressing a substantial irreversible transfer of kinetic energy into heat energy, just as above for regularized solutions. We note that in RealNSE only the momentum equation is subject to viscous regularization, since D_h expresses a penalty on the term $u\cdot\nabla u_i$ appearing in the momentum residual. A turbulent solution is identified by $D_h \sim 1$ under mesh refinement.

12.4 The Stabilization in RealNSE

We have seen that the stabilization in RealNSE is expressed by the dissipative term $D_h \sim 1$ which can be viewed as a weighted least squares control of the term $\rho u \cdot \nabla u_i$ in the momentum residual. The rationale is that least squares control of a part of a residual which is large, effectively may give control of the entire residual, and thus RealNSE gives a least squares control of the momentum residual. But the RealNSE stabilization does not correspond to an ad hoc viscosity, as in classical regularization, but to a form of penalty arising because Euler residuals of turbulent/shock solutions *cannot* be pointwise small.

In particular the dissipative mechanism of RealNSE does not correspond to a shear viscosity acting in all directions, as in standard ad hoc regularization, but rather to a form of bulk viscosity in the form of *streamline diffusion*

preventing fluid particles from colliding while allowing strong shear, connecting to the *streamline diffusion method* presented in [14].

Chapter 13

Output Wellposedness by Duality

> In love all the contradiction of existence merge themselves and are lost. Only in love are unity and duality not at variance. Love must be one and two at the same time. Only love is motion and rest in one. (Rabindranath Tagore, Literature Nobel Prize 1913)
>
> Those who have talked of "chance" are the inheritors of antique superstition and ignorance...whose minds have never been illuminated by a ray of scientific thought. (T. H. Huxley)

13.1 A Posteriori Error Estimation

Consider a mean-value output $M(\hat{u})$ in the form of a space-time integral

$$M(\hat{u}) = ((\hat{u}, \hat{\psi})) \tag{13.1}$$

defined by a smooth (positive) weight function $\hat{\psi}$ with $\|\psi\|_0 = 1$, where as above $\|\cdot\|_0$ is the $L_2(Q)$-norm. Let $\hat{u}$ and $\hat{w}$ be two RealNSE solutions on two meshes with joint maximal meshsize h. By the mean-value theorem their residual difference can be expressed as

$$R(\hat{u}) - R(\hat{w}) = R'(\hat{u}, \hat{w}) \cdot (\hat{u} - \hat{w}),$$

where $R'(\hat{u}, \hat{w})$ is a linearization of $R(\cdot)$. Letting $\hat{\varphi}$ be the solution of the dual linearized equation

$$R'^{\top}\hat{\varphi} = \hat{\psi}, \tag{13.2}$$

where $\top$ denotes transpose, we obtain the following basic *output error representation*:

$$M(\hat{u}) - M(\hat{w}) = ((\hat{u} - \hat{w}, R'^{\top}\hat{\varphi})) = ((R(\hat{u}) - R(\hat{w}), \hat{\varphi})). \tag{13.3}$$

We can thus estimate the output error in terms of the residuals as follows:

$$|M(\hat{u}) - M(\hat{w}| \leq S(\|R(\hat{u})\|_{-1} + (\|R(\hat{w})\|_{-1}) \tag{13.4}$$

or alteratively, using also *Galerkin orthogonality* as in [23]

$$|M(\hat{u}) - M(\hat{w}| \leq S(\|hR(\hat{u})\|_0 + (\|hR(\hat{w})\|_0), \tag{13.5}$$

where

$$S = S(\hat{u}, \hat{w}) = \|\hat{\varphi}\|_1 \tag{13.6}$$

with $\|\cdot\|_1$ the $H^1(Q)$-norm.

For a given output $M(\hat{u})$, we choose an error tolerance TOL and define a RealNSE-solution $\hat{u}$ to be *wellposed* up to the tolerance TOL, if

$$S\|hR(\hat{u})\|_0 \leq TOL/2, \tag{13.7}$$

where $S = S(\hat{u}, \hat{u})$. Given two wellposed RealNSE solutions $\hat{u}$ and $\hat{w}$, we may expect

$$|M(\hat{u}) - M(\hat{w}| \leq TOL, \tag{13.8}$$

up to a variation of the corresponding stability factors, which can be computed a posteriori. A solution which is not wellposed with respect to some tolerance of interest, is said to be *illposed*.

Turbulence is characterized by

$$D_h \sim \|\sqrt{h}R(\hat{u})\|_0 \sim 1 \tag{13.9}$$

under mesh refinement. We may thus expect an output $M(\hat{u})$ to be *wellposed* up to the tolerance TOL if

$$S(\hat{u}, \hat{u}) \lesssim \frac{TOL}{2\sqrt{h}}. \tag{13.10}$$

A wellposed output can be uniquely computed up to a tolerance TOL of interest, but an illposed cannot.

Computational results, some of which is presented below and more on the book web page [**?**], show that indeed $\|hR(\hat{u})\|_0 \sim \sqrt{h}$ conformiing (13.9), and that the corresponding stability factors $S(\hat{u}, \hat{u})$ for global mean-values such as drag, lift and total turbulent dissipation, can satisfy (13.10) for toleralnces TOL of interest. In short, global mean-value outputs can be wellposed, and thus represent stable computable emergent aspects of turbulent flow.

Chapter 14

Linearized Equations and Acoustics

> De Broglie, the creator of wave mechanics, accepted the results of quantum mechanics just as Schrödinger did, but not the statistical interpretation. (Born in the Born-Einstein Letters)

14.1 Linearization

Consider a solution $\hat{u} = (\rho, m, e)$ of the Euler equations:

$$\begin{aligned} \dot{\rho} + \nabla \cdot (\rho u) = \dot{\rho} + \nabla \cdot m &= 0 \quad \text{in } Q, \\ \dot{m} + \nabla \cdot (mu) + \gamma \nabla e &= f \quad \text{in } Q, \\ \dot{e} + \nabla \cdot (eu) + \gamma e \nabla \cdot u &= g \quad \text{in } Q, \\ u \cdot n &= 0 \quad \text{on } \Gamma \times I \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{14.1}$$

where $u = \frac{m}{\rho}$. The corresponding *linearized Euler equations*, linearized at $\hat{u}$, take the following form in $\underline{\hat{u}} = (\underline{\rho}, \underline{m}, \underline{e})$ representing a perturbation of $\hat{u}$ with corresponding perturbation of data, with $\underline{u}$ defined by $\underline{m} = \underline{\rho} u + \rho \underline{u}$:

$$\begin{aligned} \underline{\dot{\rho}} + \nabla \cdot (\underline{\rho} u) + \nabla \cdot (\rho \underline{u}) = \underline{\dot{\rho}} + \nabla \cdot \underline{m} &= 0 \quad \text{in } Q, \\ \underline{\dot{m}} + \nabla \cdot (\underline{m} u) + \nabla \cdot (m \underline{u}) + \gamma \nabla \underline{e} &= \underline{f} \quad \text{in } Q, \\ \underline{\dot{e}} + \nabla \cdot (\underline{e} u) + \nabla \cdot (e \underline{u}) + \gamma \underline{e} \nabla \cdot u + \gamma e \nabla \cdot \underline{u} &= \underline{g} \quad \text{in } Q, \\ \underline{u} \cdot n &= 0 \quad \text{on } \Gamma \times I, \\ \underline{\hat{u}}(\cdot, 0) &= \underline{\hat{u}}^0 \quad \text{in } \Omega, \end{aligned} \tag{14.2}$$

or alternatively with $Dw = \dot{w} + u \cdot \nabla w$ the u-convective derivative:

$$\begin{aligned} D\underline{\rho} + \underline{\rho}\nabla \cdot u + \nabla \cdot (\rho \underline{u}) &= 0 \quad \text{in } Q, \\ D\underline{m} + \underline{m}\nabla \cdot u + \nabla \cdot (m\underline{u}) + \gamma \nabla \underline{e} &= \underline{f} \quad \text{in } Q, \\ D\underline{e} + \underline{u} \cdot \nabla e + (\gamma + 1)\underline{e}\nabla \cdot u + (\gamma + 1)e\nabla \cdot \underline{u} &= \underline{g} \quad \text{in } Q, \\ \underline{u} \cdot n &= 0 \quad \text{on } \Gamma \times I \\ \underline{\hat{u}}(\cdot, 0) &= \underline{\hat{u}}^0 \quad \text{in } \Omega. \end{aligned} \tag{14.3}$$

This is a linear convection-reaction system with coefficients depending on the base flow $\hat{u}$ and the stability properties of this system govern the perturbation growth.

14.2 Wave Equation

Let $\hat{u}$ be a constant ground state with velocity $u = 0$ and temperature $\tau = 1$. If τ is not subject to perturbation, so that $\underline{e} = \underline{\rho}$, then (14.2) reduces to (leaving out initial and boundary conditions):

$$\begin{aligned} \dot{\underline{\rho}} + \nabla \cdot \underline{m} &= 0 \quad \text{in } Q, \\ \dot{\underline{m}} + \gamma \nabla \underline{\rho} &= \underline{f} \quad \text{in } Q, \end{aligned} \tag{14.4}$$

which leads to the *wave equation* for the density perturbation, assuming $\nabla \cdot \underline{f} = 0$:

$$\ddot{\underline{\rho}} - \gamma \Delta \underline{\rho} = 0 \quad \text{in } Q. \tag{14.5}$$

Small density-pressure variations at constant temperature around a constant ground state thus obeys the linear wave equation with wave speed $c = \sqrt{\gamma}$.

Allowing a non-zero velocity of the ground state, but still constant unit temperaure, the linearized equations reduce to (assuming $\underline{e} = 0$):

$$\begin{aligned} \dot{\underline{\rho}} + \nabla \cdot \underline{m} &= 0 \quad \text{in } Q, \\ \dot{\underline{m}} + 2\nabla \cdot (\underline{m}u) - \nabla \cdot (\underline{\rho} uu) + \gamma \nabla \underline{\rho} &= \underline{f} \quad \text{in } Q, \end{aligned} \tag{14.6}$$

which is a generalized wave equation with contributions from convection with velocity u.

14.3 Acoustics

Acoustics concerns propagation of (usually small) density-pressure variations in a fluid. Acoustics simulation can be based on (i) directly solving the Euler

equations for the full state (ground state plus perturbation), or (ii) solving the Euler equations for the ground state followed by solution of the linearized Euler equations for the perturbation with the ground state given. We just saw that the simplest model for acoustics is the wave equation (14.5) as a basic case of (b). In general, (ii) may be expected to give higher accuracy at the expense of solving both the Euler and linearized Euler equations.

The wave equation can be used for progagation of sound waves in in a fluid at rest, but for the generation of sound in e.g. human speech or musical instruments, fluid-structure interaction is used, which requires the Euler equations for the (usually turbulent) flow and an elasticity model for the structure (e.g, vocal chords).

14.4 Linearized Stability

The nature of the linearized system (14.2) depends on the ground state and ranges from its simplest incarnation in the form of the wave equation (14.5) for a constant ground state, to a highly complex problem for a turbulent ground state. Below we analyze the linearized problem for a shocks and rarefaction waves in a model case, and in [23] we discuss the case of turbulent incompressible flow. For now, we only observe that the equation for $\underline{\rho}$ contains the term $\underline{\rho}\nabla\cdot u$, and the same for $\underline{m}$ and $\underline{e}$, which indicates exponential perturbation growth along particle paths under compression with $\nabla\cdot u<0$, which occurs when a compression shock is forming.

Chapter 15

Dual Linearized Problem

> Those who do not understand the nature of sin and virtue are attached to duality; they wander around deluded. (Sri Guru Granth Sahib)

Formally multiplying the linearized equation (14.2) by the dual variable $\hat{\varphi} = (\rho_d, \varphi, e_d)$ and integrating by parts and varying $\underline{\hat{u}}$, we obtain the following dual problem

$$\begin{array}{rcl} -\dot{\rho}_d + \sum_j (u \cdot \nabla \varphi_j)\, u_j + \frac{e}{\rho} u \cdot \nabla e_d + \frac{\gamma}{\rho} u \cdot \nabla (e e_d) & = & \psi_\rho, \\ -\dot{\varphi} - u \cdot \nabla \varphi - \sum_i u_i \nabla \varphi_i - \nabla \rho_d - \frac{e}{\rho} \nabla e_d - \frac{\gamma}{\rho} \nabla (e e_d) & = & \psi, \\ -\dot{e}_d - u \cdot \nabla e_d + \gamma \nabla \cdot u e_d - \gamma \nabla \cdot \varphi & = & \psi_e, \end{array} \tag{15.1}$$

where the data $\hat{\psi} = (\psi_\rho, \psi, \psi_e)$ defines the output. We check by writing the Euler equations alternatively in the variables $\hat{u} = (\rho, u, e)$:

$$\begin{array}{rcl} \dot{\rho} + \nabla \cdot (\rho u) & = & 0, \\ \dot{\rho} u + \rho \dot{u} + \nabla \cdot (\rho u u) + \gamma \nabla e & = & f, \\ \dot{e} + \nabla \cdot (e u) + \gamma e \nabla \cdot u & = & g, \end{array} \tag{15.2}$$

with corresponding dual linearized problem in $\hat{\varphi} = (\rho_d, \varphi, e_d)$:

$$\begin{array}{rcl} -\dot{\rho}_d - u \cdot \nabla \rho_d - u \cdot \dot{\varphi} - \sum_i u_i u \cdot \nabla \varphi_i & = & \psi_\rho, \\ -\rho \dot{\varphi} - \rho u \cdot \nabla \varphi - \rho \nabla \rho_d - e \nabla e_d - \gamma \nabla (e e_d) - \rho \sum_i u_i \nabla \varphi_i & = & \psi, \\ -\dot{e}_d - u \cdot \nabla e_d + \gamma \nabla \cdot u e_d - \gamma \nabla \cdot \varphi & = & \psi_e, \end{array}$$

which by eliminating $\dot{\varphi}$ from the first equation using the second gives

$$\begin{array}{rcl} -\dot{\rho}_d + \sum_j (u \cdot \nabla \varphi_j)\, u_j + \frac{e}{\rho} u \cdot \nabla e_d + \frac{\gamma}{\rho} u \cdot \nabla (e e_d) & = & \psi_\rho, \\ -\dot{\varphi} - u \cdot \nabla \varphi - \sum_i u_i \nabla \varphi_i - \nabla \rho_d - \frac{e}{\rho} \nabla e_d - \frac{\gamma}{\rho} \nabla (e e_d) & = & \psi, \\ -\dot{e}_d - u \cdot \nabla e_d + \gamma \nabla \cdot u e_d - \gamma \nabla \cdot \varphi & = & \psi_e, \end{array}$$

which is the same as (15.1) with a properly modified right hand side: the elimination divides the second equation by ρ (so that $\psi \to \psi/\rho$), and substituting $\dot{\varphi}$ into the first equation absorbs the data contribution $\psi_\rho \to \psi_\rho - \frac{1}{\rho} u \cdot \psi$, while the three quadratic terms in u combine to the single term $\sum_j (u \cdot \nabla \varphi_j)\, u_j$.

The dual problem (15.1) is solved backward in time on $[0, T]$ with e.g. $\hat{\varphi}(\cdot, T) = 0$, for simplicity with stabilization by artificial diffusion of size h in each equation instead of least squares stabilization. The crucial stability factor is defined by $S = \|\varphi\|_{H^1(Q)}/\|\psi\|_{L_2(Q)}$, which bounds the error in the output $(\hat{u}, \psi)_Q$ by $S\|hR(\hat{u})\|_{L_2(Q)}$.

Chapter 16

Burgers Equation

> Some physicists. among them myself, cannot believe that we must abandon, actually and forever, the idea of direct representation of physical reality in space and time; or that we must accept then the view that events in nature are analogous to a game of chance. (Einstein 1954)

16.1 A Model of the Euler Equations

As an instructive model of the Euler equations exhibiting basic aspects of *shock waves* and *rarefaction waves*, we consider the following *scalar conservation law*, referred to as *Burgers equation*: Find the scalar function $u = u(x,t)$ such that

$$\begin{aligned} \dot{u} + (f(u))' &= 0 \quad \text{in } Q, \\ u(\cdot,0) &= u^0, \end{aligned} \tag{16.1}$$

where $f(u) = u^2/2$, $Q \equiv \mathbb{R} \times I$ with $I = (0,T]$, and we assume that the initial data $u^0(x)$ vanishes for large $|x|$.

Burgers equation takes the pointwise form $\dot{u} + uu' = 0$ for a differentiable pointwise solution u, which expresses that $u(x,t)$ is constant with values $u^0(\bar{x})$ along straight lines $x = st + \bar{x}$ with slope $s = u^0(\bar{x})$), that is, along *characteristics* defined by $\frac{dx}{dt} = u(x,t)$ with $x(0) = \bar{x}$.

If $u^0(x)$ is increasing with increasing x and is smooth, then there is a pointwise solution $u(t,x)$ for all time given by the simple recipe

$$u(x,t) = u^0(\bar{x}) \quad \text{for } x = u^0(\bar{x})t + \bar{x}. \tag{16.2}$$

However, if the initial data $u^0(x)$ somewhere is strictly decreasing, which of course will always be the case if $u^0(x)$ vanishes for large $|x|$, then characteristics will cross in finite time with different values and the solution formula gives conflicting function values, which can be interpreted as *breakdown* of the pointwise solution. The result is that there is no differentiable function $u(x,t)$ satisfying Burger's equation $\dot{u} + uu' = 0$ pointwise. We shall see that this corresponds to the occurence of *shocks*, which are represented by *discontinuous* functions, which satisfy the differential equation in a weak sense, and in a pointwise sense only away from discontinuities or jumps.

What to do with an equation without pointwise solution? Of course, we regularize and consider instead the *viscous/regularized Burgers equation*:

$$\begin{aligned} \dot{u} + uu' - \nu u'' &= 0 \quad \text{in } Q, \\ u(\cdot, 0) &= u^0, \end{aligned} \tag{16.3}$$

with $\nu > 0$ a small viscosity. This problem can be solved uniquely in a pointwise sense for all initial data, and as above the solution can be viewed as an approximate weak solution to the original inviscid Burgers equation satisfying a 2nd Law of the form

$$\dot{K}(u;t) = -D_\nu(u;t) \quad \text{for } t \in I, \tag{16.4}$$

where

$$K(u;t) = \int_{\mathbb{R}} \frac{u(x,t)^2}{2}\,dx, \quad D_\nu(u;t) = \int_{\mathbb{R}} \nu(u'(x,t))^2\,dx. \tag{16.5}$$

The 2nd Law follows by multiplication of the viscous Burgers' equation by u and integration (by parts) with respect to x.

To give a perspective we now briefly recall the basics of the mathematical theory for conservation laws developed in the 1950s motivated by the Euler equations of compressible flow, in the model setting of Burgers equation. The basic idea is to introduce the concept of *weak solution* allowing discontinuous solutions, and complement by a 2nd Law. We shall see that the 2nd Law singles out *physical shocks*, which are discontinuous weak solutions satisfying the 2nd Law, from *non-physical shocks* which are discontinuous weak solutions violating the 2nd Law.

We also present an alternative approach to single out physical shocks based on a stability analysis showing that a a physical shock is stable and represents a an observable physical phenomenon (like the "bang" from a

supersonic airplane), while a non-physical shock is unstable and thus does not represent any observable physical phenomenon.

Burgers equation is formally reversible: If $u(x,t)$ is a pointwise solution in forward time on $[0,T]$, so is $-u(x,T-t)$ in reverse time. Burgers equation thus is formally reversible system, but in reality is irreversible, as a consequence of the non-existence of stable pointwise solutions. We thus use Burgers equation to illustrate that a World governed by the formally reversible Euler equations, is an irreversible World, that is a world with an Arrow of time.

16.2 Weak Solutions

A possibly discontinuous function $u(x,t)$ is said to be a *weak solution to Burgers' equation* if

$$\int_{\mathbb{R}\times\mathbb{R}_+} (-u\dot{\varphi} - f(u)\varphi')dx\,dt - \int_{\mathbb{R}} u^0(x)\varphi(x,0)\,dx = 0 \tag{16.6}$$

for all differentiable test functions φ such that $\varphi(x,t)$ vanishes for large (x,t), assuming here $I = (0,\infty)$. This equation is obtained from (16.1) by multiplication by φ and integration by parts, shifting the derivatives onto φ, thus allowing u to be discontinuous.

16.3 The Rankine-Huginiot Condition

A discontinuous function $u(x,t)$ defined by $u(x,t) = u_+$ if $x > st$ and $u(x,t) = u_-$ if $x < st$, where u_+ and u_- are two constant states and s is a constant, corresponding to a discontinuity propagating with speed s, is a weak solution to Burgers' equation if the shock speed satisfies the *Rankine-Hugoniot condition*

$$s = \frac{[f(u)]}{[u]}, \tag{16.7}$$

where $[u] = u_+ - u_-$ and $[f(u)] = f(u_+) - f(u_-)$. With $f(u) = u^2/2$ as in Burgers' equation, we have

$$s = (u_+ + u_-)/2. \tag{16.8}$$

The Rankine-Hugoniot condition expresses the conservation law (Burgers' equation) in weak form for a piecewise constant discontinuous solution candidate u.

16.4 Rarefaction wave

The solution to Burgers equation with the *increasing* discontinuous initial data $u^0(x) = 0$ for $x < 0$, and $u^0(x) = 1$ for $x > 0$, is a *rarefaction wave* given by

$$\begin{array}{rcll} u(x,t) & = & 0 & \text{for} \quad x < 0, \\ u(x,t) & = & \frac{x}{t} & \text{for} \quad 0 \le \frac{x}{t} \le 1, \\ u(x,t) & = & 1 & \text{for} \quad 1 < \frac{x}{t}. \end{array} \tag{16.9}$$

This is a continuous function for $t > 0$ which satisfies (16.1) pointwise for $t > 0$ off the lines $x = 0$ and $x = t$ and can be viewed as a pointwise solution (since it is continuous and piecewise differentiable). In a rarefaction wave, an initial discontinuity separating two constant states develops into a continuous linear transition from one state to the other of width t in space, corresponding to "fan-like" level curves in space-time as shown in Fig. 16.1.

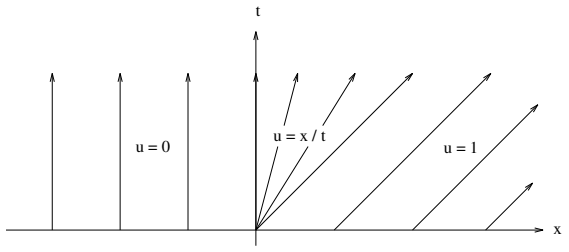


Figure 16.1: Characteristics of a rarefaction wave.

16.5 Shock

The solution with decreasing discontinuous initial data $u^0(x) = 1$ for $x < 0$, and $u^0(x) = 0$ for $x > 0$, is a discontinuous *shock wave* moving with speed $\frac{1}{2}$:

$$\begin{array}{rcl} u(x,t) & = & 1 \quad \text{for} \quad x < \frac{t}{2}, \\ u(x,t) & = & 0 \quad \text{for} \quad x > \frac{t}{2}, \end{array} \tag{16.10}$$

as shown in Fig 16.2. We shall motivate that this is a *physical shock* satifying a 2nd Law.

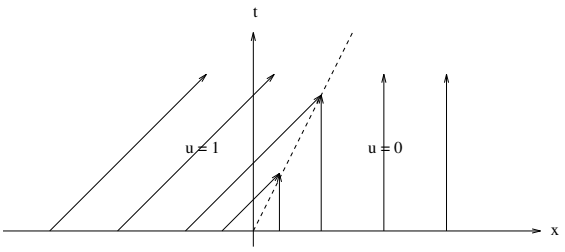


Figure 16.2: Characteristics of a shock

16.6 Weak solutions may be non-unique

The rarefaction wave initial data $u^0(x) = 0$ for $x < 0$ and $u^0(x) = 1$ for $x > 0$, also admits the alternative discontinuous weak solution

$$\begin{array}{rcl} u(x,t) & = & 0 \quad \text{for} \quad x < \frac{t}{2}, \\ u(x,t) & = & 1 \quad \text{for} \quad x > \frac{t}{2}, \end{array} \tag{16.11}$$

corresponding to a discontinuity $\{x,t) : x = st\}$ moving with speed $s = \frac{1}{2}$. This solution is obviously different from the rarefaction wave solution (16.9), which since it is a pointwise solution, also is a weak solution. Thus, we have in this case two different weak solutions, and thus we have an example of *non-uniqueness of weak solutions.* We shall see that the discontinuos solution (16.11) violates the 2nd Law and thus is a *non-physical shock solution.* The physical solution in this case is the rarefaction wave (which satisfies the 2nd Law with equality).

16.7 The 2nd Law for Burgers Equation

We now check if (16.11) satisfies the appropriate form of the 2nd Law (16.4) restricting space to $-1 \leq x \leq 1$ and time to $0 \leq t \leq 1$ assuming the boundary conditions $u(-1,t) = 0$ and $u(1,t) = 1$. In this setting, we obtain by multiplication of the viscous Burgers equation with u and integrating by parts and using the sign of the viscous term, the following form of the 2nd Law

$$-D_\nu(u;t) = \dot{K}(u;t) + \int_{-1}^{1} (\frac{u^3}{3})' = \dot{K}(u;t) + \frac{1}{3} \tag{16.12}$$

But for (16.11) we have $\dot{K} = -\frac{1}{2}\frac{d}{dt}\frac{t}{2} = -\frac{1}{4}$, which violates (16.12).

Changing to the boundary conditions to $u(-1,t) = 1$ and $u(1,t) = 0$, the 2nd Law changes to

$$-D_\nu(u;t) = \dot{K}(u;t) + \int_{-1}^{1} (\frac{u^3}{3})' = \dot{K}(u,t) - \frac{1}{3}, \tag{16.13}$$

which is satisfied by (16.10) since now $\dot{K} = \frac{1}{2}\frac{d}{dt}\frac{t}{2} = \frac{1}{4} < \frac{1}{3}$.

We conclude that a discontinuous function $u(x,t)$ with the constant states u_- for $x < st$ and u_+ for $x > st$ corresponds to a physical shock solution with shock speed $(u_- + u_+)/2$ if $u_- > u_+$, and to a non-physical shock if $u_- < u_+$.

We furher see that a shock dissipates substantial kinetic energy into heat since for a viscous profile of width ν joining two states u_- and u_+

$$D_\nu(u;t) \equiv \int_{\mathbb{R}} \nu(u')^2\,dx \approx (u_- - u_+)^2 \tag{16.14}$$

for small ν.

16.8 Burning Books

The 2nd Law states that the characteristics of a physical shock "converge into" the shock, corresponding to $u_- > u_+$. For the unphysical shock solutin corresponding to the rarefaction initial data with $u_- < u_+$, the characteristics appear to "emerge from" the discontinuity. The 2nd Law thus allows information (features of the solution) to be "destroyed" (in a shock with converging characteristics), but not "created out of nothing" (in an unphysical

rarefaction with diverging characteristics). This connects to the essence of irreversibility: Burning books can readily be done without any sophistication, while recovering the original text from the ashes is impossible.

16.9 Stability of a Rarefaction Wave

We shall now see that the 2nd Law can be replaced by a stability analysis in its main mission to distingush physical shocks from unphysical shocks. This reflects the fact that a stable process can be physically realized and observed, while an unstable process will have no permanence and thus will be very difficult (or impossible) to realize.

The stability of a solution $u(x,t)$ is governed by the linearized equation

$$\dot{w} + (uw)' = 0 \quad \text{in } \mathbb{R} \times \mathbb{R}_+ \tag{16.15}$$

where w represents a (small) perturbation (tending to zero for $|x|$ tending to infinity), and the solution u acts as a given coefficient. The growth properties of the solution w of the linearized equation determines the stability: If w grows quickly in time, then the solution u is unstable, and if w stays bounded, then u is stable. Of course, stability connects to finite precision: the level of the perturbation represents the precision.

We first consider the case of a rarefaction wave given by (16.9). Multiplying by w and integrating in space, we obtain by a simple computation using the fact that $u'(x,t) = 1/t$ for $0 \le x \le t$ and $u'(x,t) = 0$ else,

$$\frac{d}{dt}\int_{\mathbb{R}} w^2(x,t)\,dx + \int_0^t w^2(x,t)\frac{1}{t}\,dx = 0, \quad \text{for } t > 0,$$

from which follows that

$$\int_{\mathbb{R}} w^2(x,t)\,dx \le \int_{\mathbb{R}} w^2(x,0)\,dx \quad \text{for } t > 0. \tag{16.16}$$

This inequality shows that the L_2-norm in space of a perturbation from initial data does not grow with time, which proves stability of a rarefaction wave. Note that this argument builds on the fact that the rarefaction wave $u(x,t)$ is increasing in x so that u' is non-negative.

16.10 Stability of Shock

The stability proof used above to prove stability of a rarefaction wave, does not work the same way for a shock, since in this case $u(x,t)$ is decreasing with x. In fact a shock does not satisfy an L_2 stability estimate of the form (16.16). However, one can prove instead an L_1-bound of the form

$$\int_{\mathbb{R}} |w(x,t)|\,dx \leq \int_{\mathbb{R}} |w(x,0)|\,dx \quad \text{for } t > 0. \tag{16.17}$$

This follows by multiplying (16.15) by $sgn(w) = +1$ if $w > 0$ and -1 if $w < 0$, to get by integration by parts:

$$\frac{d}{dt}\int_{\mathbb{R}} |w(x,t)|\,dx + (u_- - u_+)|w(\frac{t}{2},t)| = 0, \tag{16.18}$$

and using the fact that for a shock $u_+ < u_-$. Moreover, we will below with a different type of stability estimate show that a shock is stable from computational RealNSE point of view, Thus, a shock is a stable phenomenon from both physical and computational point of view.

16.11 Instability of Non-Physical Shock

We saw above that the rarefaction wave solution is stable, and we now study the stability of the alternative weak solution (16.11). By the same argument as used to prove (16.18) we obtain

$$\frac{d}{dt}\int_{\mathbb{R}} |w(x,t)|\,dx = (u_+ - u_-)|w(\frac{t}{2},t)|, \tag{16.19}$$

where now $u_+ > u_-$. In this case, $\int_{\mathbb{R}} |w(x,t)|\,dx$ can grow arbitrarily fast, since the positive right hand side in (16.19) in no way can be controled by the left hand side. We thus conclude that an unphysical shock is unstable.

16.12 2nd Law = Stability + Finite Precision

We thus have two methods to single out physical shocks, one based on stability, and the other based on the 2nd Law. This shows that ultimately the 2nd Law expresses a stability condition, reflecting that Nature only can realize

phenomena which are properly stable, in its analog finite precision computation. We thus may view the 2nd Law to express a combination of stability and finite precision.

16.13 A Traffic Model

The equation (16.1) with $f(u) = u(1-u)$ models the flow of cars along a highway represented by the x-axis with $0 \leq u \leq 1$ the car density and $0 \leq v = (1-u) \leq 1$ the car velocity as a function of density: Sparse cars ($u \approx 0$) move fast ($v \approx 1$) and packed cars ($u = 1$) stand still ($v = 0$). The equation (16.1) then expresses conservation of mass in the sense that there are no side roads through which cars can enter or exit (and they cannot simply disappear into the sky).

We understand that if $u(x,t)$ is increasing in x so that the car density increases in the direction of motion, then the car velocity will be decreasing in the direction of motion which means that faster cars behind will approach slower cars ahead. But the x-axis is a one-lane street and does not allow a faster car to overtake a slower car, and so eventually faster cars will have to break in order to not run into slower cars ahead, that is, the law $v = (1-u)$ will have to be violated and thus the conservation law (16.1) cannot be satisfied pointwise. In the inviscid equation (16.1) this will correspond to the appearence of a shock. The corresponding regularized equation takes the form

$$\dot{u} + ((u(1-u))' - \nu u'' = 0,$$

which we formally can write as

$$\dot{u} + (uv)' = 0$$

with the modified velocity law

$$v = (1 - u - \frac{\nu}{u}u'),$$

with the effect that in the dangerous case of increasing density ($u' > 0$), faster cars will be slowed down more than slower cars and collisions from behind will thus be avoided.

In this model the shock corresponds to the well-known situation when an inexperienced driver suddenly steps on the brake to avoid collision into a

cue, which in the regularized problem corresponds to the smoother action of an experienced driver slowing down because the car density is larger ahead than behind.

16.14 Non-Existence Exact Euler Solutions

We have seen that the inviscid Burgers equation has discontinuous shock solutions, which can be viewed as weak solutions satisfying a 2nd Law. One can thus speak about a shock solution as an exact weak solution of Burgers equation satisfying a 2nd Law, and this solution is the limit of regularized solutions as the viscosity tends to zero. The goal of the classical mathematical analysis of conservation laws based on regularization, is to establish a corresponding result for the Euler equations: The goal is thus to prove the existence of a function satisfying the Euler equations in a weak sense together with a 2nd Law, as a limit of regularized solutions.

However, this goal has not been reached, and we have good reasons to believe that it cannot be reached. This is because of the appearence of turbulent solutions to the regularized Euler equations which do not seem have a limit in the form of a function. It is thus not meaningful to speak about exact solutions to the Euler equations, because such objects do not exist, but we may speak about regularized solutions because they do exist and we may view such solutions as approximate solutions to the Euler equations.

With this perspective it is reasonable to view a shock solution not as is usual as an exact weak solution to the Euler equations, but rather to view a viscous shock solution, an exact solution to the regularized Euler equations, as an approximate solution to the Euler equations. Thus only strong pointwise solutions could deserve the title of "exact solution"; a weak solution which is not a strong solution like a shock, could only be an approximate solution. This may seem to differ from the common way of looking at weak solutions of differential equations, as functions satisfying the differential equation "weakly exactly", but this is not completely natural since the notion of "weakly" has an element of "inexactly" built in. Only a strong pointwise solution can be an exact solution. I somebody offers you to buy an unknown painting by van Gogh, which you are only allowed to inspect from a distance of 10 meters, you may get suspicious and refrain from a deal.

Chapter 17

RealBurgers

> But maybe that is our mistake: maybe there are no particle positions and velocities, but only waves. It is just that we try to fit the waves to our preconceived ideas of positions and velocities. The resulting mismatch is the cause of the apparent unpredictability. (Stephen Hawking 1988)

17.1 Introduction

We will now consider RealNSE applied to Burgers' equation as a simple version of RealNSE. We recall that RealNSE is Galerkin's finite element method combined with a weighted least-squares control of the residual. RealNSE is based on piecewise polynomial approximation in space-time and offers spectrum of computational methods depending on the choice of the space-time mesh, as presented in detail in [14, 23]. RealNSE uses piecewise polynomials which are continuous in space and possibly discontinuous in time. These variants are referred to as cG(p)cG(q) or cG(p)dG(q) with cG(p) referring to continuous piecewise approximation of degree p in space, and cG(q)/dG(q) referring to continuous/discontinuous approxiamtion in time of degree q. RealNSE is *Eulerian* if the space-time mesh oriented along the space and time coordinate axis, is *Lagrangean* if the space-time mesh is oriented along particle paths in space-time, and *Arbitrary Lagrangean-Eulerian* or *ALE* if the space-time mesh is oriented according to some other feature such as space-time gradients of the solution. Lagrangean variants are also referred to as *characteristic Galerkin*, and Eulerian variants as *streamline diffusion methods*.

In all these variants the space-time mesh is usually organized in space-time slabs between discrete time levels, and RealNSE then gives a time-stepping method allowing the solution to be computed form one time level to the next progessing in time. The space mesh may be changed across the discrete time levels to avoid mesh distortion and allow mesh adaption. In dG(q) the approximation is discontinuous in time and the space mesh may vary from one slab to the next. If the space mesh is changed across a discrete time level in cG(q), then a projection from the previous mesh to the new mesh is performed. The projection is built into the Galerkin method through a jump term corresponding to a L_2 projection. The discrete solution between the discrete time levels may be viewed as an approximate *transport* step, and the whole process may be viewed as a method of the basic form *projection-transport*.

The traditional finite difference methods are of Eulerian type with the first order Lax-Friedrichs' scheme from the 50s as a prototype on conservation form and with artificial viscosity proportional to the mesh size. The next generation of classical schemes originates from Godunov's method in 1d, which is of the form projection-transport with piecewise constant (discontinuous) approximation and a Riemann solver for the transport step. The multi-dimensional finite volume schemes developed in recent decades, use discontinuous polynomial approximation with numerical fluxes often constructed using 1d Riemann solvers. All these methods may alternatively be viewed as particular RealNSE methods.

17.2 Semi-Discrete RealNSE for Burgers Equation

For the purpose of analysis displaying essential aspects, we consider a *semi-discrete* RealNSE method for Burgers equation with cG(1) discretization in space. Full discretization with cG(q) or dG(q) in time can be analyzed similarly. Let then for a given mesh on $\mathbb{R}$ with mesh size h, V_h be the set of continuous functions $v(x,t)$ which for $t \in I = [0,T]$ are piecewise linear in x on the mesh. The simplest version of semidiscrete RealNSE takes the form: Find $U \in V_h$ such that

$$((R(U),v)) + ((hU',v')) = 0 \quad \forall v \in V_h, \tag{17.1}$$

where $R(U) = \dot{U} + UU'$ is the Burgers residual, $((v, w)) = \int_Q vw\, dxdt$ with $Q = \mathbb{R} \times I$ and $U(x, 0) \in V_h$ interpolates $u^0(x)$. This is equivalent to a system of ordinary differential equations in time. We have here replaced residual stabilization by (hU', v'), and we thus consider a Galerkin method in space for a viscous Burgers equation with viscosity $\nu = h$.

17.3 Basic Energy Estimate as 2nd Law

Choosing $v = U$ in (17.1), we obtain by integration by parts over $\mathbb{R}$ the following direct analog of (16.4):

$$\dot{K}(U; t) = -D_h(U; t), \quad \text{for } t \in I.$$

In the presence of shocks, $D_h(U; t)$ will be substantial and thus U' will at shocks be large ($\sim h^{-1}$).

17.4 A Posteriori Error Estimation by Duality

We shall now prove we the following a posteriori error estimate, where u is the solution of the viscous Burgers equation with $\nu = h$:

$$\|u - U\|_Q \leq S\|hR(U)\|_Q, \tag{17.2}$$

where $\|\cdot\|_Q$ is the $L_2(Q)$-norm and

$$S = \frac{\|h\varphi''\|_Q}{\|e\|_Q}, \tag{17.3}$$

is a stability factor defined by the solution φ of the following linearized dual problem:

$$\begin{aligned} -\dot{\varphi} - a\varphi' - h\varphi'' &= e, && \text{in } Q, \\ \varphi(x, t) &\to 0, && x \to \pm\infty, t \in I, \\ \varphi(x, T) &= 0, && x \in \mathbb{R}, \end{aligned} \tag{17.4}$$

where $a = (u + U)/2$. We notice that the dual problem has a viscous term with viscsoity coefficient h.

To prove the a posteriori error estimate (17.2) we multiply (17.4) by $u-U$ and integrate in space and time to get the error representation

$$\|u-U\|_Q^2 = ((R(U),\varphi)) + ((hU',\varphi')).$$

We then use the Galerkin orthogonality (17.1) with $v=\Phi\in V_h$ an interpolant of the dual solution φ, to get

$$\|u-U\|_Q^2 = ((R(U),\varphi-\Phi)) + ((hU',\varphi'-\Phi')),$$

which combined with an interpolation error bound of the form

$$\|\varphi-\Phi\|_Q + \|h(\varphi'-\Phi')\|_Q \le C_i\|h^2\varphi''\|_Q,$$

where C_i is an interpolation constant, shows that

$$\|u-U\|_Q^2 \le C_i\|hR(U)\|_Q\|h\varphi''\|_Q,$$

where $R(U)$ has been augmented by the contribution $\|hU'\|_Q$, which proves (17.2), setting $C_i=1$ for simplicity.

By the basic energy stability estimate, we have

$$\|hU'\|_Q \le \sqrt{h} \quad \text{if } \|u^0\|_{\mathbb{R}}=1,$$

suggesting the same estimate for $\|hR(U)\|_Q$, which of course is checked a posteriori, and thus we expect

$$\|u-U\|_Q \le S\sqrt{h}. \tag{17.5}$$

We shall now prove that for a shock $S\sim 1$, which shows that a shock is computable with RealNSE, with a $L_2(Q)$ error of size $\sqrt{h}$, which is optimal from approximation point of view (if we consider the exact solution u to be discontinuous) and U is continuous in x on the mesh.

17.5 Stability Estimate for a Shock

We shall now investigate the stability properties of the dual problem (17.4) and seek to bound $h\varphi''$ in terms of the right hand side e. For simplicity we

linearize at the exact solution $u(x,t)$ (instead of the mean value $(u+U)/2$) and thus consider the dual problem

$$\begin{aligned} -\dot{\varphi} - u\varphi' - h\varphi'' &= e, \quad \text{in } Q, \\ \varphi(\cdot,T) &= 0. \end{aligned} \tag{17.6}$$

The stability properties are largely determined by the sign of u', which reflects the change of the direction u of the characteristics. If $u' \leq 0$, then the characteristics converge with increasing t, which typically occurs in the case of a shock. If $u' \geq 0$, then the characteristics diverge, which typically occurs in the case of a rarefaction. If u' is bounded below by a moderate constant, e.g. $u' \geq 0$, then we may estimate $\|\varphi\|_{L_\infty(L_2(\mathbb{R}))}$ and $\|\sqrt{h}\varphi'\|_Q$ in terms of a moderate constant times $\|e\|_Q$, which we refer to as *weak stability*. If u' is bounded above by a moderate constant, e.g. $u' \leq 0$, then we may estimate $\|h\varphi''\|_Q$ in terms of a moderate constant times $\|e\|_Q$, which we refer to as *strong stability*, because we estimate second derivatives of φ, cf. (17.3). These estimates are proved by multiplying by φ and $-h\varphi''$, respectively, bringing in the positive stabilizing terms $\frac{1}{2}u'\varphi^2$ and $-\frac{1}{2}hu'(\varphi')^2$, respectively.

We now give the details in the case of a shock with $u' \leq 0$, where we assume u is differentiable with a very large negative x-derivative close to the shock. We indicate the general nature of the characteristics of the dual problem in Fig 17.5. We shall prove that the solution φ of (17.6) satisfies

$$\|h\varphi''\|_Q + \|\dot{\varphi} + u\varphi'\|_Q + \sup_{0<t<T} \|h^{1/2}\varphi'(\cdot,t)\|_R \leq 3\|e\|_Q. \tag{17.7}$$

To see this we multiply the first equation in (17.6) by $-h\varphi''$, integrating by parts with respect to x, and integrating in time over (τ,T) with $0<\tau<T$, we get with $Q_\tau = \mathbb{R}\times(\tau,T)$

$$\begin{aligned} &\tfrac{1}{2}\textstyle\int_{\mathbb{R}} h\left(\varphi'(\cdot,\tau)\right)^2 dx + \int_{Q_\tau}\left(h\varphi''\right)^2 \, dxdt + \int_{Q_\tau}\tfrac{1}{2}\left(uh\left(\varphi'\right)^2\right)' \, dxdt \\ &\leq \tfrac{1}{2}\textstyle\int_{Q_\tau}\left(hu'\left(\varphi'\right)^2 + e^2 + \left(h\varphi''\right)^2\right) \, \mathrm{d}x\mathrm{d}t, \end{aligned} \tag{17.8}$$

which proves the desired result stating that $S \sim 1$ for a shock.

As comparison, let us now attempt to derive a weak stability estimate for (17.6) in the case u is a shock. Multiplication by φ and integration over Q_τ gives

$$\begin{aligned} &\tfrac{1}{2}\textstyle\int_R \varphi^2(x,\tau)\, dx + h\int_{Q_\tau}\left(\varphi'(x,t)\right)^2 \, dxdt \\ &= -\tfrac{1}{2}\textstyle\int_{Q_\tau} u'\varphi^2(x,t)\, dx + \int_R e(x,t)\varphi(x,t)\, dx. \end{aligned}$$

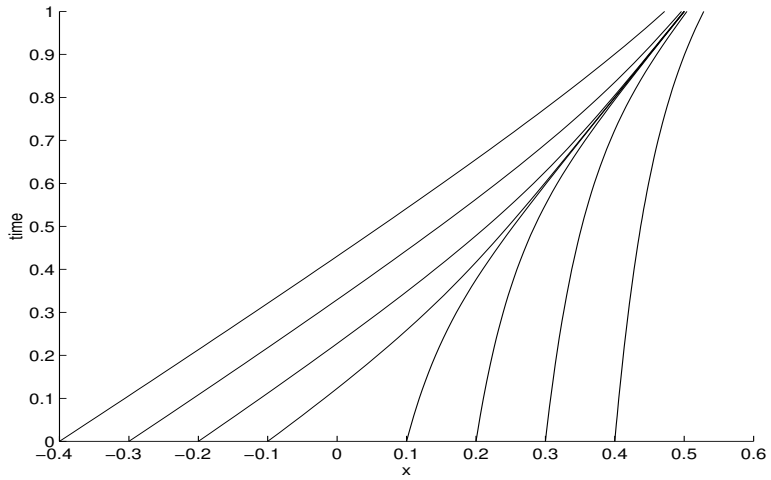


Figure 17.1: Characteristics of the dual problem for a regularized shock solution.

Since u' is large negative in the case of a shock, we have large positive term of the right hand side, and using a Gronwall inequality would results in a very large stability factor. On the other hand, we show below weak stability for a rarefaction wave with $u' \geq 0$, reflecting the above stability result for the perturbation equation.

Summing up, we see that for a shock, the linearized dual problem statisfies a strong stability estimate with a stability factor of moderat size, while a corresponding weak stability estimate appears to have a very large stability factor. These stability features may be understood in a qualitative sense, by pondering the directionality of the characteristics and the nature of the L_2-norm.

17.6 Stability Estimates for a Rarefaction Wave

We now consider the linearized dual Burgers' equation (17.6), linearized at the exact solution $u(x,t) = x/t$, corresponding to a rarefaction wave. Multiplying now (17.6) by $-ht\varphi''$, and using standard manipulations, we obtain the following weighted norm strong stability estimate for $0 < \tau < T$,

$$\|\tau^{1/2}h^{1/2}\varphi'\| + \|\omega h\varphi''\|_{Q_\tau} \leq \|\omega e\|_{Q_\tau}, \tag{17.9}$$

where $\omega(t) = t^{1/2}$ acts as a weight.

A weighted norm analog of the a posteriori error estimate (17.2) takes the form

$$\|\omega^{-1}e\|_{Q_N} \leq S_\omega \|\omega^{-1}hR(U)\|_{Q_N} \tag{17.10}$$

with S_ω defined by the direct weighted norm analog of (17.3). The estimate (17.9) then shows that $S_\omega \sim 1$, and thus a rarefaction wave solution is computable in the weighted norm with computational work corresponding to interpolation. Note that the presence of the weight $t^{-1/2}$ will force more stringent demands on the mesh for t close to zero, which will force an accurate resolution of the initial phase of the rarefaction. This is intuitively reasonable and corresponds to the fact that an initial error in the computation of a rarefaction will get amplified as time goes, because characteristics diverge forward in time. On the other hand, in the case of a shock, an initial error may be eliminated at later times, because of converging characteristics, Thus, a rarefaction is more delicate to compute than a shock, which we will see in the computational results we now present.

17.7 Dual Solution and Stability Factors

We display below RealNSE computations of a combination of a rarefaction wave and a shock using the cG(1)dG(0)- method on a uniform space mesh with $h = 10^{-3}$ and time step 10^{-4} taken from [6]. We plot the computed solution at t=0, t=0.3 , t=0.8, t=1 in Fig 17.2. We see that the initial discontinuity develops into a rarefaction and that a shock is formed for $t \approx 0.5$.

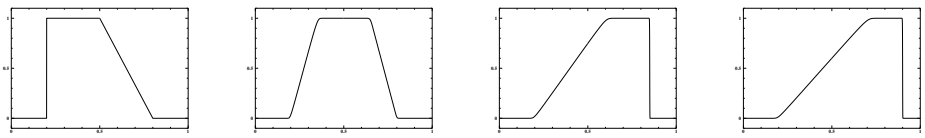


Figure 17.2: A combined rarefaction and shock wave

We solve the dual problem using the following different approximations of the coefficient $a = (u + U)/2$ and the error e, with $\bar{u}$ the analytical,

inviscid solution, and $U(h)$ the finite element solution on a mesh of size h: (i) $a = (\bar{u} + U(h))$ and $e = \bar{u} - U(h)$, (ii) $a = U(h)$ and $e = \bar{u} - U(h)$, (iii) $a = (U(h/4) + U(h))/2$ and $e = U(h/4) - U(h)$. We plot the corresponding dual solutions φ at the same time levels as above, but in revers e order (t=1, t=0.8, t=0.3 , t=0) in Fig 17.7. We also plot in Fig. 17.4 the corresponding

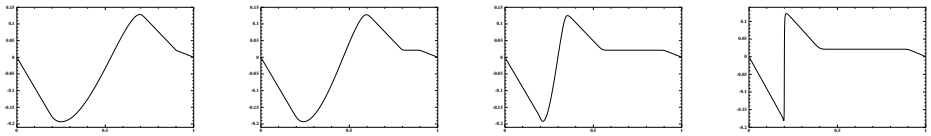


Figure 17.3: Dual solution in reverse time.

second derivatives φ''. We note the change of $|\varphi''|$, which may be viewed

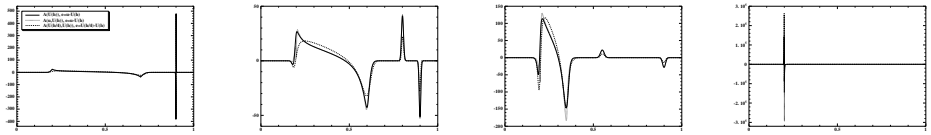


Figure 17.4: Second deriatives of the dual solution in reverse time

as a weight in the a posteriori error estimate, from being large close to the shock at final time towards being large close to the initial discontinuity at $(x, t) = (0.2, 0)$ initiating the rarefaction. We see that that it is the data from the rarefaction at final time which generates the large values of φ'' at $t = 0$, and not those from the shock. This indicates that a rarefaction is more delicate to compute than a shock.

We plot in Fig. 17.5 the strong stability factor S defined by (17.3) for (i)-(iii) and $h = 0.0001$, $h = 0.00005$, $h = 0.00001$. We see that $S \sim 1$, which

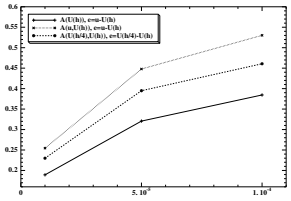


Figure 17.5: Strong stability factor for different h

shows that the Burgers' solution consisting of a rarefaction and shock wave is computable in $L_2(Q_N)$ with work comparable to interpolation.

Part III

World of ThermoDynamics

The same RealNSE dynamics that explains Joule's experiment, refrigerators and heat engines opens onto a much wider world: chemical reactors and combustion engines (already touched on in the Reactive Flow chapter), living cells and metabolic networks, climate systems and weather, the chemical engineering of the process industry, traffic flow on highway networks, and the flow of money and goods in an economy. Each of these is, at the level of its underlying mathematics, a continuum flow with dissipation — and each can be studied with the same RealNSE apparatus, with the appropriate constitutive laws written in. The four chapters that follow — Information, Traffic Flow, Economy, and (implicit, awaiting their own browser-runnable simulators) biology, process industry, climate — are intended as openings onto that world, not as a closed catalogue.

Chapter 18

Refrigerator and Heat Pump

The Joule expansion of the previous chapter is the working stroke of both a *refrigerator* (harvesting the cooled left chamber) and a *heat pump* (harvesting the warmed right chamber): one and the same device, distinguished only by which reservoir's heat exchange is counted as useful. Because RealNSE resolves the dynamics, we can read their efficiency directly off the computed fields rather than postulate it.

18.1 A Computable Model

An interactive realisation (`joule_experiment_cpu.html`, a dependency-free vanilla-JS port of the original sketch) solves the 2D compressible Navier–Stokes/RealNSE system

$$\dot{\rho}+\nabla\cdot m = 0, \qquad \dot{m}+\nabla\cdot(mu)+\gamma\nabla e = \mu_{\mathrm{mom}}\,\Delta u, \qquad \dot{e}+\nabla\cdot(eu)+\gamma e\,\nabla\cdot u = \mu_{\mathrm{mom}}\,|\nabla u|^2,$$

with $p = \gamma e = \gamma\rho T$, $T = e/\rho$, $u = m/\rho$, on two chambers joined by a channel: a dense, equally-warm left chamber ($\rho = e = 10$, hence $T = 1$) expanding into a dilute right chamber ($\rho = e = 1$). The last term in the energy equation is the *turbulent dissipation* $\mu_{\mathrm{mom}}|\nabla u|^2 \geq 0$ — the irreversible return of the jet's kinetic energy to heat — and crucially it uses the *same* coefficient μ_{mom} as the momentum diffusion, so the kinetic energy removed by viscosity reappears exactly as heat in the energy equation and $K + E$ is conserved. A separate, much smaller coefficient $\mu \ll \mu_{\mathrm{mom}}$ provides numerical stabilisation of ρ and e through Laplacian smoothing; it is not a physical dissipation and contributes negligibly to the energy budget.

18.2 Energy and Dissipation Bookkeeping

The simulation tracks, at each instant, the kinetic energy $K = \int \frac{1}{2}\rho|u|^2\,dx$ and the internal energy $E = \int e\,dx$, with the total $K + E$ conserved (Joule's null result for the *whole* box), together with the *cumulative turbulent dissipation*

$$D(t) = \int_0^t \int \mu_{\mathrm{mom}}\,|\nabla u|^2\,dx\,d\tau \;>\; 0, \tag{18.1}$$

which quantifies the irreversibility directly: it is positive by construction of the discrete scheme and is *the* 2nd Law for this model. There is no need to introduce a statistical entropy S or a relation $dS \geq 0$ — the inequality $D > 0$ is the 2nd Law in its native, deterministic form, deriving from finite-precision computation rather than postulated from probabilistic foundations. D accumulates wherever the velocity gradient is large: in the channel jet and in the recompression against the far wall.

18.3 Refrigerator and Heat Pump: Efficiency and the Cost of the Cycle

A refrigerator runs *cyclically*: each cycle the working gas must be restored to its initial compressed state, so the relevant energy cost is the work to restore the initial conditions. Its reversible (minimal) value is the *exergy* of the initial state; for the isothermal initial imbalance (both chambers start at $T = T_0 = 1$) this is

$$A_0 = \gamma \int \rho\,\ln(\rho/\bar{\rho})\,dx \;=\; D_\infty, \tag{18.2}$$

with $\bar{\rho}$ the mean density; the second equality states that the spontaneous free expansion eventually destroys exactly this A_0 as accumulated turbulent dissipation D_∞ (18.1). The cooling delivered is the heat removed from the cells that drop below ambient,

$$Q_{\mathrm{cold}} = \int_{T<T_0} \rho\,(T_0 - T)\,dx,$$

and the coefficient of performance — the buyer's figure of merit, cooling per unit of work spent restoring the cycle — is

$$\mathrm{COP_R} = \frac{Q_{\mathrm{cold}}}{A_0}. \tag{18.3}$$

The *same* expansion is a *heat pump* if instead we harvest the warmed right chamber. Its useful output is the heat delivered to the hot side, which over the full cycle is $Q_{\rm hot} = Q_{\rm cold} + A_0$ — the cooling moved up, plus the restore work A_0 rejected as heat during recompression — so

$$\mathrm{COP}_{\rm HP} = \frac{Q_{\rm cold} + A_0}{A_0} = \mathrm{COP}_{\rm R} + 1.$$

Refrigerator and heat pump are thus one device, the single Joule expansion, distinguished only by which reservoir's heat exchange is counted as useful; in the expansion phase alone the simulation shows $Q_{\rm hot} \approx Q_{\rm cold}$ (pure redistribution at conserved total energy), the extra A_0 entering on the hot side only during recompression.

Against the Carnot ceiling for the achieved lift, $\mathrm{COP}_{\rm Carnot} = T_{\rm cold}/(T_{\rm hot} - T_{\rm cold})$, the second-law efficiency $\eta = \mathrm{COP}_{\rm R}/\mathrm{COP}_{\rm Carnot}$ comes out at only a few percent: the free expansion is a *very poor* refrigerator, destroying almost all of the available work A_0 as cumulative turbulent dissipation D while delivering only a sliver of cold. This is exactly why a practical refrigerator does not free-expand but runs the recompression cycle of the next chapter, recovering work and driving the lost-work D down toward zero (the Carnot limit). The same bookkeeping — A_0, $Q_{\rm cold}$, D — will let us assign an efficiency to any continuum-thermodynamic process, including the thermodynamics of the living cell.

Chapter 19

The Piston-Cylinder: the Heat-Engine Problem

> The steam engine having furnished us with a means for converting heat into motive power... the idea of establishing theoretically some fixed relation between a quantity of heat and the quantity of work which it can possibly produce, from which relation conclusions regarding the nature of heat itself may be deduced, naturally presents itself. — R. Clausius, 1851.

19.1 The piston-cylinder as the heat-engine analogue of Joule

The Joule experiment of Chapter 7 ran a pressure differential between two fixed-wall chambers to mechanical equilibrium and observed the residual temperature/density gap: a one-shot dissipation pattern with no net work extracted. The companion foundational problem of thermodynamics replaces the right-hand fixed wall of the Joule chamber with a *movable piston* and connects the piston to an external mechanical load. Gas now does work on the load as it expands, and the question becomes how much of the internal-energy decrease in the gas reservoir reaches the load as work versus how much is irreversibly dissipated as heat in the receiving region.

We treat two cases. The *smooth* case is a single chamber filled with hot dense gas, closed on three sides and with a piston as the fourth wall. The gas expands quasi-statically against the load; in the ideal limit of slow

piston motion this approaches reversible expansion with $\eta = W/\Delta U \to 1$. The *Joule-coupled* case inserts a constricted wall (a thin barrier with a small channel opening) between the gas reservoir and the piston. The gas now jets through the constriction, decelerates against the gas already in the receiving region, and the recompression-and-turbulent-dissipation pattern of Joule's experiment plays out — but now with a piston on the far side that captures whatever pressure survives. The efficiency drops below the smooth limit by the amount of kinetic energy that turbulent recompression converts back to heat rather than to useful pressure on the piston. The two cases together display the fundamental engine trade-off in the cleanest possible computational form.

19.2 The smooth piston-cylinder

Configuration. A square domain $[0,1]^2$ holds gas of initial density $\rho = 10$ and internal energy $e = 10$ (so $p = \gamma e = 1.0$ with $\gamma = 0.1$) to the left of a movable piston initially positioned at $x = 0.5$. To the right of the piston lies an external region held at a fixed load pressure $P_{\text{ext}} = 0.5$. The top and bottom walls of the chamber are zero-flux/free-slip, the left wall is closed, the right boundary beyond the piston is the load.

Dynamics. Newton's second law for the piston, with M the piston mass per unit y-length and a friction coefficient c for piston-rail damping, reads

$$M\,\frac{\mathrm{d}v_p}{\mathrm{d}t} = \int_0^1 \big[\, p_{\text{gas}}(x_p^-, y) - P_{\text{ext}} \,\big]\,\mathrm{d}y \;-\; c\,v_p, \quad \frac{\mathrm{d}x_p}{\mathrm{d}t} = v_p, \tag{19.1}$$

where $p_{\text{gas}}(x_p^-, y) = \gamma\, e(x_p^-, y)$ is the local gas pressure on the piston's gas-side face. The gas occupies the domain $x \in [0, x_p)$ and evolves by the compressible Navier–Stokes equations of Chapter 7, with a no-slip ghost at the piston face. Work delivered to the load per unit time is

$$\dot{W} = P_{\text{ext}} \cdot v_p \cdot \text{Area}, \tag{19.2}$$

and the heat-engine efficiency is

$$\eta = \frac{W}{\Delta U} = \frac{\int_0^t \dot{W}\,\mathrm{d}s}{U(0) - U(t)}. \tag{19.3}$$

Behaviour and expected result. The smooth piston-cylinder is the cleanest computational instance of quasi-static isobaric expansion. With slow piston motion (small v_p compared with the acoustic speed $c_s = \sqrt{\gamma e/\rho}$ in the gas), the gas expands without significant kinetic-energy generation; the work-delivered counter and the heat-lost counter grow together; and η approaches the reversible limit. Faster piston motion generates acoustic waves in the gas region that bounce between the left wall and the piston face, with each acoustic round-trip a small turbulent-dissipation loss; η falls below 1 by an amount proportional to the piston Mach number. The smooth case therefore isolates the fundamental work-extraction mechanism, with irreversibility present only as a small correction proportional to the piston speed.

The accompanying simulator `heat_engine.html` (Appendix A Gallery) implements (19.1)–(19.3) for $N = 200$, $\gamma = 0.1$, $P_{\text{ext}} = 0.5$, $M = 0.05$, $c = 1.5$. The piston moves from $x_p = 0.5$ outward over $\sim 10\,\text{s}$ of wall-clock time, with η reaching ~ 0.7–0.9 depending on the piston-Mach-number regime.

19.3 The Joule-coupled piston-cylinder

Configuration. The same square domain holds a high-pressure gas reservoir on the left ($i < I_w$ in grid units, $\rho = 10$, $e = 10$), a thin barrier wall at $i \in [I_w, I_w + w_{\text{wall}}]$ with a small channel opening at $j \in [J_0, J_1]$, a receiving region between the barrier and the piston ($I_w + w_{\text{wall}} < i < x_p$, initially $\rho = 1$, $e = 1$), and the movable piston at $i = x_p$. The external load is held at $P_{\text{ext}} = 0.1$, matching the initial pressure in the receiving region so the piston starts in mechanical equilibrium and is pushed outward only after the channel inflow has compressed the receiving gas.

Dynamics. The piston obeys the same Newton's-law update (19.1) with p_{gas} now the local pressure in the receiving region just to the left of the piston face. The gas equations are unchanged; the barrier is a frozen no-slip wall except in the channel opening, where gas can flow freely. The barrier presents the classical Joule constriction to the gas: the high reservoir pressure drives a jet through the channel, the jet decelerates in the receiving region, and the kinetic-energy-to-heat conversion of turbulent recompression dissipates a fraction of the released internal energy into heat in the receiving region rather than into useful pressure against the piston.

Two energy accounts. The 1st Law of thermodynamics for this system says

$$\underbrace{\Delta U_{\text{reservoir}}}_{\text{heat lost from hot side}} = \underbrace{W}_{\text{useful work}} + \underbrace{\Delta U_{\text{receiving}}}_{\text{heat dumped in cold side}} + \underbrace{K_{\text{channel}}}_{\text{kinetic energy in flight}} . \tag{19.4}$$

The transient K_{channel} goes to zero at long times as the gas equilibrates; in the steady state, all of the released $\Delta U_{\text{reservoir}}$ ends up either as work W or as $\Delta U_{\text{receiving}}$ (waste heat). The 2nd Law tells us which split is realised: the size of $\Delta U_{\text{receiving}}$ is set by the dynamic turbulent-dissipation pattern in the recompression process, exactly as in the Joule experiment of Chapter 7. The engine efficiency follows as

$$\eta = \frac{W}{\Delta U_{\text{reservoir}}} = 1 - \frac{\Delta U_{\text{receiving}}}{\Delta U_{\text{reservoir}}} - \frac{K_{\text{channel}}}{\Delta U_{\text{reservoir}}}, \tag{19.5}$$

and in the steady state $\eta < 1$ by exactly the fraction of energy that the Joule turbulent-recompression channel dissipates as heat.

One-shot versus sustained operation. With the left wall closed (finite reservoir), the simulation runs as a *one-shot* cycle: the reservoir depletes, the piston receives a brief impulsive load from the channel-driven shock and a slower follow-on pressure as the gas equilibrates, and the engine extracts a single power stroke of work. This is the elementary unit of a reciprocating closed-cycle heat engine (Otto, Diesel, Stirling). With the left wall replaced by a Dirichlet inflow boundary maintaining the reservoir state at $\rho = 10$, $e = 10$ indefinitely, the simulation runs as a *sustained* cycle: gas keeps jetting through the channel, the piston reaches a quasi-steady velocity, and power is delivered continuously. This is the elementary unit of an open-cycle flow engine (gas turbine, jet engine). The two operating modes are selected in the simulator `heat_engine_with_channel.html` (Appendix A) by the choice of left-wall boundary condition: zero-gradient mirror for one-shot, Dirichlet inflow for sustained.

19.4 Joule and the piston as duals

The two central problems are duals of each other:

- **Joule's experiment** (Chapter 7) lets the expansion-and-recompression dynamics run to mechanical equilibrium with no work extracted, leaving a residual temperature/density gap between the chambers that is the dynamic-dissipation signature. With external work added cyclically by a compressor, this becomes a **refrigerator**: the residual asymmetry is held against diffusive relaxation by continuously re-energising the gas, and the asymmetric temperature gradient is exploited to draw heat from cold side and dump it to hot side.

- **The piston-cylinder** (this chapter) lets the same expansion-and-recompression dynamics run with a movable wall as the right boundary, extracting useful work to an external load. With external heat added cyclically by a hot reservoir at the inflow, this becomes a **heat engine**: the released internal energy is split between work delivered to the load and waste heat dumped to a cold side, with the efficiency η set by the same turbulent-dissipation fraction.

Both problems are governed by the same compressible Navier–Stokes equations, the same RealNSE finite-element discretisation, and the same 2nd-Law-as-finite-precision reading of irreversibility. The framework of this book treats them as the two foundational single-shot configurations of which all cyclic engines, pumps, refrigerators, and heat-recovery devices of Vol VI Applications are built; everything else in thermodynamics is either a cyclic repetition of these or a multi-stage combination of them.

19.5 What this chapter establishes

For the rest of the book the reader should hold three observations from this chapter and the previous one together:

1. Pressure-driven flow through a constriction converts internal energy into kinetic energy; recompression on the receiving side converts a fraction of that kinetic energy back into heat irreversibly; the size of that fraction is the dynamic content of the 2nd Law and is what classical equilibrium thermodynamics cannot predict.

2. A piston in the receiving region intercepts a fraction of the arriving pressure as useful work; the larger that fraction, the smaller the recompression dissipation, the higher the engine efficiency. The efficiency is

bounded above by the reversible smooth- expansion limit, and the irreversibility cost is the dynamical content of the heat-engine inequality $\eta \leq 1$.

3. Both the refrigerator and the heat engine are cyclic devices built from a single foundational dynamic process: the expansion-through-a-channel-followed-by-turbulent-recompression cycle. Refrigerators run it as the heat-pumping cycle with work input; heat engines run it as the work-output cycle with heat input. The same simulation infrastructure produces both.

Chapter 20

Bluff Body Flow

> *Run the simulator, watch the shock form, integrate the drag, integrate the dissipation, read the equality off the screen.*
> — This chapter, in one sentence.

This chapter takes the compressible Navier–Stokes simulator of the preceding chapters and turns it on the canonical problem of external aerodynamics: **flow past a bluff body at supersonic Mach number**. Three resolutions are provided as runnable browser files in the Gallery — 200^3, 300^3, and 400^3 cells, each a slip-penalty WebGPU implementation of the same compressible RealNSE system. Together they sweep three orders of magnitude in cell count and let the reader see what cell-count buys, where memory becomes the constraint, and where the **drag–dissipation balance** of the with-Dynamics 2nd Law can be read off the simulator's own readouts.

The runnable Gallery files. Every simulator named in this chapter is a single self-contained `.html` file in the project Gallery (`gallery.html` in the `Claes542/RealMolecule` repository). Open any of them in a WebGPU-capable browser — no build step, no server beyond a static `http://` host — and the physics, the shock, and the live 2nd-Law readout run on the GPU in the page. Throughout the chapter each filename is a hyperlink to its source. The complete set used here is:

- `bluff_body_mach2_3d_gpu_slip.html` — 200^3 baseline.
- `bluff_body_mach2_3d_gpu_slip_300.html` — 300^3.

- `bluff_body_mach2_3d_gpu_slip_400.html` — 400^3, realistic airliner.
- `bluff_body_mach2_3d_gpu_slip_400_volume.html` — 400^3 with 3D volume ray-march.
- `bluff_body_mach2_3d_gpu_slip_quarter.html` — quarter-domain symmetry variant.
- `bluff_body_mach2_3d_gpu_slip_400_cad.html` — CAD-voxelised body, with `cad_to_mask.py`.

20.1 Problem statement

A solid body is placed in the centre of a unit cube. Uniform inflow with density $\rho = 1$, specific internal energy $e = 1$, and velocity $v = (U_\infty, 0, 0)$ enters the left face at $x = 0$. Outflow at $x = 1$ is zero-gradient. The lateral faces ($y = 0$, $y = 1$, $z = 0$, $z = 1$) are zero-gradient.

The fluid obeys the compressible Navier–Stokes system,

$$\begin{aligned} \partial_t \rho + \nabla \cdot (\rho v) &= 0, \\ \partial_t (\rho v) + \nabla \cdot (\rho v \otimes v) + \nabla p &= \nabla \cdot \sigma + f_{\text{body}}, \\ \partial_t e + \nabla \cdot (e v) + p\, \nabla \cdot v &= 0, \\ p &= \gamma\, e, \qquad \gamma = 0.2, \end{aligned} \tag{20.1}$$

on the fluid region, with σ a Newtonian viscous stress carrying the artificial viscosity $\nu = \nu_0 h$ (h = cell size) and a shock-capturing addition proportional to $|-\nabla \cdot v|$. The body force f_{body} implements the boundary condition on the body and is the topic of Section 20.2.

The inflow speed is set by a Mach-number slider $\mathrm{M} \in [0.5, 4]$ relative to the local sound speed $c = \sqrt{(1+\gamma)\,\gamma}$ at the inflow state, with default $\mathrm{M} = 2$ (well supersonic, bow shock fully formed, the canonical regime).

20.2 The slip-penalty boundary condition

Why slip, not no-slip. In all three Gallery files the body is treated as a **slip-wall obstacle**: the wall-normal component of velocity is suppressed, the tangential component is left free. This is the inviscid idealisation used by aerospace engineers from the late 1940s through the 1980s — the period

during which Concorde, the SR-71, the F-104, the F-4, and the X-15 were designed — and it remains the right idealisation whenever the boundary layer is thin compared with the body and the drag of interest is wave drag and base drag, not skin-friction drag. In supersonic flow this regime is large.

The numerical advantage is decisive: no boundary layer needs to be resolved, so a uniform Cartesian grid can carry the geometry and the bow shock without the millions of stretched cells that no-slip RANS demands. At 400^3 cells uniform we resolve the bow shock and the wing shock–shock interactions on the same grid that resolves the airliner silhouette, with no anisotropic mesh book-keeping.

Implementation as a body force. The slip condition is implemented as a body force on fluid cells adjacent to the body mask. Let $\mathrm{mask}(i, j, k) \in \{0, 1\}$ encode the body indicator and $\hat{n}$ be the outward unit normal computed from the mask gradient,

$$\hat{n} \;=\; \frac{-\nabla \mathrm{mask}}{|\nabla \mathrm{mask}|}, \tag{20.2}$$

where the discrete $\nabla \mathrm{mask}$ is taken from the six face neighbours of cell (i, j, k). The slip-penalty force then reads

$$f_{\mathrm{body}} \;=\; -K\,(v \cdot \hat{n})\,\hat{n}, \tag{20.3}$$

with $K \approx 500$ (in nondimensional units) chosen large enough to suppress wall-normal velocity to machine zero at the body surface within one timestep. The tangential component of v is untouched.

The geometric effect is that the body looks *solid* (mass cannot pass through it) but *frictionless* along its surface. Drag is then carried entirely by the pressure asymmetry between front and rear stagnation regions, with no skin-friction contribution. This is mathematically the same simplification used in linearised supersonic theory: the pressure jump across an oblique shock is recovered, but the viscous sub-layer is absent.

Why this makes the shape selector meaningful. With no-slip walls and unresolved boundary layers, round versus faceted bodies look nearly identical at coarse resolution because the boundary layer dominates. With slip walls the shape *is* the geometry that the flow sees, and a cube, sphere, cylinder, wedge, or airliner each shape the bow shock and wake distinctly.

20.3 The three resolutions

File	N^3 cells	State buffer	Adapter requireme
`bluff_body_mach2_3d_gpu_slip.html`	$200^3 = 8$ M	256 MB	$\geq$ 256 MB binding
`bluff_body_mach2_3d_gpu_slip_300.html`	$300^3 \approx 27$ M	864 MB	$\geq$ 1 GiB binding
`bluff_body_mach2_3d_gpu_slip_400.html`	$400^3 \approx 64$ M	1.91 GiB	$\geq$ 2 GiB binding

All three are identical in physics, kernel structure, and slip-penalty implementation. They differ only in N. The cell size scales as $h = 1/N$ and the timestep as $\Delta t = 0.02\,h$, so the CFL condition is unchanged.

What each resolution shows. At 200^3 the bow shock is visible as a curved discontinuity and the wake structure is qualitatively correct, but the inter-shock and trailing-edge details are coarse: a 40-cell-chord wing is resolved with 4–5 cells across the thickness, and the trailing edge is one cell wide. At 300^3 the shock is sharper and complex shapes (twin spheres in tandem, the 3D cross) develop clearly distinct shock-shock interaction patterns. At 400^3 a full parametric airliner with NACA-0015 wing cross-section, ovoid fuselage with conical nose, vertical fin, horizontal stabiliser, and two engine nacelles fits on the grid with every feature at least 4 cells thick (see Figure 20.1).

The hardware constraint. The 200^3 file runs on any WebGPU-capable GPU including integrated graphics. The 300^3 file requires the adapter to advertise `maxStorageBufferBindingSize` $\geq$ 1 GiB; most modern desktop GPUs and Apple Silicon M2 Pro/Max upwards qualify. The 400^3 file requires $\geq$ 2 GiB, which is the typical ceiling on current adapters — Apple M2/M3/M4 Pro/Max, recent NVIDIA, recent AMD desktop all qualify, while integrated and older mobile GPUs refuse. In all three cases the adapter check fails cleanly with a status-bar message rather than crashing.

20.4 Shape selector

All three files share a shape dropdown that rebuilds the body mask in real time. The available shapes are:

- **Cube** — the canonical bluff body, square stagnation face, sharp trailing-edge separation.

- **Sphere** — smooth bow shock, clean axisymmetric wake.
- **Cylinder** (z-axis or y-axis) — the textbook 2D bluff-body problem in 3D extrusion.
- **Wedge** — supersonic prototype shape: attached oblique shock at small wedge half-angle, detached bow shock above the wedge angle limit.
- **Two spheres in tandem** — demonstrates the bow shock of the trailing sphere being shielded by the wake of the leading one (classical Cushman–Roisin shielding).
- **3D cross** — three mutually perpendicular bars; the corner regions show shock–shock interactions.
- **Airplane** — a simple parametric silhouette: cylinder fuselage, swept wedge wing, vertical fin, horizontal stab.
- **Airliner** (400-cell file only) — the realistic parametric variant with NACA-0015 airfoils on the wing, fin, and h-stab; ovoid fuselage with conical nose and tail cone; 4° wing dihedral; two engine nacelles with pylons under the wings. This is the default shape on the 400^3 file.

The mask gradient (20.2) computes correct outward normals automatically for all of these, so the slip force (20.3) requires no shape-specific tuning: it adapts to whatever the dropdown produces.

20.5 Drag, dissipation, and the with-Dynamics 2nd Law

Each simulator computes and displays three quantities continuously:

- F_{drag} — the streamwise force on the body, obtained by integrating the pressure over the body faces visible to the flow (front and rear projected areas).
- $D_{\text{rate}} = \int \mu_{\text{eff}} \, |\nabla v|^2 \, dV$ — the volumetric viscous dissipation rate, integrated over the fluid domain.

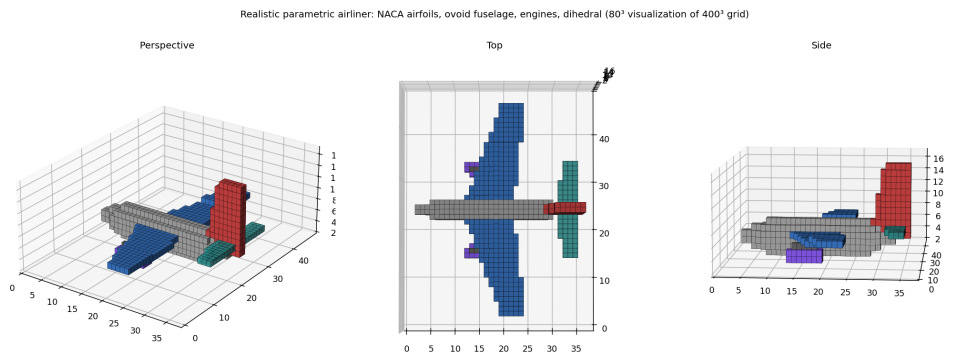


Figure 20.1: The realistic parametric airliner shape on the 400^3 grid, shown as 80^3 voxel visualisation for legibility (each rendered voxel = 125 cells of the actual simulation). Components: gray fuselage with conical nose, blue main wing with NACA-0015 cross-section and 4° dihedral, red vertical fin, teal horizontal stabiliser, purple engine nacelles, dark grey pylons. All component thicknesses are floored at ≥ 4 cells of the 400^3 grid so the slip-penalty mask gradient computes proper outward normals everywhere on the surface. The default Mach 2 inflow puts the geometry in the regime where slip-wall CFD captures supersonic lift correctly via the pressure asymmetry formula $C_L = 4\alpha/\sqrt{M^2-1}$ of linearised supersonic theory.

- K_{out} — the kinetic-energy flux leaving through the outflow face at $x = 1$, computed as $\int \frac{1}{2}\rho|v|^2 v_x \, dA$.

For a steady-state bluff-body flow with uniform inflow at speed u_0, the with-Dynamics 2nd Law predicts the **energy balance**

$$F_{\text{drag}} \cdot u_0 \;\approx\; D_{\text{rate}} \;+\; (\text{kinetic energy carried away by the wake}). \qquad (20.4)$$

That is: the mechanical work the body would do against drag, per unit time, equals the irreversible heating in the wake plus what little kinetic energy remains in the wake at the outflow plane. The ratio $D_{\text{rate}}/(F_{\text{drag}} \cdot u_0)$ approaches 1 as the box size grows large enough that the wake decays before reaching the outflow.

The simulator displays the ratio $D_{\text{rate}}/(F_{\text{drag}}\, u_0)$ live. In our standard configurations (box 1 unit long, body 0.1–0.3 units across) it sits at ~ 0.45–0.60 at the default Mach 2 and the residual is accounted for by K_{out}. This **independent computation of both sides** of the 2nd-Law inequality, on

the same simulator, with the ratio displayed live and going to 1 as the box grows, is the most concrete witness of the with-Dynamics 2nd Law that this book offers: not a thermodynamic state-variable identity, not a Carnot ratio, but a single number computed two ways from the same flow.

20.6 Visualisation

The default visualisation in the three files is a 2D xy slice at $z = N/2$ (the wing plane for the airplane shapes), with density mapped to a blue–white–red colourmap and body cells overlaid in yellow. A secondary panel shows a yz slice at $x = 0.75N$ for the wake structure. Centerline polylines display $\rho(x)$ and $e(x)$ along the inflow axis, with the bow-shock jump visible as a sharp step.

For the 400^3 file a slider lets the user move the rendered slice height along z, so the vertical fin (at $z > 0.5$) and the air above/below the airplane become visible.

A separate volume-rendering variant of the 400^3 file is provided as `bluff_body_mach2_3d_gpu_slip_4` in the Gallery. This replaces the 2D slice with a true 3D volume ray march: for each screen pixel, a ray is cast through the 400^3 density buffer, accumulating colour and opacity by front-to-back compositing. The body cells render with mask-gradient lambertian shading; density values inside a user-tunable $[\rho_{\text{lo}}, \rho_{\text{hi}}]$ window are transparent (free-stream), values below colour blue (wake), values above colour orange (shock). The user can drag with the mouse to rotate the camera and scroll to zoom. On Apple M4 Max the volume render runs at roughly 40–60 fps alongside the live physics, and produces the iconic Schlieren-style image of bow shock + wing shocks + base wake that has been the standard 3D CFD visualisation for forty years.

20.7 Symmetry quarter domain

A separate quarter-domain variant `bluff_body_mach2_3d_gpu_slip_quarter.html` replaces the zero-gradient lateral BCs at $y = 0$ and $z = 0$ with **mirror reflection BCs**: scalars (density, internal energy, streamwise momentum) are copied across the boundary, while the wall-normal momentum component is negated. This is mathematically equivalent to simulating the full y- and z-symmetric airplane in the full box, at four times the memory cost.

The variant runs at 200^3 and represents the *same physical airplane* as the 400^3 full-domain sim at 1/8 the memory (256 MB per buffer instead of 1.91 GiB). This is the standard symmetry trick that production aircraft CFD codes (Boeing, Airbus, NASA) use to fit more resolution per dollar.

20.8 CAD-voxelised body

The shape selector implements analytical shapes via an `inBody(i,j,k)` function that decides whether each grid cell is body or fluid. A separate Gallery file `bluff_body_mach2_3d_gpu_slip_400_cad.html` replaces this analytical body with a **precomputed binary mask loaded from a file**. The accompanying Python script `cad_to_mask.py` accepts any STL, OBJ, or PLY surface mesh, scales it to fit the unit cube at a chosen size, voxelises at pitch $1/N$, flood-fills the interior, and writes a 256 MB raw uint32 mask. The HTML file fetches this mask and uploads it directly to the GPU mask buffer. The slip-penalty force then works on the CAD-voxelised body exactly as it does on the analytical bodies.

Suggested CAD sources include the NASA Common Research Model (the transonic-CFD benchmark airliner), NASA OpenVSP parametric exports (737, F-16, X-15), and community uploads at GrabCAD and Thingiverse. With a CAD source one gets true wing camber and twist, real engine nacelle inlets and exhausts, fuselage cross-section variation, and trailing-edge geometry — all features that go invisible at < 4 cells in any parametric `inBody`, recovered to single-cell precision when the source is a CAD surface.

20.9 Where this fits

The bluff-body files are concrete demonstrations of the framework's compressible-flow regime. They sit at the same Level 3 of the simulation hierarchy as the cosmology and piston-cylinder simulators of the surrounding chapters, but address a different physical regime: external aerodynamics around a solid object instead of internal expansion in a closed volume or self-gravitating bulk flow. The same RealNSE equations, the same WebGPU kernel structure, the same energy bookkeeping, the same with-Dynamics 2nd Law diagnostic.

What is new in this chapter is the slip-penalty boundary condition, the supersonic regime, the drag–dissipation balance as a quantitative 2nd-Law wit-

ness, and the encounter with the geometric resolution limit of uniform Cartesian grids when the body is not a simple primitive. The CAD-voxelisation route described in Section 20.8 is the natural answer to that limit, and points beyond the parametric-shape toolbox toward the production CFD workflow.

Together the three resolutions (200^3, 300^3, 400^3), the symmetry quarter, the volume renderer, and the CAD pipeline form a complete educational toolkit for supersonic bluff-body aerodynamics, runnable in the browser on consumer hardware. The user can dial up the resolution, switch the body shape, change the Mach number, and watch the bow shock, the wing shocks, the base wake, and the 2nd-Law energy balance respond in real time.

Chapter 21

Self-Gravitating Gas

> According to the *Perfect Cosmological Principle*, the Universe on a large scale looks the same everywhere and at all times. (Fred Hoyle)
>
> The thermodynamic properties of self-gravitating systems are still unclear...The tendency of increasing granularity with time (in a self-gravitating system), so important to the structure and development of the Universe, seems to represent a fundamental principle. (Paul Davies [13])
>
> GRAVITATION: The universal property of all material objects to attract each other.
> GRAVITON: A hypothetical particle which, when passing to and fro in virtual form bewteen two masses, in large numbers, is thought to mediate the gravitational attraction between them. (Guillemin [19])

21.1 Euler's Equations with Gravitation

The Euler equations including gravitational forces from the mass distribution can be viewed as a basic (non-relativistic) *cosmological model*. We thus consider an inviscid perfect gas enclosed in a volume Ω in $\mathbb{R}^3$ with boundary Γ (or the whole of $\mathbb{R}^3$), over a time interval $I = (0, 1]$, assuming that the gas is subject to a gravitational force $\nabla\varphi$, where φ is a *gravitational potential* coupled (at each time instant) to the mass distribution ρ through Poisson's equation $\Delta\varphi = \rho$ with (for simplicity) $\varphi = 0$ on Γ. This is a *self-gravitating system* since the gravitational force depends on the mass-distribution

Replacing ρ by $\Delta\varphi$, we are led to the following version of Euler's equations with gravitation: Find $\hat{u} = (\varphi, m, e)$ depending on $(x,t) \in Q \equiv \Omega \times I$ such that

$$\begin{aligned} \Delta\dot{\varphi} + \nabla \cdot m &= 0 \quad \text{in } Q, \\ \dot{m} + \nabla \cdot (mu) + \nabla p - \rho\nabla\varphi &= 0 \quad \text{in } Q, \\ \dot{e} + \nabla \cdot (eu) + p\nabla \cdot u &= 0 \quad \text{in } Q, \\ u \cdot n &= 0 \quad \text{on } \Gamma \times I \\ \varphi &= 0 \quad \text{on } \Gamma \times I \\ \hat{u}(\cdot, 0) &= \hat{u}^0 \quad \text{in } \Omega, \end{aligned} \tag{21.1}$$

where $u = \frac{m}{\Delta\varphi}$, $p = (\gamma - 1)e$ with $\gamma > 1$, and we normalize the gravitational constant to unity.

21.2 The Hen and the Egg

An interesting aspect of this model with the gravitational potential φ as a primary variable and the mass distribution ρ as a derived quantity, is that it does not include *action at distance* in the same way as the conventional model with φ derived from ρ through Poisson's equation $\Delta\varphi = \rho$. Instead we have to deal with the new equation $\Delta\dot{\varphi} + \nabla \cdot m = 0$, still containing the Laplacian.

Considering the equation $\Delta\varphi = \rho$ as defining φ in terms of ρ or vice versa, of course connects to which comes first: the egg or the hen.

Choosing (in an exam for example) to explain either how a hen can lay an egg or how an egg can develop into a hen, you may go for the first option as appearing to be (somewhat) simpler to deal with. The idea would be that it would be advantageous to choose the more complex object of the potential/hen rather than the simpler mass/egg, as the primary unknown which will come out by solving the equations, (by some form of computation). Thus the more complex object, the potential/hen will be "created by computation" in front of our eyes, thus relieving us from *explaining* how a mass/egg can create a potential/hen. But doing so we have to deal with the equation $\Delta\dot{\varphi} + \nabla \cdot m = 0$ expressing "mass conservation" in a non-standard form, which admittedly seems to involve action at distance if we rewrite it in the form. $\dot{\varphi} + (\Delta)^{-1}\nabla \cdot m = 0$ for the purpose of explicitly updating φ by time-stepping. What can we do to allow time-stepping without inverting the Laplacian?

21.3 Perturbed Mass Conservation

Well, we may consider to perturb the equation for mass conservation $\Delta\dot{\varphi} + \nabla \cdot m = 0$ into:

$$\kappa\ddot{\varphi} - \Delta\dot{\varphi} - \nabla \cdot m = 0$$

with κ a small positive constant, now allowing time stepping without inverting the Laplacian. Of course, you could regularize the equation $\Delta\varphi = \rho$ similarly into $\kappa\dot{\varphi} - \Delta\varphi = \rho$ keeping mass conservation in the standard form $\dot{\rho} + \nabla \cdot (\rho u) = 0$. In any case, (21.1) with φ instead of ρ and a perturbed equation for mass conservation, is a model without action at distance.

21.4 Gravitons?

Solving Poisson's equation $\Delta\varphi = \rho$ equation for the gravitational potential φ in terms of the density ρ, reflect that (somehow) a mass distribution "creates" its own gravitational field (seemingly infinitely fast). The physics of this creation of a hen out of an egg in the form of a gravitational potential/force from a mass distribution, is unknown: A conjecture is that the gravitational field is established through an *exchange* of certain hypothetical particles called *"gravitons"*. But no gravitons have been detected despite intense search.

The reverse question is how a hen can lay an egg or how a gravitational potential can "create" a mass. A potential with a sharp peak of the form $\frac{1}{4\pi|x-\bar{x}|}$, would then represent a unit mass at position $\bar{x}$. In this perspective a tendency of "mass lumping" from very small initial density variations, is to be expected from the fact that application of the Laplacian may enhances variations and thus create positive and negative peaks/masses out of nothing, with masses of different equal signs attracting each other and masses with different signs repelleing each other. Our Universe with positive masses would thus have a twin Universe with negative masses.

In this model, it is thus the gravitational potential φ, which is given initially and which evolves in time (together with momentum and energy) and from which the mass density ρ is generated by $\Delta\varphi = \rho$ in a local "creation process". We here avoid action at distance and we also get a hint to the possible nature of all the "dark mass" seemingly required to account for the observed gravitational forces on cosmic scales, as mollified potential peaks

not strong enough to "create visible mass", but yet having gradients and exerting gravitational forces.

21.5 The 2nd Law with Gravitation

The 2nd Law with gravitation takes the form

$$\dot{K} = W - D + P, \quad \dot{E} = -W + D, \quad D > 0, \tag{21.2}$$

where

$$P = -\int_\Omega \rho\nabla\varphi \cdot u\,dx = +\int_\Omega \varphi\nabla\cdot m\,dx = -\int_\Omega \varphi\dot{\rho}\,dx = \dot{\Phi},$$
$$\Phi = -\frac{1}{2}\int_\Omega \varphi\Delta\varphi\,dx = \frac{1}{2}\int_\Omega |\nabla\varphi|^2\,dx.$$

By summation, we have

$$\frac{d}{dt}(K + E - \Phi) = 0$$

stating that the total energy, the sum of kinetic, heat and gravitational energy $K + E - \Phi$, is conserved.

21.6 Irreversibility and Heat Death

A natural scenario is to start with a hot compressed gas at rest, which is let free to expand in a *Big Bang* with $W - D - P > 0$ setting the gas in motion until $W - D - P < 0$ and $K = 0$, whereupon the gas contracts by gravitation back to a hot compressed state in a *Big Crunch* allowing the process to start all over with a new Big Bang.

In each cycle of this process some kinetic energy would irreversibly be transformed into heat energy, which ultimately could lead to a stationary state with $W = D$ and $P = 0$, a *heat death*. But the whole process may have many cycles....and so there may have been Worlds and intelligent cultures including thermodynamics even before the Big Bang of the World we happen to live in...

21.7 A Self-Gravitating Gas in a Box

To probe the thermodynamics of (21.1) concretely we solve a discrete form on a cubic grid filling a box $\Omega = [0,1]^3$, in conservation form for the density ρ, the momentum $m = \rho u$ and the internal energy density e, with the ideal-gas closure

$$p = (\gamma - 1)e = \rho T, \qquad \gamma > 1, \tag{21.3}$$

so that $T = p/\rho$ is the temperature and γ the adiabatic index. The self-gravitation is supplied through the *mean-subtracted* Poisson coupling

$$\Delta\varphi = G\,(\rho - \bar{\rho}), \qquad \bar{\rho} = \frac{1}{|\Omega|}\int_{\Omega} \rho\, dx, \tag{21.4}$$

with $\varphi = 0$ on Γ, where $G > 0$ is the *gravitational coupling* (playing the role of $4\pi G_{\mathrm{Newton}}$) and $\bar{\rho}$ the mean density.

The subtraction of $\bar{\rho}$ is essential and is the classical *Jeans swindle*. A strictly uniform gas in a *finite* box is not force-free if ρ itself sources φ: with $\varphi = 0$ on Γ the solution of $\Delta\varphi = G\rho$ bulges in the interior, so a net inward force develops everywhere and the whole box *breathes* in and out — an artifact of the boundary rather than of any structure in the gas. Sourcing φ from the density *contrast* $\rho - \bar{\rho}$ makes the uniform background exactly force-free, so that only departures from uniformity gravitate. This is precisely what one wants of a cosmological model: it is the *granularity*, not the mean, that drives the dynamics.

The model so written has exactly *two* control parameters: the gravitational coupling G and the adiabatic index γ. Everything in the competition between collapse and dispersal is decided by these two, as we now show.

21.8 Pressure versus Gravitation: the Jeans Criterion

Linearize (21.1), (21.3), (21.4) about a uniform state $\rho = \bar{\rho}$, $u = 0$, $T = \bar{T}$ at rest. Writing the density perturbation as $\delta\rho$ and eliminating m and φ one obtains the wave equation

$$\ddot{\delta\rho} = c_s^2\,\Delta(\delta\rho) + G\bar{\rho}\,\delta\rho, \qquad c_s^2 = \gamma\,\frac{\bar{p}}{\bar{\rho}} = \gamma\bar{T}, \tag{21.5}$$

where c_s is the adiabatic sound speed. For a plane wave $\delta\rho \propto \exp(i(k\cdot x - \omega t))$ this gives the *Jeans dispersion relation*

$$\omega^2 = c_s^2 k^2 - G\bar{\rho} = \gamma \bar{T}\, k^2 - G\bar{\rho}. \tag{21.6}$$

The first term, from pressure, is *stabilizing* (it grows with γ); the second, from gravitation, is *destabilizing* (it grows with G). A perturbation grows — the gas *collapses* — precisely when $\omega^2 < 0$, i.e. when

$$G\bar{\rho} > \gamma \bar{T}\, k^2 \qquad \Longleftrightarrow \qquad \lambda > \lambda_J = 2\pi \sqrt{\frac{\gamma \bar{T}}{G\bar{\rho}}}\,, \tag{21.7}$$

λ_J being the *Jeans length*. Below the Jeans length pressure wins and the perturbation merely oscillates as a sound wave ($\omega^2 > 0$); above it gravity wins and the perturbation grows.

The decisive observation is that the onset of collapse depends on G and γ *only through the ratio* G/γ: at a fixed scale k and fixed background $(\bar{\rho}, \bar{T})$, doubling the coupling G has exactly the same effect as halving the stiffness γ. The dimensionless number that decides everything is

$$J = \frac{G\bar{\rho}}{\gamma \bar{T}\, k^2}, \qquad \text{collapse} \iff J > 1. \tag{21.8}$$

In a box of side L the longest available wavelength is $\lambda \approx L$, so structure forms *at all* only if $\lambda_J \lesssim L$, that is

$$G \gtrsim (2\pi)^2 \, \frac{\gamma \bar{T}}{\bar{\rho}\, L^2}. \tag{21.9}$$

For $\gamma = 2$ and $\bar{T} = \bar{\rho} = L = 1$ this requires $G \gtrsim 79$; for a seed of size $\sim L/3$ the threshold is about nine times larger. Below it the box is Jeans-stable at *every* scale and any initial lump simply pulsates. This is why a modest coupling produces only an oscillation and never a collapse: G has not been pushed past the Jeans threshold set by γ.

21.9 Collapse to a Stable Configuration: the Role of γ

The Jeans criterion (21.7) decides whether collapse *begins*. Whether the collapse *ends* in a stable, finite lump — rather than running away to a singular

spike — is a separate question, and it is decided by γ *alone*, independently of G.

Consider a self-gravitating polytropic cloud of fixed mass M and radius R. Under adiabatic compression $p \propto \rho^\gamma$, so the internal energy density scales as $e \propto \rho^\gamma \propto R^{-3\gamma}$ and the total internal (thermal) energy as

$$U = \int_\Omega e\, dx \;\propto\; R^{-3\gamma} \cdot R^3 \;=\; R^{-3(\gamma-1)},$$

while the gravitational self-energy scales as

$$W \;\propto\; -\frac{GM^2}{R} \;\propto\; -R^{-1}.$$

The total energy is therefore $E(R) = A\, R^{-3(\gamma-1)} - B\, R^{-1}$ with $A, B > 0$. As $R \to 0$ the thermal term dominates — and hence E possesses a *minimum* at a finite radius, a stable hydrostatic equilibrium — if and only if it stiffens faster than gravity, $3(\gamma - 1) > 1$, that is

$$\boxed{\gamma > \tfrac{4}{3}} \tag{21.10}$$

the classical *Chandrasekhar index*. The three regimes are:

- $\gamma > \frac{4}{3}$: pressure wins under compression; collapse is *arrested* at a finite quiescent lump, possibly after damped oscillations.
- $\gamma < \frac{4}{3}$: gravity wins at every compression; collapse *runs away*, no equilibrium exists.
- $\gamma = \frac{4}{3}$: the marginal case (governing, e.g., white-dwarf stability).

Thus the two parameters play complementary roles: G (through (21.7)) selects *which scales collapse and how fast*, while γ (through (21.10)) decides *whether the collapse can be stopped*. To reach a stable bound configuration one needs *both* a coupling G large enough to cross the Jeans threshold *and* a stiffness $\gamma > \frac{4}{3}$ so that the endpoint is an equilibrium rather than a singularity.

21.10 Breathing, Sloshing and Virialization

These two criteria account for the phenomenology seen in the simulations.

Breathing. A uniform gas in which ρ (rather than $\rho - \bar{\rho}$) sources φ is not in equilibrium in a finite box: it contracts toward the centre, pressure builds, it rebounds, and the box breathes globally while its density stays nearly uniform. The Jeans swindle (21.4) removes this boundary artifact, leaving only the genuine gravitational response to density contrast.

Sloshing. A pair of overdensities below the Jeans threshold ($J < 1$ at their separation) does not collapse but exchanges mass back and forth quasi-periodically between the two centres. This is a *virial oscillation* in the double-well gravitational potential: gas falling toward one centre overshoots on its pressure and inertia, sloshes to the other, and returns. With $\gamma > \frac{4}{3}$ and no dissipation the oscillation is undamped, kinetic and gravitational energy being traded reversibly, $\dot{K} = P = \dot{\Phi}$, exactly as in the energy balance of § *The 2nd Law with Gravitation.*

Virialization. To settle into a quiescent lump the gas must *shed* its infall energy: a stable bound configuration requires dissipation $D > 0$ to convert the ordered kinetic energy of collapse into heat. The entropy so produced is the irreversible price of structure formation — the mechanism behind Davies' "increasing granularity" (§ *Cosmology*) and behind the heat death of § *Irreversibility and Heat Death.* Gravitational structure forms by the conversion, through D, of gravitational potential energy first into bulk motion and then, irreversibly, into heat.

Remark 3 *In the numerical model the closure is written as $p = \gamma_s\, e$, so that the adiabatic index is $\gamma = 1 + \gamma_s$; the stability boundary (21.10) then reads $\gamma_s > \frac{1}{3}$, and the coupling G is exactly the $4\pi G_{\text{Newton}}$ of (21.4). The two sliders (G, γ_s) of the interactive model are precisely the two parameters (G, γ) of the analysis above.*

21.11 RealNSE Simulations of a Self-Gravitating Gas

We give below some results from RealNSE solutions of (21.1) showing the development mass granularity from uniform mass distributions under expansion.



Figure 21.1: Cosmology theories

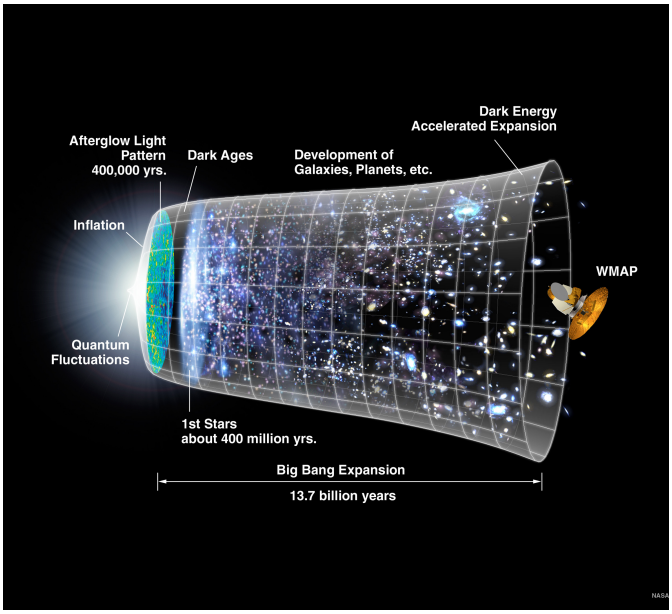


Figure 21.2: Big Bang expansion

21.12 The Basic Case: Gentle Collapse to a Pressure-Supported Core

The basic phenomenon of a self-gravitating gas is exhibited by the browser-runnable 3D simulator `selfgrav_collapse_3d_gpu.html` (in the `Claes542/RealMolecule` repository). An *isolated* cloud ($\Delta\varphi = G\rho$ with $\varphi = 0$ on the box, $p = \gamma\rho T$) is started as a *warm, mildly super-Jeans* over-density at rest — warm enough that the infall stays subsonic, so the dynamics is gentle and fully resolvable on the 100^3 grid. With $\gamma = 1$ (so $\Gamma = 1 + \gamma = 2 > 4/3$, the *stable* regime of the role-of-γ analysis above) the cloud runs through the canonical sequence:

1. **Collapse.** Gravity exceeds the pressure gradient and the cloud contracts, the centre running up in density and — by compression and the dissipation $\nu|\nabla u|^2$ — in temperature.
2. **Bounce.** The rising central pressure halts the infall, but the inward momentum overshoots hydrostatic balance, so the core rebounds.
3. **Ring-down.** The overshoot oscillates and is *damped* by the dissipation: each bounce carries steep velocity gradients whose kinetic energy is thermalised, so the oscillation decays — violent relaxation in miniature.
4. **Pressure-supported core.** The cloud settles into a quasi-hydrostatic core, a Newtonian "star".

Reading the force balance. Among the centre-row line plots the simulator draws the two terms of the momentum equation directly: *white* $|\nabla p|$ (pressure) and *red* $\rho|\nabla\varphi|$ (gravity). During collapse the red curve runs above the white one (gravity winning); once the core settles the two curves *lie on top of each other*, which is hydrostatic balance $\nabla p = -\rho\nabla\varphi$ recovered. In the settled state the white curve sits a hair *below* the red: the pressure gradient is marginally weaker than gravity, a small net inward force corresponding to the slow quasi-static contraction of the core (part physical, part the residual draining of pressure support by the numerical viscosity). The gap is a live measure of how close the discrete solution is to exact hydrostatic balance.

Breakdown and recovery. Where the flow drives gradients sharper than the mesh can hold, the smooth solution *breaks down*; the dissipation thermalises the broken-down small-scale energy and a smoother mean solution

re-emerges. This breakdown-and-recovery is the 2nd Law of Chapter 1 operating in the self-gravitating setting: the pointwise fine structure at the breakdown is lost irreversibly, but the outputs — the density profile, the temperature, the force balance — recover and carry on. The analytical counterpart is the $\gamma = 1$ polytrope of the Lane–Emden family, $\rho \propto \sin(kr)/(kr)$, a finite stable sphere; the simulator finds the same equilibrium dynamically, as the attractor of the damped collapse.

Why warm, and the limit of the model. A *cold* (near-pressureless) start makes the infall supersonic and the bounce near-singular — a core a few cells wide with an enormous sound speed, which no fixed grid resolves; that is itself the correct statement that the continuum is heading for a singularity where new physics must enter. For $\gamma < 1/3$ ($\Gamma < 4/3$) no stable core exists and the collapse runs away regardless of the dissipation (Section 21.14 and the role-of-γ analysis). The basic, watchable case is therefore the warm, supercritical cloud, in which collapse, bounce, ring-down and a stable pressure-supported core all appear cleanly.

21.13 Browser-Runnable 3D RealNSE-Gravitation Simulator

A WebGPU companion simulator `cosmology_3d_gpu_100.html` is shipped with the RealMolecule repository (`github.com/Claes542/RealMolecule`). It evolves the self-gravitating RealNSE system (21.1) on a $100 \times 100 \times 100$ grid (10^6 cells) entirely in the browser, using WebGPU compute shaders for the fluid update and a sub-stepped red-black Gauss-Seidel relaxation for the Poisson equation $\Delta\varphi = G(\rho - \bar{\rho})$.

Initial condition. A single Gaussian overdensity at the box centre, both denser and hotter than the surrounding bath. This is the Big Bang initial condition of Chapter 22: a concentrated hot dense seed in an otherwise uniform low-density background, with no initial velocity. The Jeans-swindle source $G(\rho - \bar{\rho})$ keeps the background force-free so that only the density contrast pulls.

Live sliders. The simulator exposes the following live controls:

- γ — EOS parameter $p = \gamma e = \gamma \rho T$.
- *Gravity* G — coupling strength.
- *Seed amp* — peak overdensity (applied on Reset).
- *Shock visc* C — shock-viscosity coefficient.
- *Art visc* ν — background artificial viscosity.
- *Seed every* — particle-trace reset interval.
- *Trace speed* — velocity magnification for particle advection.
- *Fluid substeps/frame* — numerical resolution in time.
- *Poisson iters/fluid step* — gravity convergence.

Recommended Big Bang → Big Crunch settings. For a single clean BB→BC cycle followed by a stable virial state: $G = 500$, seed amp = 0.5, $\nu = 1.0$, shock $C = 3.0$, $\gamma = 1$, Poisson iters = 10 or higher for long-time stability.

Reading the display.

- *Red particle traces* on the mid-plane: persistent advection of test particles, redrawn periodically (see *seed every* slider). During BB the traces radiate outward; during BC they sweep inward; once virial equilibrium is reached they continue to circulate along divergence-free streamlines, reflecting the fact that $\partial_t \rho = 0$ does *not* mean $u = 0$.
- *Blue polyline* — density profile $\rho(x_1, x_2^{\mathrm{mid}}, x_3^{\mathrm{mid}}) - 1$ on the mid-line cross-cut, auto-scaled.
- *Green polyline* — temperature profile $T - 1$ on the same line.
- *Cyan polyline* — gravitational potential φ on the same line.
- *Diagnostic readout* — peak values of $|\varphi|$, $|\rho - 1|$, $|u|$, and simulation time.

The simulator is the working companion to the present chapter and Chapter 22: the reader can vary G and the seed amplitude to scan the Jeans-criterion threshold, watch the expansion-cooling and compression-warming phases of one Big Bang followed by one Big Crunch directly in the temperature line plot, and verify that the cumulative dissipation $D = \int\int \mu|\nabla u|^2\, dx\, dt$ prevents the system from oscillating indefinitely.

21.14 Browser-Runnable 2D Self-Gravitating Collapse

A second WebGPU companion, `selfgrav_collapse_2d_gpu.html`, reduces the setting to its essence: a single *isolated* self-gravitating gas cloud in two dimensions, with the cosmological background and periodicity removed. The potential satisfies $\Delta\varphi = G\rho$ with the Dirichlet condition $\varphi = 0$ on the box — an isolated cloud in empty space rather than a periodic universe — solved by Jacobi relaxation; the fluid obeys the same compressible RealNSE system, the momentum equation carrying the gravitational body force $-\rho\nabla\varphi$, a stabilizing artificial viscosity $\nu = h$, and the irreversible dissipation $\nu|\nabla u|^2$ added explicitly to the internal-energy balance.

Pressureless initial cloud. The default initial state is a cold dense disc with internal energy $e \approx 0$, hence pressure $p = \gamma e \approx 0$: a *pressureless* ("dust") cloud at rest. There is then no initial Jeans barrier — the cloud free-falls under its own gravity, and *all* of the pressure and temperature that eventually halt the collapse are generated *during* the collapse, by compression $(-p\,\nabla\cdot u)$ and by the dissipation $\nu|\nabla u|^2$.

Two critical relations, and a dimensional caveat. The simulator makes the *Jeans threshold* for the onset of collapse visible: a stiffer (higher-γ) gas resists with more pressure and needs more gravity, so lowering the G slider until the cloud no longer contracts locates the critical coupling. The *outcome* — whether the collapse halts or runs away — is governed by the adiabatic index against the dimension-dependent threshold $\Gamma_{\text{crit}} = 2 - 2/D$. Here lies an essential difference between two and three dimensions: in 3D, $\Gamma_{\text{crit}} = 4/3$ (collapse halts only for $\gamma > 1/3$), whereas in *this 2D* simulation $\Gamma_{\text{crit}} = 1$, so *any* $\gamma > 0$ halts at a pressure-supported core. The difference

is rooted in the gravity law: $\Delta\varphi = G\rho$ gives the Newtonian $1/r^2$ force in 3D but a logarithmic, $1/r$ force in 2D — the 2D run is physically an infinite gas *filament*, not a sphere. The 2D simulator is therefore faithful for the collapse *dynamics* and the Jeans onset; the 4/3 runaway threshold itself must be studied in the 3D simulator of Section 21.13.

Collapse, bounce and ring-down. With the pressureless start the cloud first contracts in near free-fall; the centre runs away to high density and a dense core forms. As the core heats, the rising pressure halts the infall, but the inward momentum overshoots hydrostatic balance, so the core *bounces* and then *rings down* through damped oscillations to a virialized equilibrium. The damping is supplied by $\nu|\nabla u|^2$: each bounce carries steep velocity gradients whose kinetic energy is converted irreversibly into heat, so the oscillations decay and the core settles. The dense core thus forms not by a single smooth contraction but by overshoot followed by a damped ring-down.

Four cross-cut line plots. Along the centre row the simulator draws four profiles on a *logarithmic* scale (about ten octaves, so the curves keep climbing through a collapse rather than saturating): density (cyan), temperature (orange), pressure (yellow), and the gravitational well depth $|\varphi|$ (magenta). The density and pressure curves spike at the centre, overshoot, dip, and ring down; the potential well deepens; the temperature climbs from the compression and dissipation heating.

Three-dimensional companion. A fully three-dimensional version, `selfgrav_collapse_3d`, runs the identical model on a 100^3 grid with the seven-point Poisson solve and the $1/r^2$ Newtonian force, rendering a movable z-slice with the same four log-scale cross-cuts. It is the simulator in which the genuine $\Gamma_{\rm crit} = 4/3$ threshold lives: with the pressureless dust sphere it free-falls, forms a dense core, bounces, and rings down for $\gamma > 1/3$, while $\gamma < 1/3$ gives runaway — the behaviour the 2D filament cannot show.

What halts the collapse — and what dissipation does not do. It is tempting to suppose that, with no radiative cooling, the irreversible heating $\nu|\nabla u|^2$ traps enough energy to halt the collapse for any γ. This is *not* so in general, and the distinction is instructive. What sets whether the collapse halts is the adiabatic index against the threshold $\Gamma_{\rm crit} = 2 - 2/D$ of the

preceding paragraph: a *super-critical* gas ($\Gamma > \Gamma_{\rm crit}$) builds pressure faster than gravity demands and settles into a pressure-supported core, while a *sub-critical* gas ($\Gamma < \Gamma_{\rm crit}$) does not. The dissipation $\nu|\nabla u|^2$ supplies the *damping* that lets the bounce ring down to equilibrium and the entropy production of the 2nd Law — but it does not by itself lift a sub-critical gas to stability. Energy conservation explains why: with no cooling the dissipated infall energy gives an internal energy $U \sim |W| \sim GM^2/R \propto R^{-1}$, which is *exactly* the marginal $\Gamma_{\rm crit}$ scaling — dissipation drives the system *onto* the critical knife-edge, not above it.

The consequence is dimensional. In *this 2D* run $\Gamma_{\rm crit} = 1$, so *every* $\gamma > 0$ is super-critical and the cloud halts at a core — which is why the 2D simulator always forms a stable core. In *3D*, $\Gamma_{\rm crit} = 4/3$: for $\gamma < 1/3$ the gas is sub-critical and *collapses anyway*, dissipation heating notwithstanding (the 3D simulator's readout flips to runaway there). Physically this is correct: the protostellar first core halts because the opaque, adiabatic gas has $\Gamma > 4/3$, not because of turbulent dissipation; and real collapse resumes — toward a star, and ultimately, with General Relativity, a *black hole* — precisely when endothermic microphysics (molecular dissociation, ionization) drags the effective Γ back below 4/3, the same heat now absorbed by the phase change rather than building pressure. What this Newtonian model cannot produce is the black hole itself: a horizon requires General Relativity (a Schwarzschild radius, the Chandrasekhar/TOV mass limits), absent here. The model thus marks cleanly where this level of the hierarchy ends and General Relativity must take over.

Chapter 22

ThermoDynamics of Big Bang and Big Crunch

The Big Bang is the simplest thermodynamic event in the universe. A hot, dense, expanding fluid cools. The pressure-work term $-p\,\nabla\cdot u$ of §2.3 is positive, the expansion is everywhere, and the cooling is universal.

22.1 Expansion Cools, at Cosmic Scale

The thermodynamic content of the Big Bang follows directly from the two primitives of Chapter 2, applied to the self-gravitating compressible fluid of Chapter 21. The early universe is a hot dense plasma in which every comoving fluid element has positive divergence: $\nabla\cdot u > 0$ everywhere, at every time. By the expansion-cools primitive (§2.3), the internal-energy equation gives

$$\dot{e} \;=\; -\,p\,\nabla\cdot u + \mu_{\mathrm{mom}}|\nabla u|^2 \;<\; 0$$

in regions where the pressure work dominates the dissipation, i.e. where the bulk expansion of the universe is the dominant dynamical process. Internal energy decreases, temperature drops, and the universe cools. There is no fine-tuning in this; it is the same $-p\,\nabla\cdot u$ term that cools the gas in the left chamber of Joule's experiment (Chapter 7), now applied to a gas whose container is the universe itself.

The cooling timeline is fixed once the equation of state and the initial conditions are. The radiation-dominated era ($T \gg 10^4\,\mathrm{K}$, $\nabla\cdot u \propto t^{-1}$) cools

as $T \propto a^{-1}$ with $a \propto t^{1/2}$; the matter-dominated era cools as $T \propto a^{-2}$ with $a \propto t^{2/3}$; and the present-day cosmic microwave background sits at $T \approx 2.7\,\mathrm{K}$, the remnant of the radiation field after ~ 13.8 billion years of expansion-cooling. No part of this is statistical; all of it is the expansion-cools primitive applied at cosmic scale and integrated over cosmic time.

22.2 The CMB as Blackbody Cutoff Signature

The most direct experimental signature of the Big Bang — the cosmic microwave background — is the radiative half of the story. At the recombination epoch ($z \approx 1100$, $T \approx 3000\,\mathrm{K}$) the electrons and protons combined into neutral hydrogen, the universe became transparent, and the radiation field decoupled from the matter field. From that moment on, the radiation field expanded with the universe at fixed photon number, with the spectrum red-shifted as $\omega \to \omega/a$ and the temperature dropping as $T \to T/a$. The Planck spectrum maintains its functional form under this transformation — a remarkable feature of the blackbody spectrum, and a non-trivial test of the with-Dynamics reading of Chapter 25: the cutoff at $\omega \sim k_{\mathrm{B}}T/\hbar$ rescales together with T, so a Planck distribution at $T_{\mathrm{rec}} = 3000\,\mathrm{K}$ at recombination becomes a Planck distribution at $T_{\mathrm{today}} = 2.7\,\mathrm{K}$ today, observed as the CMB.

The high-frequency tail of the early radiation field — the part that, before recombination, was being thermalised by the free-electron bath at the blackbody cutoff — has its present-day counterpart in the small fraction of the CMB energy that even now continues to be absorbed and re-emitted by intervening matter, with a cumulative entropy production whose continuum-mechanics realisation is the cosmological dissipation $D = \int\int \mu_{\mathrm{mom}} |\nabla u|^2 \, dx\, dt$ of the gravity-coupled RealNSE flow.

22.3 The Joule Pattern at Cosmic Scale

The three-stage Joule pattern that organises this volume applies to the Big Bang at cosmic scale:

1. **Internal energy reservoir.** The initial hot dense plasma holds an enormous internal energy density. This is the cosmic analogue of the high-pressure gas in Joule's left chamber.

2. **Conversion to bulk kinetic energy.** The Hubble expansion converts internal energy into the bulk kinetic energy of the comoving fluid through the pressure-work term, exactly as in Joule's free expansion — only with the geometry of an isotropic three-dimensional expansion rather than a channel between two chambers.

3. **Dissipation into the bath.** The bulk kinetic energy is in turn dissipated, at every scale at which structure forms, by the turbulent and shock-front dynamics of the gravity-coupled RealNSE flow. This dissipation is what allowed galaxies, stars, and planets to condense out of an initially smooth distribution — the D-generating processes of the previous chapter, played out in the cosmological setting.

The asymmetry between the two halves of the Joule experiment is here asymmetry between cosmic expansion (bulk, smooth, universal) and structure formation (local, dissipative, gravitationally-driven). The universe as a whole is in the expansion-cools half of the cycle. The local condensations — galaxy clusters, galaxies, stars, planets — are sites where the structure-formation half has run its D-generating dynamics far enough to produce stable, locally bound configurations of the kind discussed in Chapter 21.

22.4 Big Crunch: Compression Warms, at Cosmic Scale

The hypothetical *Big Crunch*, in which a closed universe whose present expansion is eventually reversed by its own gravity collapses back to a hot dense state, is the cosmic-scale realisation of the compression-warms half of the primitive of §2.3. The pressure-work term that drives the universal cooling in the expansion-dominated era,

$$-\,p\,\nabla\cdot u,$$

reverses sign once $\nabla\cdot u$ does. In a contracting universe the pressure work is positive, internal energy is increasing, and the universe *warms*. The same mechanism that delivered the present-day 2.7 K CMB by 13.8 billion years of expansion-cooling would, run in reverse over the symmetric timescale, deliver an arbitrarily hot final state by the same number of e-foldings of compression-warming.

Big Bang and Big Crunch are therefore not two different physical mechanisms but the two signs of the same one — the cosmic-scale $-p\,\nabla\cdot u$ term, with $\nabla\cdot u > 0$ in the expansion era and $\nabla\cdot u < 0$ in the contraction era. The companion `cosmology_2d_cpu.html` simulator can be run in either mode by choice of initial conditions: with hot dense central patch and zero initial velocity (Big Bang); or with initially cool dispersed matter assembled under its own gravity into a single collapsing mass (Big Crunch). The diagnostics readout shows the expansion-cooling in the first case and the compression-warming in the second.

A point worth noting: the irreversible dissipation $D \geq 0$ does *not* reverse sign with the expansion. Even in a contracting universe, the local turbulent and shock-front processes that generate D continue to convert organised kinetic energy into heat. The Big Crunch, in the with-Dynamics framework, is not a time-reversal of the Big Bang — it is the Big Bang's structural mirror in the bulk pressure-work term, accompanied by the same unidirectional irreversibility of D in the small-scale dynamics. What classical state-function thermodynamics would describe by a "maximum entropy at the moment of maximum expansion" the with-Dynamics framework describes by: the integral $\int D\,dt$ over the whole history continues to grow, even as the bulk pressure work reverses direction.

22.5 Is the Big Bang/Crunch Inherently 3D?

The expansion-cools / compression-warms mechanism itself is *dimension-free*. The pressure-work term $-p\,\nabla\cdot u$ is a continuum-mechanics object that lives in any dimension d, with the divergence simply summing more terms: $\nabla\cdot u = \sum_{i=1}^{d} \partial u_i/\partial x_i$. A 1D Big Bang (a slab expanding along one axis), a 2D Big Bang (a sheet expanding radially in a plane — what `cosmology_2d_cpu.html` simulates), and the 3D Big Bang all obey the same primitive. The cumulative dissipation $D = \int\int \mu_{\text{mom}} |\nabla u|^2\,dx\,dt$ is also dimension-free.

But the *specific structure* of our observed universe is 3D-specific:

- **Gravity.** Newton's law in d dimensions falls off as $1/r^{d-2}$. In $d = 1$ gravity grows linearly with distance (collapse is inevitable, no structure formation regime); in $d = 2$ it is logarithmic (marginal); only in $d = 3$ does it fall off fast enough to give the Jeans criterion that al-

lows the bulk expansion to coexist with local condensation into bound structures.

- **Stable bound orbits.** Bertrand's theorem singles out $d = 3$ as the dimension in which closed bound orbits exist under an inverse-square central force. 2D galaxies and stars would form qualitatively differently; in $d \geq 4$ they would not form stable bound orbits at all.

- **Radiation density of states.** Photon modes per unit frequency go as ω^{d-1}. The CMB spectrum is a Planck distribution with ω^3 density of states; in 2D the corresponding spectrum would have ω^1 scaling, with a different Stefan–Boltzmann constant and a different expansion-cooling trajectory for the radiation temperature.

- **Jeans criterion.** The condition for gravitational collapse depends on dimension through both the gravitational coupling and the pressure response. Only in $d = 3$ does the classical Jeans length come out at the scale that organises structure formation as we observe it.

The 2D simulator `cosmology_2d_cpu.html` is therefore a qualitative illustration: it captures the bulk pressure-work term faithfully and demonstrates expansion-cooling and compression-warming, but does not reproduce the $d = 3$-specific structure-formation regime.

22.6 Computational Big Bang and Big Crunch

For quantitatively faithful Big Bang / Big Crunch simulations the 3D companions are required. The repository ships two WebGPU-accelerated 3D self-gravitating RealNSE simulators:

- `cosmology_3d_gpu_100.html` — $100 \times 100 \times 100$ grid (10^6 cells), suitable for desktop GPUs.

- `cosmology_3d_gpu_200.html` — $200 \times 200 \times 200$ grid (8×10^6 cells), suitable for high-end GPUs.

Both implement the same compressible-flow RealNSE apparatus coupled to a Poisson-solved gravitational potential, with sliders for the central patch density and temperature (Big Bang initial condition), for the gravitational

coupling G, and for the EOS parameter γ that sets the pressure response. The reader can observe the expansion-cools phase, the formation of localised gravitational condensations, the D-generating turbulent and shock dynamics that allow those condensations to virialise into stable bound configurations, and — by choice of initial conditions and G — the compression-warming of a Big-Crunch-like trajectory.

The Cosmology chapter (Ch. 21) develops the self-gravitating RealNSE apparatus and uses the simpler `cosmology_2d_cpu.html` as its working example. The full 3D companions are referenced there and used here for the explicitly cosmological Big Bang / Big Crunch initial conditions of the present chapter.

The cosmological extension of the framework therefore requires no new physics beyond what Chapters 2, 25, and 21 already establish. The Big Bang, read in the with-Dynamics framework, is the universe as a single instance of the same compressible-flow apparatus that organises Joule, refrigerators, heat engines, chemical bonding, and blackbody radiation. The 2nd Law of the universe is the inevitability of its dissipation D, and the cosmological expansion-cools is the inevitability of its pressure-work term.

Chapter 23

RealNSE with Chemical Reactions

To every action there is always opposed an equal reaction: or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts. (Newton's 3rd Law)

President Bush gave his first-ever presidential radio address in both English and Spanish. Reaction was mixed, however, as people were trying to figure out which one was which. (Dennis Miller)

23.1 Reactive Euler Equations

We now generalize the Euler equations (9.1) to a *mixture* of N gas species, undergoing M chemical reactions: We seek the *species densities* $\rho_1, ..., \rho_N$, the *mixture momentum* $m = \rho u$ with $u = (u_1, u_2, u_3)$ the *mixture velocity* and $\rho = \sum_i \rho_i$ the *mixture density*, and the *mixture internal energy* e satisfying

$$\begin{array}{rcll} \dot{\rho}_i + \nabla \cdot (\rho_i u) & = & s_i, & \text{in } Q, \quad i = 1, ..., N, \\ \dot{m} + \nabla \cdot (mu) + \nabla p & = & f & \text{in } Q, \\ \dot{e} + \nabla \cdot (eu) + p\nabla \cdot u & = & \sum_i s_i r_i & \text{in } Q, \\ u \cdot n & = & 0 & \text{on } \Gamma \times I, \\ \hat{u}(\cdot, 0) & = & \hat{u}^0 & \text{in } \Omega, \end{array} \tag{23.1}$$

where s_i is the net reaction rate of species i from all the reactions, and we assume the following constitutive relations

$$e = \sum_{i=1}^{N} \rho_i r_i + \rho\tau$$

with τ the *mixture temperature* and r_i the *heat of formation* of species i, together with a *mixture pressure* p given as the sum of the partial pressures from the different species:

$$p = C\tau \sum_i \frac{\rho_i}{W_i}.$$

where W_i is the molecular weight of species i and C a positive constant. Evidently, we assume here that the species velocities are all equal to the mixture velocity and that all species take on the mixture temperature.

23.2 Reaction Rates

We assume that the reaction rate s_i of species i is given by an *Arrhenius law* of the form

$$s_i = W_i \sum_{k=1}^{M} (\nu''_{i,k} - \nu'_{i,k}) B_k T^{\alpha_k} e^{-\frac{E_{\alpha,k}}{RT}} \prod_{j=1}^{N} (\frac{X_j p}{RT})^{\nu'_{j,k}},$$

where the ν are stochiometric coefficients, R is the gas cosntant, the B_k are reaction constants, $E_{\alpha,k}$ is an activation energy, and

$$X_i = \frac{\frac{\rho_i}{\rho W_i}}{\sum_j \frac{\rho_j}{\rho W_j}}$$

is the mole fraction of species i.

23.3 The 2nd Law for Reactive Euler

The new 2nd Law takes the form:

$$\dot{K} = W - D, \tag{23.2}$$

$$\dot{E} = -W + D + \sum_i s_i r_i, \tag{23.3}$$

where $D \geq 0$ is the turbulent dissipation, and

$$k = \frac{\rho|u|^2}{2}, \quad K = \int_\Omega k\,dx, \quad E = \int_\Omega e\,dx, \quad W = \int_\Omega p\nabla\cdot u\,dx.$$

The 2nd Law shows irreversibility of turbulent mixing and heating, while the chemical reactions as such, are reversible.

23.4 Simulation of Combustion

Chapter 24

ThermoDynamics of Chemical Bonding

> The bond is not a state. It is the trace, on the equilibrium surface, of a process that took the system from one RealQM minimum to another and dissipated the difference into the surrounding bath.
>
> *The really fundamental thing is the interaction of electrons and nuclei, which is governed by Coulomb's law. (Schrödinger, 1926)*

24.1 The Bond as State vs the Bond as Event

The classical thermodynamic description of a chemical bond consists of a few state-function numbers — a bond enthalpy ΔH_{bond}, a bond entropy ΔS_{bond}, a free energy $\Delta G = \Delta H - T\Delta S$ and the equilibrium constant $K = \exp(-\Delta G/RT)$ that decides whether the bond exists at temperature T. The framework is silent on the *process* by which the bond forms. It encodes the released energy as a state-function difference and the multiplicity loss as a state-function difference, with the actual bonding event reduced to a discontinuous jump on the equilibrium surface. There is no mechanism in it for how the energy is released, or where it goes, or why an entropy decreases when two free atoms become one bound molecule.

The view taken in this volume, and consistent with the with-Dynamics framing of Chapter 6, is different. The bond is not a state but an *event*: a process that takes the system from one RealQM minimum (two separated atoms with their own electron territories) to another RealQM minimum (a

bonded molecule with overlapping electron territories between the two nuclei), with the energy gap between the two minima dissipated into the bath through the same molecular-scale Joule pattern that organises the rest of this book. The state-function bookkeeping of classical thermodynamics records the result; the dynamics records the process.

24.2 The Atomic-Scale Energy Gap (RealQM Input)

The energy gap between the two configurations is supplied by Volume VI of this programme, *Real Quantum Mechanics* (RealQM) [1]. In the RealQM formulation, an N-electron system is described by a sum of one-electron wave functions on non-overlapping spatial domains separated by free (Bernoulli) boundaries, with ground states minimising the ordinary Coulomb energy — no configuration space, no antisymmetry postulate.

For the two-atom problem, two minima of the Coulomb energy functional typically exist:

- A *separated* minimum: the two atoms sit at large internuclear distance $R \to \infty$ with their electron territories disjoint and spherically symmetric about each nucleus. The total energy is $E_{\text{sep}} = E_A + E_B$, the sum of the isolated-atom energies.

- A *bonded* minimum: the two atoms sit at the equilibrium bond distance R_{eq} with their electron territories rearranged so as to overlap in the internuclear region, screening the proton–proton repulsion and pulling both nuclei into the shared electron density. The total energy is E_{bond}, below E_{sep} by the *bond dissociation energy*

$$D_e \;=\; E_{\text{sep}} - E_{\text{bond}} \;>\; 0. \tag{24.1}$$

For H_2 the RealQM calculation reproduced in Vol VI gives $D_e \approx 0.17\,\text{Ha}$ at $R_{\text{eq}} \approx 1.4\,\text{au}$, in agreement with the experimental value $4.75\,\text{eV}$. The gap is a deterministic property of the Coulomb energy functional — it requires no statistics, no probability interpretation, no microstate counting. Whatever "bond energy" the classical thermodynamic tables list for H–H, RealQM computes from first principles.

24.3 The Joule Pattern at Molecular Scale

The bonding event, viewed as a dynamical process, follows the same three-stage Joule pattern that organises Chapters 7, 18 and 19:

1. **Internal energy reservoir.** The two separated atoms together hold an electronic configuration energy E_{sep}. This is the RealQM analogue of the high-pressure gas in Joule's left chamber: a stored internal energy waiting to be released.

2. **Conversion to bulk kinetic energy.** As the atoms approach and the electron territories rearrange, the energy gap D_e is released by the electronic system into the *nuclear* degrees of freedom: first as relative translational kinetic energy of the two approaching nuclei, then, once they fall into the bonded minimum, as oscillatory kinetic energy of the newly formed vibrational mode of the diatomic molecule. The energy is not yet heat — it is coherent kinetic motion of a single oscillator. This is the analogue of the bulk jet through the channel in Joule's experiment: internal energy converted into directed bulk flow.

3. **Dissipation into the bath.** The vibrational mode of the newly formed molecule does not stay coherent. It couples to surrounding molecules through collisions, transferring quanta of vibrational energy into translational kinetic energy of the bath in a sequence of irreversible inelastic collisions. After a few mean-free-path times, the coherent vibration has been thermalised: the originally localised energy D_e has become a tiny incremental heating of the surrounding gas. This is the analogue of the turbulent dissipation in the receiving chamber of Joule's experiment: directed bulk kinetic energy degraded into incoherent thermal motion.

The result is the same as the result reported by classical thermodynamics — a heat release of magnitude D_e per bond formed — but the mechanism is now explicit. The energy was not magically transferred from one column of a table to another; it was released by the electronic system, channelled into a single coherent mode of nuclear motion, and then dissipated into the bath by collisional coupling. The Joule pattern — internal energy $\rightarrow$ bulk kinetic energy $\rightarrow$ irreversible dissipation $\rightarrow$ heat — applies at molecular scale exactly as it applies at macroscopic scale.

24.4 Macroscopic Heat Balance and the Reactive RealNSE Apparatus

For a gas in which the bonding event is happening many times per unit volume per unit time, the bookkeeping is the business of the reactive RealNSE apparatus developed in Chapter 23 (Chemistry: RealNSE with Chemical Reactions). Each elementary reaction $\mathrm{A} + \mathrm{B} \to \mathrm{AB}$ contributes:

- A source term in the species equation for AB and matching sink terms for A and B, with rate set by collision frequency and the Arrhenius factor for any activation barrier.
- A heat source in the internal-energy equation equal to D_e per reaction event, contributing to the local temperature rise.
- A momentum redistribution (typically small for a near-equilibrium reaction) reflecting the change in mean molecular weight of the gas.

The viscous dissipation $D = \int\int \mu_{\mathrm{mom}} |\nabla u|^2 \, dx \, dt$ of the bulk flow is now joined by a *reactive dissipation* from the local relaxation of species gradients — the chemical analogue of viscous shear. Together they form the irreversibility of a reacting gas: the 2nd Law in this setting is the statement that both contributions are positive and that their sum equals the rate at which the chemical free energy stored in unreacted species is degraded to heat.

24.5 The Mysteries of Bonding Entropy Dissolved

Several classical puzzles about bonding now have natural resolutions:

- *Why is ΔS_{bond} negative?* In the classical framework this is mysterious: the bonded molecule has fewer "microstates" than two free atoms, so ΔS must be negative, but the framework gives no reason *why* the bonded molecule is the natural endpoint despite the entropic penalty. In the with-Dynamics framework the question dissolves: bond formation is the dissipative dynamics relaxing toward the deeper RealQM

minimum, with the released energy D_e dissipated into the bath. There is no microstate counting. The bond exists because the D_e-deep minimum exists, and the dissipative dynamics finds it.

- *Why is the equilibrium constant $K = \exp(-\Delta G/RT)$?* In the with-Dynamics framework this is the steady-state condition for the reactive RealNSE: at temperature T, the forward reaction rate (driven by D_e) and the backward rate (driven by thermal kinetic energy of bonded pairs sufficient to climb out of the bonded minimum) balance when the ratio of populations is set by the Boltzmann factor evaluated at the bonded-minimum depth. No postulate is needed; the steady state of the reactive flow gives the same answer that the classical state-function bookkeeping was forced to postulate.

- *Where does the bond energy "go"?* Into the bath, by the three-stage Joule pattern: into vibrational kinetic energy of the newly formed molecule, and from there by inelastic collisions into translational kinetic energy of surrounding molecules. The same dissipation $D > 0$ that organises the rest of this book carries the bonding event from RealQM to the macroscopic temperature.

The chemical bond, the most basic non-trivial chemical event, is therefore the cleanest single illustration of why ThermoDynamics needs Dynamics. The classical state-function description captures the bookkeeping; the with-Dynamics description captures the bookkeeping *and* the mechanism, with RealQM supplying the energy gap and the Joule pattern at molecular scale supplying the channel through which the gap is dissipated.

Chapter 25

ThermoDynamics of Blackbody Radiation

A perfectly stable cavity, at rest at temperature T in empty space, fills with electromagnetic radiation of a spectrum that depends only on T. The mystery is not the spectrum — Planck wrote it down in 1900. The mystery is the cutoff: why the high-frequency tail does not diverge.

The whole of the energy of the radiation is degraded into heat at a rate that is finite, never infinite. (Planck, paraphrased)

25.1 The Mystery of the Cutoff

The classical electromagnetic theory of a cavity in equilibrium with its walls assigns equal energy $k_{\mathrm{B}}T$ to every standing-wave mode. The number of modes between frequencies ω and $\omega + d\omega$ grows as $\omega^2 d\omega$, so the predicted spectral energy density grows as $\omega^2 k_{\mathrm{B}}T$ and the total energy $\int_0^\infty \omega^2 k_{\mathrm{B}}T\, d\omega$ diverges — the *ultraviolet catastrophe*. Real cavities of course hold finite energy. The puzzle, in 1900, was where the high-frequency end of the spectrum gets cut off.

Planck's resolution introduced an inseparable quantum of energy $\hbar\omega$ per mode, with the Boltzmann factor $\exp(-\hbar\omega/k_{\mathrm{B}}T)$ suppressing modes for which $\hbar\omega \gg k_{\mathrm{B}}T$. The spectrum becomes

$$u(\omega, T) \;=\; \frac{\hbar\omega^3/\pi^2 c^3}{\exp(\hbar\omega/k_{\mathrm{B}}T) - 1},$$

finite for every T, with a sharp drop at $\omega \sim k_{\mathrm{B}}T/\hbar$ that places the cutoff in the right physical place. The mathematics works. The physical picture — *what happens to the high-frequency radiation that would have been there without the cutoff* — was less clear at the time and is still treated in many textbooks as merely a statistical consequence of mode counting.

25.2 The Cutoff as Dissipation

Read in the with-Dynamics framing of this volume, the Planck cutoff is the radiative analogue of the viscous cutoff in turbulent flow.

In turbulent compressible flow at high Reynolds number, the deterministic continuum equations would describe a flow with structure on all scales down to zero. The flow that is actually realised, in finite resolution, has a cutoff at the viscous scale — the scale below which the molecular viscosity becomes comparable to the inertial forces and the energy that would have cascaded to still smaller scales is instead converted to internal energy. The dissipation $D = \int\int \mu_{\mathrm{mom}} |\nabla u|^2 \, dx \, dt$ of this volume is the cumulative rate of that conversion. Without the cutoff the cascade would run to infinity; with the cutoff it terminates by turning small-scale kinetic energy into heat.

The same pattern governs the radiation field in a cavity. The classical (unquantised) Maxwell theory predicts modes at arbitrarily high frequency, each carrying its equipartition $k_{\mathrm{B}}T$. The field that is actually realised has a cutoff at the quantum scale — the scale at which a single quantum $\hbar\omega$ exceeds the thermal energy $k_{\mathrm{B}}T$ and the corresponding modes can no longer be populated. The high-frequency radiation that would have been there without the cutoff is absorbed by the walls and converted to heat — exactly the radiative counterpart of the small-scale turbulent kinetic energy being absorbed at the viscous scale and converted to heat.

Turbulence (RealNSE)	**Blackbody radiation**
inertial flow at small scales	high-frequency modes
viscous cutoff at $\ell \sim (\nu^3/\epsilon)^{1/4}$	quantum cutoff at $\omega \sim k_{\mathrm{B}}T/\hbar$
small-scale kinetic energy $\to$ heat	above-cutoff radiation $\to$ heat
dissipation $D = \int \mu \lvert\nabla u\rvert^2 \, dx \, dt \geq 0$	absorption rate ≥ 0
2nd Law as inevitability of $D > 0$	2nd Law as inevitability of absorption

The cutoff is what makes both phenomena *finite*. Without it, the cascade

would run unbounded and the field would carry infinite energy. The mechanism is the same in spirit: a structural barrier below which the small-scale degrees of freedom can no longer be supported, and above which the energy that would have populated them is delivered instead to the thermal bath.

25.3 Computational Blackbody

The same computational framing that organises this volume applies. A finite-precision solver of the Maxwell equations on a cavity at temperature T has a natural cutoff set by its spatial resolution: it cannot represent modes with wavelength shorter than its mesh spacing, and the radiation energy that would have lived in those modes is re-deposited, by the discretisation, as a heating source in the bath. The Planck cutoff is, in this reading, not a quantum-mechanical postulate but the cutoff that any finite-resolution representation of the field must impose — realised in Nature by whatever physical process limits the resolution at very short wavelength, and realised on a computer by the mesh.

This view is developed in detail in the author's earlier *Computational Black-Body Radiation* [27]. The present chapter records that the Planck cutoff and the viscous cutoff of turbulent RealNSE are two instances of the same with-Dynamics mechanism: small-scale degrees of freedom that the underlying continuum equations *would* support, and that finite resolution *does not*, deliver their energy as heat. The 2nd Law, in this joint reading, is the inevitability of that delivery — in turbulence, in radiation, and in every other context in which a continuum field is realised at finite resolution.

25.4 The Planck Cutoff as the Real 2nd Law

The connection just stated can be made precise in the language of the **Real 2nd Law** of §1.1. Recall the three load-bearing clauses:

- **(i) Necessary / unavoidable.** In a cavity at temperature T, the high-frequency modes $\omega \gg k_{\mathrm{B}}T/\hbar$ are unstable to absorption by the walls — the available thermal energy per mode falls exponentially with ω, and any quantum deposited in such a mode is rapidly returned to the bath as incoherent thermal motion of the wall material. The destruction of the high-frequency tail is not optional: it is what *must*

happen given the available resolution (the wall's microscopic degrees of freedom) and the energy budget.

- **(ii) Instability of fine-scale structure.** The "fine-scale structure" here is the standing-wave field at very short wavelength — the radiative analogue of small-scale turbulent eddies. As ω increases past $k_{\mathrm{B}}T/\hbar$, the wave profile sustains gradients that the available resolution (the photon-quantum scale $\hbar\omega$) cannot support coherently, and the wave is absorbed.

- **(iii) Irreversibility from precision deficit.** Once the energy that would have populated the high-frequency tail has been deposited as heat in the walls, the precision required to re-emit it back into a coherent high-frequency mode is not available — neither in the analog physical apparatus of the cavity walls, nor in any digital simulator of the same field. Re-population of the suppressed modes would require coordinating the microscopic wall motion to produce coherent above-cutoff radiation, which the precision of the wall material does not support.

The Planck distribution is therefore not just an empirical curve fitted to spectroscopic data, nor a quantum-statistical postulate; it is the spectral signature of the Real 2nd Law applied to a continuum electromagnetic field: the high-frequency suppression is the necessary, unavoidable, irreversible destruction of unstable fine-scale field structure that the available resolution cannot support.

Chapter 26

Transformations of Energy

God has chosen that which is the most simple in hypotheses and the most rich in phenomena. (Leibniz, Discours de métaphysique, VI, 1686)

The sight of day and night, and the months and the revolutions of the years, have created number and have given us conception of time, and the power of inquiring about the nature of the Universe. (Plato in *Timaeus*)

... in a purely mechanical world, the tree could become a shoot and a seed again, the butterfly turn back into a caterpillar, and the old man into a child. No explanation is given by the mechanistic doctrine for the fact that it does not happen. (Ostwald)

... a complete explanation of the arrow requires explaining why the universe started out as it did. It is a problem in cosmology. (Hawking)

26.1 Kinetic Energy and Heat Energy

Thermodynamics is fundamental in a wide range of phenomena from macroscopic to microscopic scales. Thermodynamics essentially concerns the interplay between *kinetic energy* and *heat energy* in a *gas* or *fluid*, where heat energy is a form of small scale kinetic energy. Thermodynamics thus concerns transformations between large scale kinetic energy and small scale kinetic energy in the form of heat. Large scale kinetic energy, or *mechanical energy*, may generate heat energy by *compression* or *turbulent dissipation*.

Heat energy may generate (large scale) kinetic energy by *expansion*, but not through a *reverse* process of turbulent dissipation. Thermodynamics is closely connected to the dynamics of *slightly viscous* and *compressible* gases, since substantial compression and expansion can occur in a gas, but less in fluids (and solids).

26.2 Cosmology and Big Bang

Cosmology is the scientific study of the large scale properties of the *Universe* as a whole, including its origin, evolution and ultimate fate. The *Big Bang* model postulates that 12 to 14 billion years ago, the universe was in a very hot very dense state, which expanded along with galaxies and stars being formed by gravitation from variations in density, see Fig. **??**. It is believed that we can see remnants of the hot dense matter as the now very cold cosmic microwave background radiation, which still pervades the universe as a slightly varying glow across the entire sky, see Fig. 26.1, a discovery made by the Cosmic Background Explorer COBE-telescope, which gave the Nobel Prize in 2006 to John Mather and George Smoot. Cosmology is basically a very (very) large scale application of thermodynamics as an interplay of mass, momentum and energy governed by gravitational and inertial forces and driven by nuclear reactions.

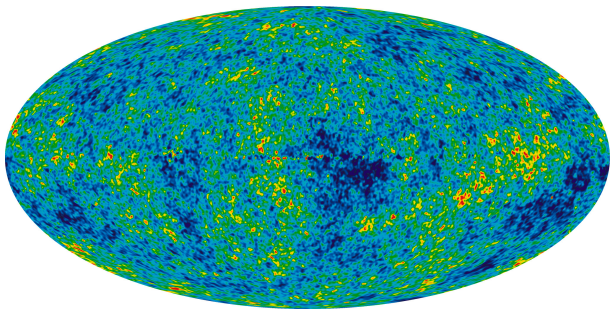


Figure 26.1: Cosmic back-ground radiation according to COBE.

26.3 Astronomy and Nuclear Energy

The *stars* were formed in the early Universe by accretion of mass by gravitational collaps increasing the temperature of a central core and thus igniting the star in a nuclear fusion process with first deuterium and then hydrogen being transformed to helium under intense release of energy (according to Einsteins famous formula $E = mc^2$) increasing the pressure and counterbalancing the gravitational collaps. *Planets* may be formed from heavier atoms and molecules created in nuclear processes in stars, and under favorable conditions give rise to (intelligent) *life* and theories of thermodynamics. Stars of 0.4-10 times the mass of our own Sun expand into red giant very bright stars, when the supply of hydrogen in the core is empty and fusion starts in an outer core. Our own Sun thus eventually will swallow the Earth into a fusion process, but this is estimated to be a couple of billion years away. The thermodynamics of the Sun thus is the basis of both life and death of the Earth.

26.4 Geology and Global Warming

Geology or earth science contains the subfields of geophysics, glaciology, volcanology, climatology and meteorology connecting to the the central issue of the thermodynamics of *global warming*. Computational simulation of global climate change has a very important role to play to find out what actions are beneficial for sustainable development and long-time survival.

26.5 Black Body Radiation and the Sun

A *black body* absorbs incoming white light of all frequencies, from high frequency ultraviolet to low frequency infrared, but only radiates low frequency light (with color depending on the temperature of the body). This is the basis of the gross energy balance of the Earth absorbing white light from the hot Sun and radiating low frequency light into empty space. One can view black body absorption/emission as an (irreversible) thermodynamic process transforming high frequency light energy to low frequency (long wave) light energy [27].

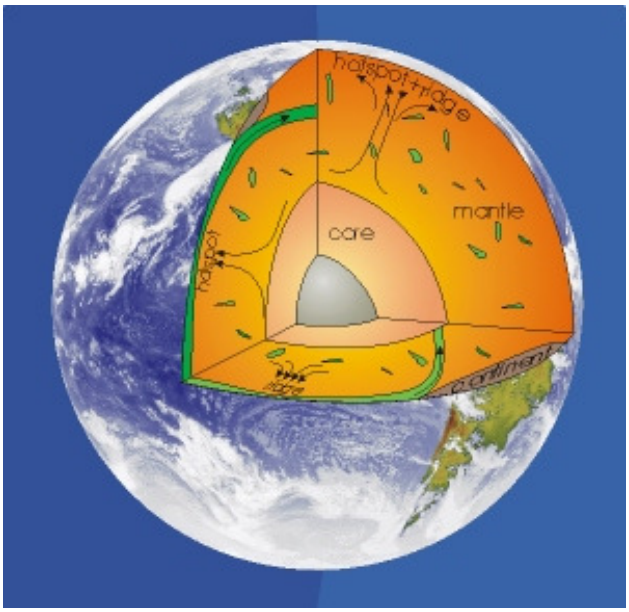


Figure 26.2: The Earth

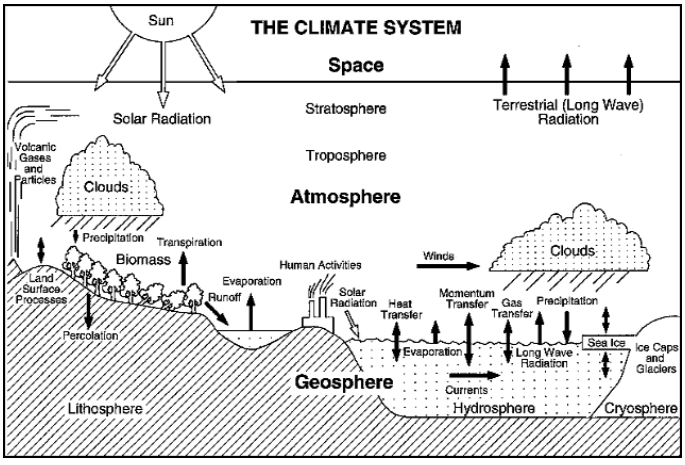


Figure 26.3: Global heat balance.



Figure 26.4: Life in action.

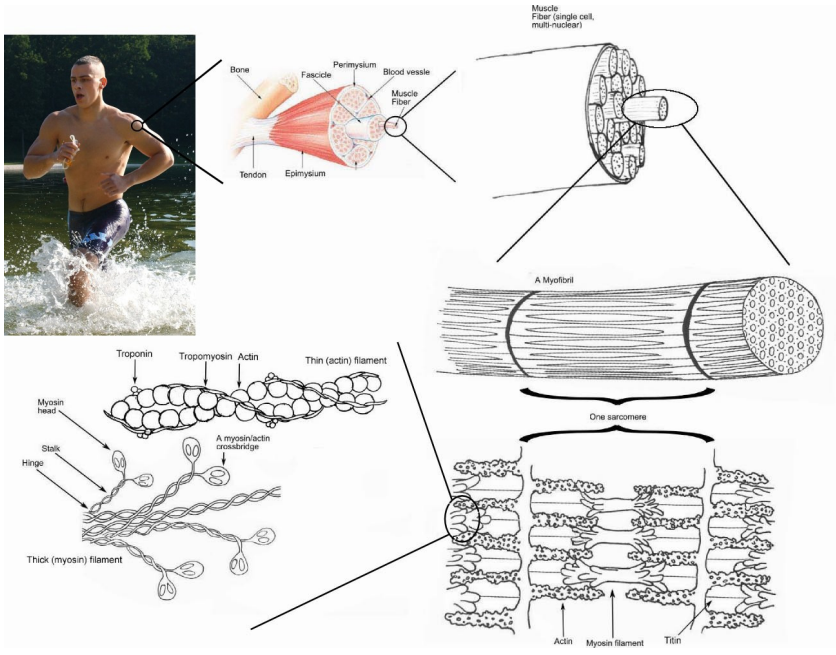


Figure 26.5: Muscle contraction by actin-myosin twitch fueled by ATP.

26.6 Biology and Life

Most forms of biological processes build on conversion of chemical energy into heat and kinetic energy, and thus ultimately on Solar energy converted to chemical energy in the *photo-synthesis*.

Muscle contraction occurs when a muscle fiber generates tension through *actin and myosin cross-bridge cycling* shortening the fiber fueled by *Adanenosine TriPhosphate ATP*. Locomotion in most higher animals is possible only through the repeated contraction of many muscles at the correct times controlled by the central nervous system, the brain and spinal cord. Voluntary muscle contractions are initiated in the brain, while the spinal cord initiates involuntary reflexes.

The emergence of biological life may naively appear to contradict the 2nd Law in its classical form, by creating ordered structures out of disordered, while the death of a living organism when order dissolves into disorder, reflects the classical 2nd Law very well. We shall see that in the new formulation of the 2nd Law not relying on notions of ordered/unordered, there is nothing (in principle) that prevents ordered structures to develop from unordered (for example in Irak), assuming there is some input of energy (oil).

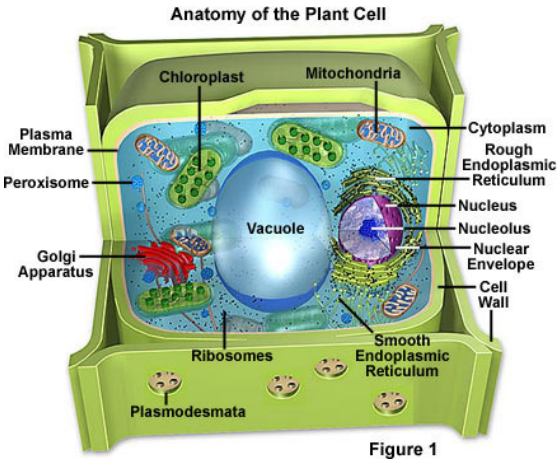


Figure 26.6: Inside a plant cell.

26.7 Molecular Biology and Protein Folding

Biology builds on cell microbiology based on the thermodynamics of chemical reactions of *protein synthesis* from amino acid molecules according to the instructions of *DNA*. Computational thermodynamics of the cell is becoming an increasingly important tool in medicine and drug design.

26.8 Quantum Physics and Chemistry

The thermodynamics of chemical reactions and radiation build on the *quantum mechanics* of electrons, and the thermodynamics of nuclear reactions build on quantum electrodynamics of elementary particles. Computational solution of the equations of quantum mechanics presents a tough challenge, because of the richness of the the wave function solution with in principle three independent space coordinates for each particle (electron) involved. Methods reducing the dimensionality are today routinely used in massive computations in *quantum chemistry*, and could open to new insights of the nature of the quarks of elementary particle physics.

26.9 Energy Generation and Consumption

Our *industrial society* is based on production and consumption of energy in thermodynamic energy conversion processes of the form:

- nuclear/fossile/biological/chemical → heat/kinetic/electric,
- wind/wave kinetic → electric,
- solar radiation → heat/electric.
- electric → heat/kinetic.

In all conversion of energy, minimization of *losses* into useless heat is of prime concern, for which computational thermodynamics opens new possibilities.

26.10 Heat Engines and Industrialization

The industrial society of the 19th century was built on the use of *steam engines*, and the initial motivation to understand thermodynamics came from a need to increase the efficiency of steam engines for conversion of heat energy to useful mechanical energy. A steam engine is a particular form of a *heat*

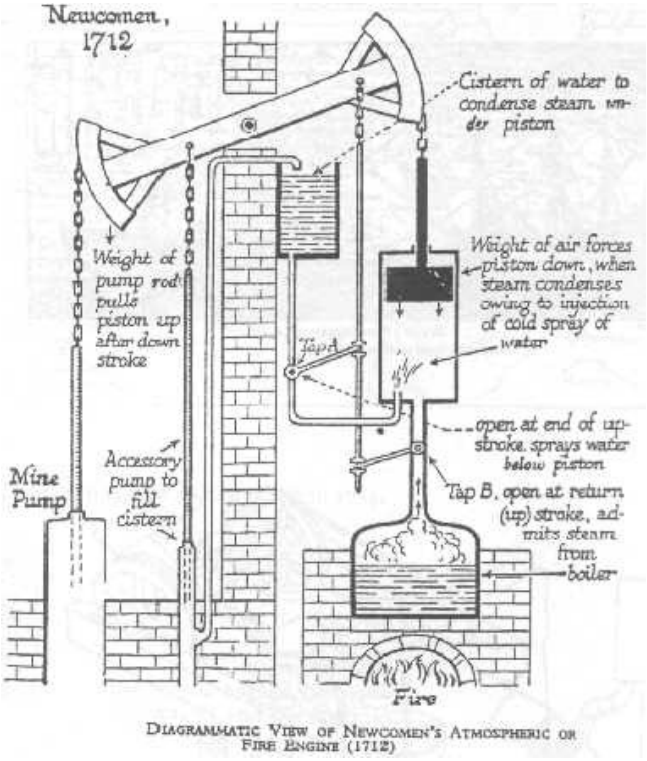


Figure 26.7: Converting heat to kinetic (mechanical) energy in 1712.

engine converting heat to mechanical work in a thermodynamical process. The modern *combustion engine* in a car is a form of heat engine generating kinetic energy from explosive burning of fuel. The efficiency of a heat engine is of prime importance, and can be studied by computational thermodynamics. The necessary *cooling* of a car engine represents a loss of energy which is not transformed into useful mechanical work.

26.11 Heat Pumps and Oil Crisis

The *heat pump* is replacing the increasingly expensive oil for heating of houses. A heat pump converts low temperature heat energy from the earth or the air to higher temperature useful heat energy, by supplying the necessary conversion energy from electricity. The efficiency is today typically around 2/3, that is, for 1/3 unit of electric energy, you get out 1 unit of useful heat energy. You can thus reduce your electrical bill by a factor of 3 by changing from full electrical heating of your house to heating by an efficient heat pump. Of course, it is of interest to seek to increase the efficieny even more. Can we reach 99% efficiency, and if not why? We will address this question below.

26.12 Refrigerators and Urban Civilization

A *refrigerator* is a form of heat pump taking heat from the inside of the refrigerator and delivering it to the exterior at higher temperature. The refrigerator is an absolute necessity of modern urban civilization.

The standard design builds on cycling a fluid alternating between *expansion/evaporation* to gas phase with temperature drop, and *compression/condensation* to liquid phase with temperature increase, with the cold gas in contact with the interior of the refrigerator and the warm liquid phase in contact with the exterior. The cooling process with heat flowing from cold to hot (contrary to its natural tendency to flow from hot to cold) is thus maintained by an (electric) *compressor* consuming energy.

The first design of an *absorption refrigerator*, without compressor and moving parts and thus silent, was invented by the two young Swedish engineering students Carl Munters and Baltzar von Platen during their class in thermodynamics at the Royal Institute of Technology in 1922 (although they slept through the lectures working all night on their invention). This invention together with a vacuum cleaner gave the impetus of the quick expansion of the Electrolux company. The production of the Munters-Platen refrigerator has continued at Motala in Sweden uninterrupted since 1925 with today a total of 10 million units being produced. We hope this book may be useful for young minds of today for new inventions.



Figure 26.8: Inside a refrigerator: The principle of preservation of food.

26.13 Information and Communication

Information theory was created by Shannon [47] in the 1940s borrowing the concept of *entropy* from *thermodynamics*, with the idea of measuring communication channel capacity requirements for different types of information. Random information would then represent information with maximal entropy, in principle requiring maximal capacity (although random information would seem like useless information).

Alternatively, one can view communication of information as a thermodynamic flow with the heat energy representing loss or destruction of information during the communication. More generally, not only creation but also destruction of information represent fundamental aspects of physical and biological processes.

26.14 Economy and Welfare

One can also view an evolving economy as a thermodynamic flow process transforming resources into utilities. In (a free) economy turbulent dissipation could be viewed as a measure of necessary losses during the process, in the form of interest rates, taxes, bribes, thefts, which cannot (fully) be avoided. The flow of the World economy represents a complex process of prime importance to all of us. An economist with (some) understanding of

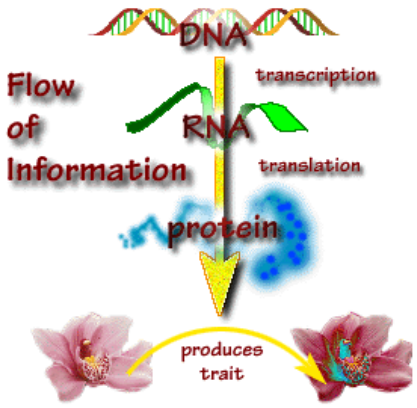


Figure 26.9: Flow of information in a flower.

the process can be raised to fame and fortune.

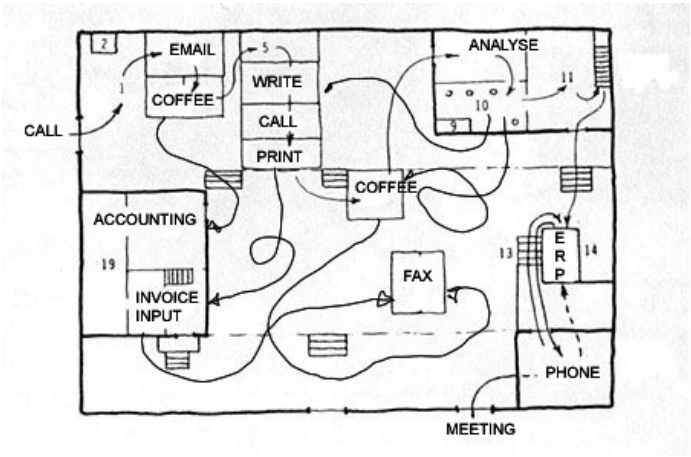


Figure 26.10: Flow of information in an office of the last century.

26.15 Emergence: A New Approach to Physics

Ilya Prigogine received the Nobel Prize in Chemistry in 1977 for his work on *non-equlibrium thermodynamics* proposing a new (statistical) approach to the 2nd Law allowing ordered structures to develop from non-ordered, see Fig.

26.11 and 26.12. Even if Prigogine did not convince the physics community at large, he at least showed the severe short-coming of the classical 2nd Law seemingly preventing order from develping from chaos.

Emergence is the process of complex pattern formation from more basic constituent parts or "particles", that is the formation of order out of chaos. Turbulence shows phenomena of both emergence and featureless chaos, none of which can be understood by considering only a single fluid particle, only by considering the interaction of many. The (new) 2nd Law expresses a collective behavior of many fluid particles to form turbulent flow with an irreversible transfer of large scale kinetic energy to heat energy. Thus thermodynamics naturally connects to the new approach to physics based on emergence, which is now developing (emerging) with input from e.g. 1988 Nobel Laurate Robert Laughlin [29], and which is radically different from the *reductionist* approach of elementary particle physics completely dominating the scene since the birth of quantum mechanics in the beginning of the last century.

Since emergence concerns the interaction of many particles, analytical methods fall short (which lies behind the reductionist obsession with a single particle hopefully allowing analytical mathematical description), while computational methods open new possiblities of simulating emergent and turbulent phenomena, as is shown in this book.

26.16 The Battle between Growth and Decay

We shall see that turbulent flow and thus thermodynamics (the World) can be viewed as a battle between a growth process of *differentiation by advection* and a decay process of *mixing by diffusion* according to:

- *growth: increase difference-differentiate-separate: advection*
- *decay: decreasse difference–uniformize-mix: diffusion.*

The classical 2nd Law captures the diffusion process with mixing and decreasing difference. But it does not capture the advection process allowing increasing difference and separation, which is a severe short-coming. The new 2nd Law captures both processes. In the thermodynamics of Prigogine the two processes are represented by "Unstable Dynamical Systems" and "Dissipative Structures", but Prigogine sticks to "Randomness", see Fig. 26.11.

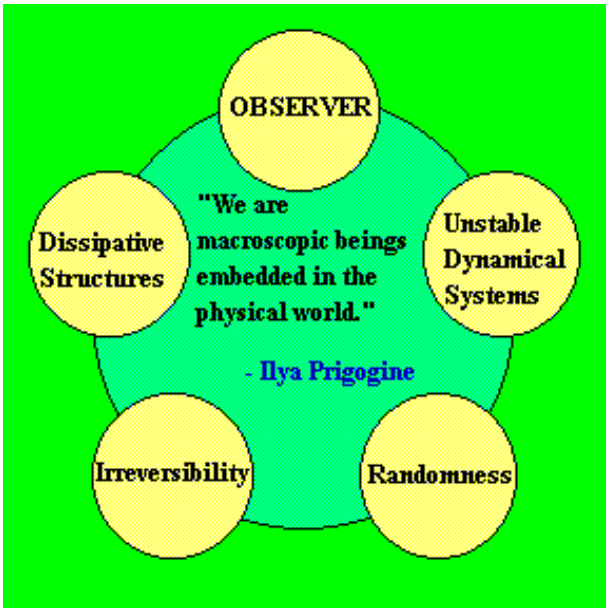


Figure 26.11: The World according to Ilya Prigogine.



Figure 26.12: Ilya Prigogine receiving the Nobel Prize in Chemistry in 1977 from the hands of King Karl XVI Gustav.

26.17 Physics: Flow of Information: Computation

The basic idea of this book is to view thermodynamics as a form of analog computation, which can be simulated by digital computation. Both analog physics and digital computation can be seen as flows of information reflecting the flow of mass, momentum and energy in a physical system and the flow of digits from input to output in the execution of a subroutine of the computational code.

For a computer programmer the information flow is represented in the flow chart of a computer code, which is essential for the understanding of the code. Likewise, a flow chart of the analog computation of a physical process can help the physicist to understanding.

26.18 Thermodynamics as a "Best of Worlds"

The above survey indicates that thermodynamics is a fundamental part of science and technology and thus serves a very important role in our society. Thermodynamics is maybe the scientific field closest to Leibniz' definition of a *Best of Worlds* as a world, which is richest in variation of ordered complex structures or emergence, such as different forms of life and conscience, while being governed by simplest possible principles or laws.

> I say that I am strongly inclined to believe that heat is of the same kind, and that material things which make us hot and feel hot (such as we call by the general name *fire* are a multitude of tiny corpuscles of such and such a shape, and moving at such and such a speed. When they meet our bodies, they penetrate them with their extreme subtlety, and the way they touch our substance as they pass through it is sensed by us as the affection we call "heat". It is pleasant or unpleasant in proportion to the number and speed of the particles as they prick and penetrate it. This penetration is pleasant if it is beneficial to our insensible vital functions, and unpleasant when it causes too much of a separation and dissolution of our substance. (Galieo in *The Assayer*, 1623)

Chapter 27

Information

> The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. (Shannon in [47], 1948)

> You should call it entropy for two reasons; in the first place your uncertainty function has been used in statistical mechanics under that name, so it already has a name. In the second place, and more important, no one knows what entropy really is, so in a debate you will always have an advantage. (von Neumann to Shannon)

27.1 Introduction

Information theory was created by Shannon [47] in the 1940s borrowing the concept of entropy from thermodynamics. We now explore if the new foundation of thermodynamics presented may open to a new approach to information theory. We thus consider a model of *information flow* in the form of the Euler equations for a variable density incompressible fluid flow with the distribution in space of *mass density* representing information like a message, image or movie, which is being transported or *communicated* from one location to another by the fluid flow. The key question is what aspects of the information which can be communicated to what precision. We expect to see fine details of an image being distorted or destroyed by the communication, while gross features may remain unaltered. We introduce a measure of *irrecoverable loss of information*, which couples to increase of entropy in the classical setting, which corresponds to information which cannot be retrieved.

We thus distinguish between *recoverable information*, which can be recovered from a distorted image by some form of image processing, and *irrecoverable information* which cannot be recovered by image processing.

27.2 Entropy vs Loss of Information

In Shannon's information theory the notion of entropy is used to measure the randomness or complexity of the information to be communicated, with random complex information requiring more work to communicate than organized simple information. Shannon's entropy thus is a measure of the work to communicate certain information, assuming that the communication itself does not distort the information. In this setting information is coded into a digital representation, which is communicated without loss of any digits and then is decoded after reception. The main question then concerns the number of digits required to represent the information, which directly couples to the work of accurate communication of the digits. In Shannon's theory the digital coding in general introduces a certain loss of information, with more loss for complex information, while the communication of the digital representation is assumed to be without loss. Shannon's entropy thus may be viewed as a measure of loss of information arising from the digital representation but not from the communication.

Our notion of entropy is a bit different and measures the *total loss of information* taking into account not only the loss from digital representation, but also from the communication process itself. In our setting, the digital representation of the information as a mass density, is a finite element approximation on a mesh of certain mesh size h, which can be viewed as generalized Fourier or wavelet representation, and the communication is realized through a finite element solution of the Euler equations. The mesh size h directly couples to the work required for the communication.

The main aspects influencing the total loss of information, and thus the error in the communication, relate to (a) the complexity of the information and (b) the complexity of the communication process, where (b) can be viewed as a measure of the distortion from the communication process. We identify the following basic cases:

(i) simple information and simple communication,

(ii) complex information and simple communication,

(iii) simple information and complex communication,

(iv) complex information and complex communication.

We will see that in (i) the loss of information typically is small, in case (ii) and (iii) it is of medium size, and in case (iv) it is large. We will see that (i) requires little work (h is not small), (ii) and (iii) require medium work and (iv) large work (h small), for the same accuracy. We shall see that if the communication is partly *turbulent*, which is the generic case, then the loss of information cannot be made arbitrarily small by decreasing h, relecting that the turbulent dissipation has a non-zero limit as h tends to zero.

27.3 A Model Problem

As a simple model of communication, we consider the following 1d transport equation:

$$\begin{aligned} \dot\rho + \rho' &= 0 \quad \text{for } 0 < x < 1\,, 0 < t \le 2, \\ \rho(x,0) &= 0 \quad \text{for } 0 < x < 1 \end{aligned} \tag{27.1}$$

where $\dot\rho = \frac{\partial\rho}{\partial t}$, $\rho' = \frac{\partial\rho}{\partial x}$, and $\rho(0,t) = \rho^-(t)$ with $0 \le t \le 1$ represents an input signal and $\rho^+(t) = \rho(1,t)$ with $1 \le t \le 2$ the corresponding output signal. Since the solution to the convection equation is given by $\rho(x,t) = \rho(0,t-x)$ for $t \ge x$, we simply have $\rho^+(t) = \rho(1,t) = \rho(0,t-1) = \rho^-(t-1)$ for $1 \le t \le 2$, that is, the output signal is equal to the input signal with a unit time delay.

Let now ρ_h be a finite element computational solution of the convection problem representing a realization of the communication from input to output with a certain imprecision due to the numerics. More precisely, let $\rho_h \in V_h$ be a cG(1)-solution determined by

$$((\dot\rho_h + \rho_h', v)) + (\rho(0,\cdot) - \rho^-, v(0,\cdot)) + ((h\rho_h', v')) = 0, \quad \text{for all } v \in V_h, \tag{27.2}$$

where V_h is the set of continuous piecewise linear functions on a space-time mesh of $Q = [0,1] \times [0,2]$ with mesh size h, $((v,w)) = \int_Q vw\,dxdt$ and $(v,w) = \int_0^1 vw\,dx$. Here, the first two terms come out of Galerkin's cG(1)-method and boil down to a centered difference scheme on a regular mesh and the the third term is a stabilizing artifical viscosity term. More generally, we may apply the RealNSE method with a residual-based artificial viscosity.

We observe observe the reverse communication process taking ρ^+ to ρ^-, would read: Find $\rho \in V_h$ such that

$$((v, -\dot{\rho}_h - \rho_h')) + (\rho(1,\cdot) - \rho^+, v(1,\cdot)) + ((h\rho_h', v')) = 0, \quad \text{for all } v \in V_h,$$

where $(v, w) = \int_1^2 vw\, dx$. This means that neglecting the viscous terms, we could retrieve ρ_h^- exactly from ρ_h^+ by time stepping backwards in time. In contrast, we cannot "undo" the effect of the viscous term in a process from ρ_h^- to ρ_h^+, since time stepping backwards with the artificial viscosity present would just add more diffusion to the process.

To study the resulting output error in the communication from ρ^- to ρ^+, let $\psi(t)$ with $1 \le t \le 2$ be a given weight function, and let $\varphi(x,t)$ solve the dual problem

$$\begin{aligned} -\dot{\varphi} - \varphi' &= 0 \quad \text{for } 0 < x < 1\,, 0 < t \le 2, \\ \varphi(x,2) &= 0 \quad \text{for } 0 < x < 1, \\ \varphi(1,t) &= \psi(t) \quad \text{for } 1 < t < 2. \end{aligned} \tag{27.3}$$

Multiplying the dual equation by $e = \rho - \rho_h$, integrating by parts and using (27.2) assuming $\rho_h(0,\cdot) = \rho^-$, we obtain the following error representation:

$$E_h \equiv \int_1^2 (\rho^+ - \rho_h(1,\cdot))\psi dt = \int_1^2 e(1,t)\psi(t)\, dt = ((R(\rho_h), \varphi - \varphi_h)) + ((h\rho_h', \varphi_h')),$$

where $R(v) = \dot{v} + v'$, and $\varphi_h \in V_h$ is a finite element approximation φ.

It is now natural to define $I = (R(\rho_h), \varphi - \varphi_h))$ to represent a *recoverable* error, and $II = ((h\rho_h', \varphi_h'))$ an *irrecoverable* error. In particular, choosing φ_h to be a cG(1)-approximation, we can make the I vanish, which may be viewed as retrieving ρ^- by backwards time stepping. With this motivation we now focus on the irrecoverable part II of the error, neglecting here the recoverable part I, and thus consider the error representation

$$E_h \approx ((h\rho_h', \varphi_h')). \tag{27.4}$$

We can now rephraze the above basic cases as follows: (i) ρ_h and φ_h smooth, (ii) ρ_h smooth and φ_h non-smooth , (iii) ρ_h non-smooth and φ_h smooth, and (iv) ρ_h non-smooth and φ_h non-smooth. We will then typically have in case (i): $E_h \approx h$, in cases (ii) and (iii): $E_h \approx \sqrt{h}$ and in case (iv): $E_h \approx 1$. We may refer to (i) and (ii) as laminar cases and (iii) and (iv) as

turbulent cases. With a smooth φ, we measure the output error in a weak mean value sense, and with a non-smooth φ_h we measure the error in a more pointwise sense. We see that in the case of non-smooth ρ_h, only mean-value outputs can be meaningful.

Viewing entropy as a measure of loss of information, we may say that (i) has small entropy production or little loss of information, (ii) and (iii) has medium entropy production and loss of information while (iv) has massive entropy production and loss of information.

27.4 Simulations in Model Problem

27.5 Euler's Equations

We now generalize to the Euler equations for a variable density incompressible fluid over a time interval $I = (0, T)$, in a cylinder $\Omega = \{x = (x_1, x_2, x_3) = (x_1, \bar{x}) : 0 \leq x_1 \leq L,\ |\bar{x}| \leq R\}$ of length L and radius R, with boundary Γ with outward unit normal n, take the form: Find the *density* $\rho(x,t)$, the *flow velocity* $u(x,t)$ and *pressure* $p(x,t)$, such that

$$\begin{aligned}
\dot{\rho} + \nabla\cdot(\rho u) &= 0 \quad \text{in } \Omega\times I,\\
\dot{u} + u\cdot\nabla u + \frac{\nabla p}{\rho} &= 0 \quad \text{in } \Omega\times I,\\
\nabla\cdot u &= 0 \quad \text{in } \Omega\times I,\\
u\cdot n &= u^- \quad \text{on } \Gamma^-\times I,\\
\rho &= \rho^- \quad \text{on } \Gamma^-\times I,\\
\rho(\cdot,0) = \rho^0,\ u(\cdot,0) &= u^0 \quad \text{in } \Omega,
\end{aligned} \tag{27.5}$$

where the fluid enters Ω through $\Gamma^- = \{x \in \Gamma : x_1 = 0\}$ with a given inflow velocity $u\cdot n = g < 0$ and leaves through $\Gamma^+ = \{x \in \Gamma : x_1 = L\}$ with $u\cdot n > 0$, and $u\cdot n = 0$ on the remaining (inpenetrable) part of the boundary. Here $\rho(0,\cdot,\cdot) = \rho^-$ represents a given input signal, or rather movie, consisting of a sequence of time dependent 2d images $\rho^-(\cdot,t)$ with $t \in I$, defined on the cylinder cross section $\omega = \{\bar{x} : |\bar{x}| \leq R\}$, which is transported by the flow through the channel to an output movie $\rho^+(\cdot,t) = \rho(L,\cdot,t)$ for t in a relevant interval. The communication thus concerns the transformation of ρ^- to ρ^+. The basic problem concerns the *distortion* $d = \rho^+ - \rho^-$, which measures the *accuracy* of the communication.

It is natural to assume that $\rho = \bar{\rho} + \tilde{\rho}$ where $\bar{\rho}$ is a constant given "background" and $\tilde{\rho}$ contains the information of interest. Replacing ρ by $\bar{\rho}$ the second equation (momentum equation) takes the form $\dot{u} + u \cdot \nabla u + \nabla \bar{p} = 0$ with the modified pressure $\bar{p} = p/\bar{\rho}$. This decouples ρ from the last two equations, which determine $(u, \bar{p})$ as a solution to the (unit density) incompressible Euler equations. In this case, ρ thus can be determined from the first equation alone, which with $(u, \bar{p})$ given is a linear convection equation. For simplicity we consider this decoupled situation below, in which the image itself is not influencing the flow, and thus is transported "passively".

The solution $(u, \bar{p})$ to the (unit density) incompressible Euler equations in general is (partly) *turbulent* with a complex flow velocity u, and thus the image $\rho(\cdot, t)$ in general is convected in a (partly) complex convection field with a corresponding distortion. For short time also a *laminar* smooth flow velocity u is possible. Thus the image can be convected by a laminar or turbulent flow velocity, with vastly different distortion effects.

27.6 Finite Precision Euler Solution: RealNSE

We solve the Euler equations (27.5) using as above a stabilized cG(1)-method on a finite element mesh in space-time with mesh size $h(x, t)$, and denote the finite element solution by $\hat{u}_h = (\rho_h, u_h, p_h)$ with output ρ_h^+. We will compare with a non-distorted output $\rho^+ = \rho^-$ convected by a constant velocity field $\bar{u}(x, t) = (\bar{u}_1, 0, 0)$, where $\bar{u}_1$ is a representative inflow velocity. We will thus as a measure of the distortion, use the mean value

$$E_h = \int_\omega (\rho^- - \rho_h^+)\psi \, d\bar{x} \, dt. \tag{27.6}$$

where $\psi(\bar{x}, t)$ is a weight function.

27.7 Error Estimation by Duality

As in the model problem, we may assume that (see [23] for details)

$$E_h \approx ((h\nabla\rho_h, \nabla\varphi_h)),$$

where $((v,w)) = \int_0^T \int_\Omega vw\,dx\,dt$, and φ_h is an approximate solution of the dual problem

$$-\dot{\varphi} - u_h \cdot \nabla\varphi = 0 \quad \text{in } \Omega \times I,$$

$$\varphi(L,\cdot,\cdot) = \psi \quad \text{in } \omega \times I,$$

,

$$\varphi(\cdot,T) = 0 \quad \text{in } \Omega,$$

,

27.8 Laminar and Turbulent Communication

As in the model problem, E_h depends both on $\nabla\rho_h$ and $\nabla\varphi_h$. If u is turbulent, then E_h may be small if ψ is smooth with large support, corresponding to a gross feature of interest. If u is laminar, then E_h may be small also for a small feature of interest.

27.9 The 2nd Law

The basic energy balance for ρ_h reflecting the structure of the stabilized cG(1) method, reads:

$$\int_{I^+}\int_\omega \rho_h(L,\bar{x},t)^2\,dx\,dt + D(\rho_h) = \int_{I^-}\int_\omega \rho_h(0,\bar{x},t)^2\,dx\,dt$$

where

$$D(u_h;\rho_h) = \int_0^T \int_\Omega h|\nabla\rho_h|^2\,dx\,dt,$$

and I^- and I^+ are relevant time intervals for input and output, respectively. As above, $D(u_h;\rho_h)$ represents an *irrecoverable loss of information*. Reversing the process letting ρ_h convect backwards in time starting from $\rho_h(\cdot,T)$, we have an analogous energy balance with a new dissipation term of positive sign, showing that the initial data $\rho(\cdot,0)$ cannot be fully recovered.

27.10 Dissipation of Information

As a rough measure of the amount of irrecoverable information we may use the *total dissipation of information*:

$$D_h = \int_0^T \int_\Omega h|\nabla \rho_h|^2 \, dx \, dt.$$

27.11 Simulations

We consider communication in cylindrical channels of different length. If the channel is short enough, then the flow may stay laminar all along the channel and the image suffer little distortion, while if the flow gets turbulent, then the image may get more distorted. Because of boundary layer effects the distortion will usually be bigger close to the boundary.

We also study the effects of different obstructions in the channel perturbing the flow, modeling a communication system with bottle-necks.



Figure 27.1: Flow of information at New York Stock Exchange

Chapter 28

Traffic Flow

> I think the idea of getting out of a traffic jam and getting out of work each week and going and doing all this stuff would be really exhausting.(Paul McCarthy)

28.1 A Model of Traffic Flow

We return to the simple traffic model which we considered above as a variant of Burgers equation: Find the density $\rho(x,t)$ and the velocity $u(x,t)$ such that

$$\begin{aligned} \dot{\rho} + (u\rho)' &= 0 \quad \text{for } x \in \mathbb{R},\ t > 0, \\ \rho(x,0) &= \rho^0(x) \quad \text{for } x \in \mathbb{R}, \end{aligned} \tag{28.1}$$

combined with the constitutive law $u = (1-\rho)$, assuming $0 \leq \rho^(x) \leq 1$ for $x \in \mathbb{R}$.

28.2 Shocks and Congestions

We have seen that a shock corresponds to the abrupt changes of density and velocity encountered when the end of que of packed cars at rest is propagating in the opposite direction to the direction of the traffic.

28.3 Rarefactions and Traffic Lights

A rarefaction solution corresponds to the situation when at a traffic light turning green, the cars gradually accelerate one after the other. A nonphysical shock solution corresponds to the familiar situation when the driver in the first car at the traffic light, does not notice the light turning green, with the instability of this situation being illustrated by the fact that a slight "honk" usually will alert the driver to get going and allowing the rarefaction to develop. Instead of a honk a little bit of viscosity will get the car going.

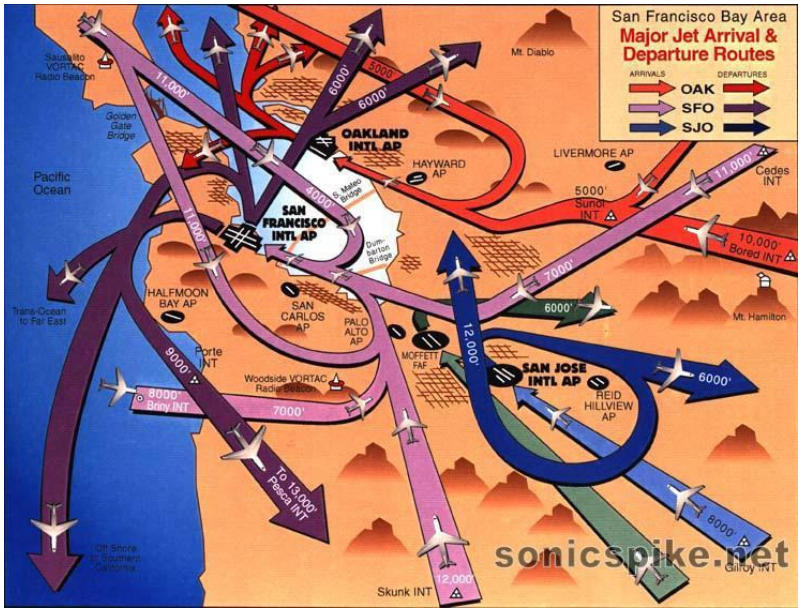


Figure 28.1: Traffic flow in the Bay Area

Chapter 29

Economy

> As every individual, therefore, endeavours as much as he can both to employ his capital in the support of domestic industry, and so to direct that industry that its produce may be of the greatest value; every individual necessarily labours to render the annual value of society as great as he can. He generally, indeed, neither intends to promote the public interest, nor knows how much he is promoting it. By preferring the support of domestic to that of foreign industry, he intends only his own security; and by directing that industry in such a manner as its produce may be of the greatest value, he intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention. Nor is it always the worse for the society that it was no part of it. By pursuing his own interest he frequently promotes that of society more effectually than when he really intends to promote it. I have never known much good done by those who affected to trade for the public good. It is an affectation, indeed, not very common among merchants, and very few words need be employed in dissuading them from it. (Adam Smith)

29.1 A Flow Model

We model an economy as a form of of reactive flow in a network.

Chapter 30

Biology: RealNSE with Metabolism

A living cell sustains itself far from equilibrium by a continuous turnover of matter and energy — intake of fuel, oxidation in mitochondria, export of waste heat. At the level of continuum mechanics this is a reacting, diffusing, dissipating flow on a complex internal geometry: RealNSE with metabolism. Glycolysis and the citric acid cycle deliver chemical free energy at a controlled rate; ATP turnover couples that energy to mechanical work in muscle and ion-pumping work across membranes; the by-products diffuse out as low-grade heat. Schrödinger's "negative entropy fed upon by the organism" is, in the RealNSE reading, simply the maintained gradient that the organism must keep open to avoid the equilibration to which the ambient dissipation would otherwise drive it. A companion browser simulation `cell_metabolism_cpu.html` *is forthcoming.*

Chapter 31

Process Industry: RealNSE with Unit Operations

The chemical and process industry — distillation columns, reactors, heat exchangers, separators, fluidised beds, combustion chambers — realises ThermoDynamics with explicit engineering constraints: throughput, selectivity, energy integration, environmental footprint. Each unit operation is a controlled continuum flow with dissipation and (often) chemical reaction; the plant as a whole is a network of such units coupled by streams and heat-recovery loops. The RealNSE apparatus of this volume — a compressible reacting flow with positive dissipation $D > 0$ resolved at finite precision — is the same apparatus that the discipline of process systems engineering applies under names like CFD-coupled flowsheet simulation, pinch analysis, and exergy accounting. A companion sequence of unit-operation simulations is forthcoming.

Chapter 32

Climate: RealNSE with Radiation and the Hydrologic Cycle

The Earth's atmosphere is a thin, rotating, stratified compressible flow driven by an equator-to-pole insolation gradient, with phase change of water as its dominant internal energy reservoir and longwave radiation to space as its ultimate sink. At the level of continuum mechanics this is RealNSE coupled to a radiative transfer model and a moist-thermodynamics constitutive law — a substantially richer system than dry compressible flow, but the same kind of object: an irreversible continuum flow whose 2nd Law is the cumulative dissipation generated by the resolved-scale turbulence and the latent-heat release in convective and frontal systems. Where classical equilibrium thermodynamics gives only the gross radiative balance between solar input and outgoing longwave, RealNSE gives the dynamic process by which that balance is realised — with the same fifteen-minute browser-runnable codes scaled up. A companion simulation `climate_2d_cpu.html` *is forthcoming.*

Appendix A

Loschmidt's and Gibbs' Paradoxes

> How wonderful that we have met with a paradox. Now we have some hope of making progress.
> (Nils Bohr)

> There is apparently a contradiction between the law of increasing entropy and the principles of Newtonian mechanics, since the latter do not recognize any difference between past and future times. This is the so-called reversibility paradox (Umkehreinwand) which was advanced as an objection to Boltzmann's theory by Loschmidt 1876-77. (Translators foreword to Lectures on Gas Theory by Boltzmann).

> Neither Herr Boltzmann nor Herr Planck has given a definition of W... Usually W is put equal to the number of complexions. In order to calculate W, one needs a *complete* (molecular-mechanical) theory of the system under consideration. Therefore it is dubious whether the Boltzmann principle has any meaning without a *complete* molecular-mechanical theory or some other theory which describes the elementary processes (and such a theory is missing). (Einstein)

A.1 Scientific Paradoxes

We now present new resolutions of *Loschmidt's paradox* on irreversibility in reversible Hamiltonian systems and *Gibbs' paradox* on inextensivity of

Boltzmann's entropy. We first give some give some general comments on the role of paradoxes in science.

The above citation by the physicist Nils Bohr illustrates the role of a *paradox* in the development of a physical theory, as a striking formulation of a contradiction within the theory, or between a prediction of the theory and a factual observation. Scientific progress can be made by showing that the contradiction is only *apparent*, thus improving the understanding of the meaning of a theory remaining correct, or that the contradiction is *real*, thus showing that the theory is not correct and thereby opening for a new better theory to be developed. A theory predicting that an apple let free to fall, will stay where it is, will be contradicted by Newton's observation that the apple falls to the ground, and thus will have to be abandoned.

The first known paradoxes were formulated by Zeno of Elea (490-430 BC) on the (still) puzzling nature of motion: Zeno asked e.g. how it can be that a flying arrow, which at each instant looks just the same as a motionless arrow, can be moving? Modern physics harbors many unresolved paradoxes including the wave-particle duality of quantum mechanics and time dilation of special relativity.

Since an unresolved paradox is a deadly threat to a theory, it has to be resolved by showing that the contradiction is only apparent and not real, "at any price, no matter how high that might be..." from the above citation of the famous physicist Max Planck facing in 1900 the paradox of the ultra-violet catastrophy in the classical theory of black-body radiation.

If a paradox cannot be resolved by showing that the contradiction is only apparent and not real, it can be *covered up* by presenting new information. When asked where the tortoises can be seen, which according to a certain (ancient) theory support the Earth, the cover up would be to claim that they are invisible (but still there).

In Loschmidt's paradox formulated in 1876 [40], mathematics of Hamiltonian systems (with zero viscosity) predicts that time reversal and a perpetuum mobile is possible. But everybody (except possibly a physics specialist) knows that time is always moving forward and that a perpetuum mobile is impossible, even though ultimately the World is based on Hamiltonian (quantum) mechanics.

In Gibbs' paradox formulated in 1875, statistical mechanics predicts that removing a membrane separating two volumes of equal gases at rest in the same state (same density, temperature and pressure), will result in a significant increase of entropy. But everybody (except a statistical mechanics

specialist) understands that removing (or inserting) a membrane in such a situation changes nothing.

A.2 Cover Up

The cover up of Loschmidt's paradox by Boltzmann [7] is to introduce statistical mechanics based on microscopic games of roulette. Statistical mechanics poses difficulties of *falsification* required by Popper in a scientific theory, since the validity of Boltzmann's basic microscopic assumption of statistical independence in a gas with each mole consisting of $6 \cdot 10^{23}$ molecules, seems to be beyond the possibility of any kind of conceivable experiment or mathematics; only indirect evidence in the form of macroscopic observations seem to be possible, which is far from enough. In fact, it is known that Boltzmann's assumption can only be (nearly) true in the very special case of a very dilute gas with rare collisions, and the derivation of Boltzmann equations for more general situations seems to pose unsurmountable problems.

The cover up of Gibbs' paradox by Gibbs' himself [**?**], is to change the counting of microstates underlying Boltzmann's definition of entropy in statistical mechanics, to give no change of entropy for equal gases, but retaining a substantial entropy change as soon as the two gases differ slightly. We will below argue that such a discontinuous change is not scientifically convincing, and present a different resolution based on computational turbulence avoiding tricky statistics.

A.3 Resolution of Loschmidt's Paradox

We have seen that the Euler equations are formally reversible but in reality are irreversible, because pointwise solutions to the Euler equations are non-existing. The irreversiblity arises because pointwise conservation of momentum is impossible to achieve, while the flow cannot cease to exist. The only way out is to violate momentum balance and pay a penalty in the form of turbulent dissipation with an irreversible transfer from kinetic to heat energy.

In other words, pointwsie solutions of the Euler equations would have been reversible had they only existed, but they do not exist, and the weak dissipative solutions which do exist are irreversible and thus define an Arrow

of time.

More generally, Hamiltonian systems sufficiently complex to allow turbulence, have an Arrow of time defined without any reference to a concepts of entropy and statistical mechanics.

A.4 Gibbs' Paradox

On Boltzmann's tombstone the famous formula

$$S = k \log(W) \tag{A.1}$$

is engraved, where k is Boltzmann's constant ($k \approx 10^{-23}$ joules/kelvin) and W denotes the probability of a certain state representing the number of "complexions" or "microstates" corresponding to a "macrostate". Increasing entropy would then reflect that Nature would tend to move from less probable to more probable states or towards states with more complexions. However, the crucial questions *why* and *how* Nature would seek to always increase entropy, was left without any answer.

The definition $S = k \log(W)$ is different from the classical expression $S = S = \log \tau\rho^{-\gamma} = \log(\tau V^{\gamma})$ derived above, which illustrates some of confusion surrounding the classical concept of entropy in classical thermodynamics.

Boltzmann's statistical mechanics poses serious difficulties from scientific point of view and paradoxes line up. One of them is *Gibbs' paradox* [**?**] illustrating the difficulty of defining entropy by counting micro-states: Consider a volume V divided by a membrane into two equal volumes $V/2$ filled with two different types of gas at rest with the same density and temperature. Remove the membrane and let both gases expand to the double volume and come to rest in a mixed state. Gibbs recalls that according to Boltzmann the entropy of each gas would then increase by the factor $\gamma(\log(V) - \log(V/2)) = \gamma \log(2)$, since for constant temperature $S = \gamma \log(V)$ with V the volume of the gas. The entropy of the system would then also increase by a $\gamma \log(2)$ factor.

Gibbs then compares with the case with the gases being of the same type in which case removal of a membrane would change nothing and in particular the entropy should not change, in contrast to the above case. In other words, the entropy should be *extensive* in the sense that the entropy of a gas over a volume V should be equal to the sum of the entropies over two volumes $V/2$. To achieve this, Gibbs suggested to change the counting of microstates in the case of equal gases, taking into account permutations of equal molecules,

but was then faced with a paradoxical discontinuos behavior of the entropy: No change for the same type of gas, and a log(2) change as soon as the types of gas differ only the slightest.

A.5 New Resolution of Gibbs'Paradox

With our experience from the Joule-Thomson experiment, we can easily resolve this paradox. It suffices to note that an initial pressure difference resulting from different state equations for two types of gas in the two volumes, would drive a process towards equal pressure in the full volume, with the associated turbulent dissipation (entropy production) being small if the initial pressure difference is small, that is, if the type of gas is nearly the same. In particular, for equal gases nothing would happen and there would be no sudden entropy jump under a slight change of gas, as in Gibbs' resolution by recounting microstates.

We understand that the Gibbs'paradox results from Boltzmann's idea to define entropy by counting microstates, and thus is (one of many) indications that statistical mechanics creates more problems than it solves, and from scientific point of view therefore is questionable.

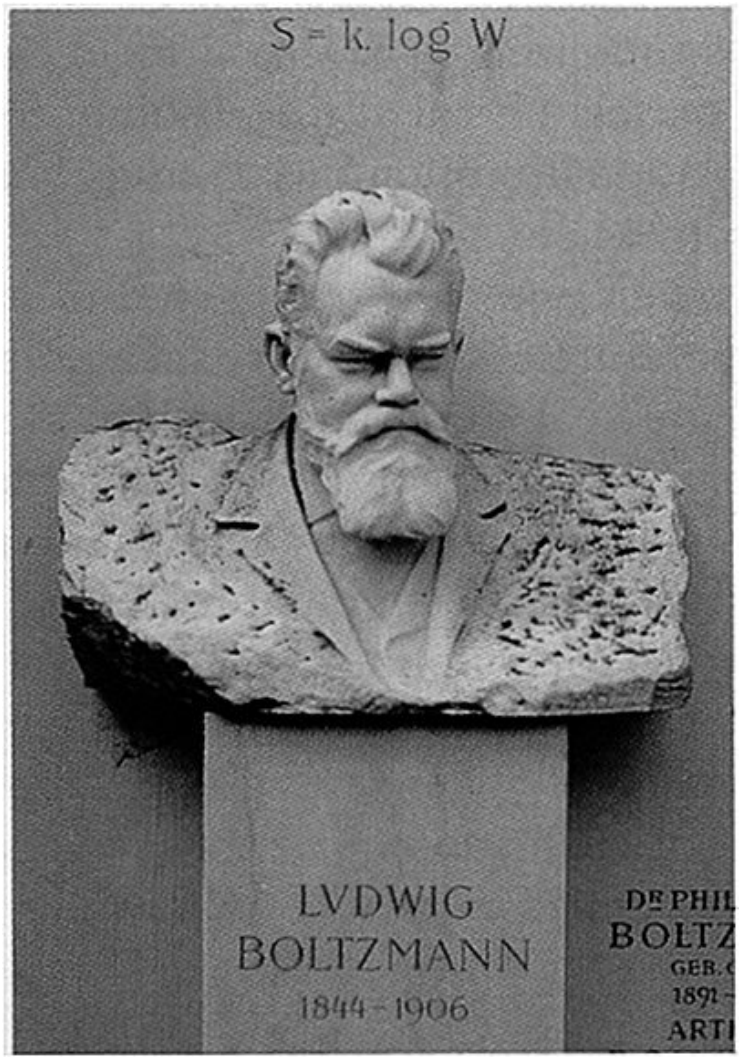


Figure A.1: Boltzmann's tombstone with $S = k \log(W)$.

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